

VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



DATABASE SYSTEMS (CO2013)

Assignment 2

“Mentor Finder for Students”

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Class: L06

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Danh sách thành viên và nhiệm vụ

STT	Họ và tên	MSSV	Nhiệm vụ	Hoàn thành	Ký tên
1	Dương Gia Bảo	2310207	- Tạo bảng dữ liệu và thêm dữ liệu mẫu. - Hiện thực thủ tục chèn dữ liệu vào bảng. - Quản lý source code.	100%	
2	Trần Minh Đăng	2310735	- Hiện thực các trigger liên quan tới ràng buộc ngữ nghĩa. - Hiện thực giao diện thông tin Mentor.	100%	
3	Phạm Thành Trí	2313621	- Hiện thực thủ tục cập nhật và xóa dữ liệu vào bảng. - Hiện thực giao diện quản lý Slot cho Mentor.	100%	
4	Nguyễn Minh Thái	2313101	- Hiện thực hàm tìm mã giảm giá và hàm tính thời gian phản hồi trung bình. - Hiện thực giao diện thông tin mã giảm giá ở trang Booking.	100%	



STT	Họ và tên	MSSV	Nhiệm vụ	Hoàn thành	Ký tên
5	Phan Thị Cẩm Vân	2313873	- Hiện thực thủ tục tìm kiếm Mentor và thống kê Dashboard. - Hiện thực giao diện tìm kiếm Mentor và giao diện thống kê của Admin.	100%	

Bảng 1: Danh sách thành viên và nhiệm vụ

1 Tạo bảng và dữ liệu mẫu

Dựa trên thiết kế EERD và lược đồ quan hệ đã phân tích ở Bài tập lớn 1, nhóm tiến hành hiện thực hóa cơ sở dữ liệu trên hệ quản trị SQL Server. Phần này trình bày các câu lệnh DDL để tạo bảng, thiết lập các ràng buộc toàn vẹn và DML để thêm dữ liệu mẫu.

1.1 Tạo bảng và các ràng buộc

Các bảng dữ liệu được tạo theo trình tự hợp lý nhằm đảm bảo toàn vẹn tham chiếu trong toàn bộ hệ thống. Các ràng buộc miền giá trị được triển khai trực tiếp bằng từ khóa CHECK. Đối với những ràng buộc ngữ nghĩa vượt ngoài phạm vi kiểm tra của CHECK, nhóm đã xây dựng và cài đặt các trigger tương ứng để đảm bảo dữ liệu luôn tuân thủ đúng quy tắc nghiệp vụ.

```
1 CREATE TABLE users (  
2     user_id INT IDENTITY(1,1) PRIMARY KEY,  
3     first_name NVARCHAR(50) NOT NULL,  
4     last_name NVARCHAR(50) NOT NULL,  
5     sex NVARCHAR(10) CHECK (sex IN (N'Male', N'Female')),  
6     created_at DATETIME DEFAULT GETDATE(),  
7     updated_at DATETIME NULL,  
8     email NVARCHAR(100) UNIQUE NOT NULL CHECK (  
9         email LIKE '%_@_%._%'  
10    ),  
11    password NVARCHAR(255) NULL,  
12    avatar_url NVARCHAR(255) NULL CHECK (  
13        avatar_url IS NULL OR avatar_url LIKE 'http://%' OR avatar_url  
14        LIKE 'https://%'  
15    ),  
16    country NVARCHAR(100) NULL,  
17    role NVARCHAR(50) CHECK (role IN (N'Mentee', N'Mentor', N'Admin'))  
    DEFAULT N'Mentee',  
    timezone NVARCHAR(50) NULL,
```

```
18     status NVARCHAR(20) CHECK (status IN (N'Active', N'Inactive', N'  
19     Banned', N'Pending')) DEFAULT N'Pending',  
20     is_email_verified BIT DEFAULT 0,  
21     otp NVARCHAR(6) NULL,  
22     otp_expiration DATETIME NULL,  
23     reset_password_token NVARCHAR(255) NULL,  
24     reset_password_token_expiration DATETIME NULL,  
25     google_id NVARCHAR(255) NULL,  
26     provider NVARCHAR(50) CHECK (provider IN (N'Local', N'Google'))  
27     DEFAULT N'Local',  
28     CHECK (  
29         (provider = N'Local' AND password IS NOT NULL AND LEN(password  
30         ) >= 60) OR  
31         (provider = N'Google' AND password IS NULL)  
32     )  
33 );  
34  
35 CREATE TABLE user_social_links (  
36     user_id INT NOT NULL,  
37     link NVARCHAR(255) NOT NULL CHECK (  
38         link LIKE 'http://%' OR link LIKE 'https://%'  
39     ) ,  
40     platform NVARCHAR(100) NOT NULL,  
41     PRIMARY KEY (user_id, link, platform),  
42     FOREIGN KEY (user_id) REFERENCES users(user_id)  
43     ON DELETE CASCADE  
44     ON UPDATE CASCADE  
45 );  
46  
47 CREATE TABLE notifications (  
48     no_id INT IDENTITY(1,1) PRIMARY KEY,  
49     title NVARCHAR(100) NOT NULL,  
50     content NVARCHAR(MAX) NOT NULL  
51 );  
52  
53 CREATE TABLE senders (  
54     sender_id INT NOT NULL,  
55     sender_name NVARCHAR(100) NOT NULL,  
56     sender_email NVARCHAR(100) NOT NULL,  
57     sender_phone NVARCHAR(20) NOT NULL,  
58     sender_address NVARCHAR(255) NOT NULL,  
59     sender_status NVARCHAR(20) CHECK (sender_status IN (N'Active', N'Inactive', N'  
60     Banned', N'Pending')) DEFAULT N'Pending',  
61     is_email_verified BIT DEFAULT 0,  
62     otp NVARCHAR(6) NULL,  
63     otp_expiration DATETIME NULL,  
64     reset_password_token NVARCHAR(255) NULL,  
65     reset_password_token_expiration DATETIME NULL,  
66     google_id NVARCHAR(255) NULL,  
67     provider NVARCHAR(50) CHECK (provider IN (N'Local', N'Google'))  
68     DEFAULT N'Local',  
69     CHECK (  
70         (provider = N'Local' AND password IS NOT NULL AND LEN(password  
71         ) >= 60) OR  
72         (provider = N'Google' AND password IS NULL)  
73     )  
74 );
```

```
51     u_user_id INT NOT NULL ,
52     n_no_id INT NOT NULL ,
53     sent_time DATETIME DEFAULT GETDATE() ,
54     PRIMARY KEY (u_user_id, n_no_id),
55     FOREIGN KEY (u_user_id) REFERENCES users(user_id)
56         ON DELETE CASCADE
57         ON UPDATE CASCADE ,
58     FOREIGN KEY (n_no_id) REFERENCES notifications(no_id)
59         ON DELETE CASCADE
60         ON UPDATE CASCADE
61 );
62
63 CREATE TABLE mentees (
64     user_id INT PRIMARY KEY ,
65     goal NVARCHAR(MAX) ,
66     FOREIGN KEY (user_id) REFERENCES users(user_id)
67         ON DELETE CASCADE
68         ON UPDATE CASCADE
69 );
70
71 CREATE TABLE mentors (
72     user_id INT PRIMARY KEY ,
73     bio NVARCHAR(MAX) ,
74     headline NVARCHAR(255) ,
75     response_time NVARCHAR(100) NOT NULL CHECK (
76         response_time IN (N'Within 1 hour', N'Within 3 hours', N'Within
77         6 hours', N'Within 12 hours', N'Within 24 hours', N'Within 36 hours
78         ', N'Within 48 hours', N'More than 48 hours', N'No responses yet')
79     ),
80     cv_url NVARCHAR(255) NOT NULL CHECK (
81         cv_url LIKE 'http://%' OR cv_url LIKE 'https://%'
82     ),
83     bank_name NVARCHAR(255) NULL CHECK (
84         bank_name IS NULL OR LEN(LTRIM(RTRIM(bank_name))) >= 2
85     ),
86     account_number NVARCHAR(50) NULL CHECK (
```

```
85         account_number IS NULL OR (LEN(account_number) >= 8 AND
account_number NOT LIKE '%[~0-9]%')
86     ),
87     account_holder_name NVARCHAR(255) NULL CHECK (
88         account_holder_name IS NULL OR LEN(LTRIM(RTRIM(
account_holder_name))) >= 2
89     ),
90     bank_branch NVARCHAR(255) NULL CHECK (
91         bank_branch IS NULL OR LEN(LTRIM(RTRIM(bank_branch))) >= 2
92     ),
93     swift_code NVARCHAR(50) NULL CHECK (
94         swift_code IS NULL OR (
95             LEN(swift_code) IN (8, 11) AND
96             swift_code COLLATE Latin1_General_BIN LIKE '[A-Z][A-Z][A-Z][A-
Z][A-Z][A-Z]%'
97         )
98     ),
99     FOREIGN KEY (user_id) REFERENCES users(user_id)
100         ON DELETE CASCADE
101         ON UPDATE CASCADE
102 );
103
104 CREATE TABLE mentor_languages (
105     mentor_id INT NOT NULL,
106     mentor_language NVARCHAR(100) NOT NULL,
107     PRIMARY KEY (mentor_id, mentor_language),
108     FOREIGN KEY (mentor_id) REFERENCES mentors(user_id)
109         ON DELETE CASCADE
110         ON UPDATE CASCADE
111 );
112
113 CREATE TABLE companies (
114     company_id INT IDENTITY(1,1) PRIMARY KEY,
115     cname NVARCHAR(255) NOT NULL
116 );
117
```

```
118 CREATE TABLE job_title (  
119     job_title_id INT IDENTITY(1,1) PRIMARY KEY,  
120     job_name NVARCHAR(255) NOT NULL  
121 );  
122  
123 CREATE TABLE work_for (  
124     mentor_id INT NOT NULL,  
125     c_company_id INT NOT NULL,  
126     current_job_title_id INT NOT NULL,  
127     PRIMARY KEY (mentor_id, c_company_id),  
128     FOREIGN KEY (mentor_id) REFERENCES mentors(user_id)  
129         ON DELETE CASCADE  
130         ON UPDATE CASCADE,  
131     FOREIGN KEY (c_company_id) REFERENCES companies(company_id)  
132         ON DELETE CASCADE  
133         ON UPDATE CASCADE,  
134     FOREIGN KEY (current_job_title_id) REFERENCES job_title(  
135         job_title_id)  
136         ON DELETE CASCADE  
137         ON UPDATE CASCADE  
138 );  
139  
139 CREATE TABLE messages (  
140     message_id INT IDENTITY(1,1) PRIMARY KEY,  
141     content NVARCHAR(MAX) NOT NULL,  
142     sent_time DATETIME DEFAULT GETDATE(),  
143     sender_id INT NOT NULL,  
144     receiver_id INT NOT NULL,  
145     FOREIGN KEY (sender_id) REFERENCES users(user_id)  
146         ON DELETE NO ACTION  
147         ON UPDATE NO ACTION,  
148     FOREIGN KEY (receiver_id) REFERENCES users(user_id)  
149         ON DELETE NO ACTION  
150         ON UPDATE NO ACTION  
151 );  
152
```

```
153 CREATE TABLE feedbacks (  
154     mentee_id INT NOT NULL,  
155     mentor_id INT NOT NULL,  
156     stars INT CHECK (stars BETWEEN 1 AND 5),  
157     content NVARCHAR(MAX) NOT NULL CHECK (  
158         LEN(LTRIM(RTRIM(content))) > 0  
159     ),  
160     sent_time DATETIME DEFAULT GETDATE(),  
161     PRIMARY KEY (mentee_id, mentor_id),  
162     FOREIGN KEY (mentee_id) REFERENCES mentees(user_id)  
163         ON DELETE NO ACTION  
164         ON UPDATE NO ACTION,  
165     FOREIGN KEY (mentor_id) REFERENCES mentors(user_id)  
166         ON DELETE NO ACTION  
167         ON UPDATE NO ACTION  
168 );  
169  
170 CREATE TABLE categories (  
171     category_id INT IDENTITY(1,1) PRIMARY KEY,  
172     category_name NVARCHAR(100) NOT NULL,  
173     super_category_id INT NULL,  
174     FOREIGN KEY (super_category_id) REFERENCES categories(category_id)  
175         ON DELETE NO ACTION  
176         ON UPDATE NO ACTION  
177 );  
178  
179 CREATE TABLE skills (  
180     skill_id INT IDENTITY(1,1) PRIMARY KEY,  
181     skill_name NVARCHAR(100) NOT NULL  
182 );  
183  
184 CREATE TABLE set_skill(  
185     mentor_id INT NOT NULL,  
186     skill_id INT NOT NULL,  
187     PRIMARY KEY (mentor_id, skill_id),  
188     FOREIGN KEY (mentor_id) REFERENCES mentors(user_id)
```

```
189         ON DELETE CASCADE
190         ON UPDATE CASCADE,
191     FOREIGN KEY (skill_id) REFERENCES skills(skill_id)
192         ON DELETE CASCADE
193         ON UPDATE CASCADE
194 );
195
196 CREATE TABLE own_skill(
197     category_id INT NOT NULL,
198     skill_id INT NOT NULL,
199     PRIMARY KEY (category_id, skill_id),
200     FOREIGN KEY (category_id) REFERENCES categories(category_id)
201         ON DELETE CASCADE
202         ON UPDATE CASCADE,
203     FOREIGN KEY (skill_id) REFERENCES skills(skill_id)
204         ON DELETE CASCADE
205         ON UPDATE CASCADE
206 );
207
208 CREATE TABLE plans(
209     plan_id INT IDENTITY(1,1) PRIMARY KEY,
210     plan_description NVARCHAR(MAX) NOT NULL,
211     plan_charge DECIMAL(10,2) NOT NULL,
212     plan_type NVARCHAR(50) NOT NULL,
213     mentor_id INT NOT NULL,
214     FOREIGN KEY (mentor_id) REFERENCES mentors(user_id)
215         ON DELETE CASCADE
216         ON UPDATE CASCADE
217 );
218
219 CREATE TABLE plan_sessions(
220     sessions_id INT PRIMARY KEY,
221     sessions_duration INT NOT NULL CHECK (sessions_duration >= 15 AND
sessions_duration <= 120),
222     FOREIGN KEY (sessions_id) REFERENCES plans(plan_id)
223         ON DELETE CASCADE
```

```
224         ON UPDATE CASCADE
225 );
226
227 CREATE TABLE plan_mentorships(
228     mentorships_id INT PRIMARY KEY,
229     calls_per_month INT NOT NULL CHECK (calls_per_month > 0),
230     minutes_per_call INT NOT NULL CHECK (minutes_per_call >= 15 AND
minutes_per_call <= 120),
231     FOREIGN KEY (mentorships_id) REFERENCES plans(plan_id)
232         ON DELETE CASCADE
233         ON UPDATE CASCADE
234 );
235
236 CREATE TABLE mentorships_benefits(
237     mentorships_id INT NOT NULL,
238     benefit_description NVARCHAR(255) NOT NULL,
239     PRIMARY KEY (mentorships_id, benefit_description),
240     FOREIGN KEY (mentorships_id) REFERENCES plan_mentorships(
mentorships_id)
241         ON DELETE CASCADE
242         ON UPDATE CASCADE
243 );
244
245 CREATE TABLE discounts(
246     discount_id INT IDENTITY(1,1) PRIMARY KEY,
247     discount_name NVARCHAR(100) NOT NULL,
248     discount_type NVARCHAR(50) CHECK (discount_type IN (N'Percentage',
N'Fixed')),
249     discount_value DECIMAL(10,2) NOT NULL,
250     start_date DATETIME NOT NULL,
251     end_date DATETIME NOT NULL,
252     status NVARCHAR(20) CHECK (status IN (N'Active', N'Inactive'))
DEFAULT N'Active',
253     usage_limit INT NOT NULL,
254     used_count INT DEFAULT 0
255 );
```



```
256
257 CREATE TABLE plan_registrations(
258     registration_id INT IDENTITY(1,1) PRIMARY KEY,
259     message NVARCHAR(MAX) NOT NULL,
260     discount_id INT NULL,
261     FOREIGN KEY (discount_id) REFERENCES discounts(discount_id)
262         ON DELETE CASCADE
263         ON UPDATE CASCADE
264 );
265
266 CREATE TABLE bookings(
267     mentee_id INT NOT NULL,
268     plan_registrations_id INT NOT NULL,
269     plan_id INT NOT NULL,
270     PRIMARY KEY (mentee_id, plan_registrations_id),
271     FOREIGN KEY (mentee_id) REFERENCES mentees(user_id)
272         ON DELETE CASCADE
273         ON UPDATE CASCADE,
274     FOREIGN KEY (plan_registrations_id) REFERENCES
275     plan_registrations(registration_id)
276         ON DELETE CASCADE
277         ON UPDATE CASCADE,
278     FOREIGN KEY (plan_id) REFERENCES plans(plan_id)
279         ON DELETE NO ACTION
280         ON UPDATE NO ACTION
281 );
282
283 CREATE TABLE slots(
284     start_time DATETIME NOT NULL,
285     end_time DATETIME NOT NULL,
286     date DATE NOT NULL,
287     mentor_id INT NOT NULL,
288     status NVARCHAR(20) CHECK (status IN (N'Available', N'Booked'))
289     DEFAULT N'Available',
290     plan_id INT NOT NULL,
291     PRIMARY KEY (mentor_id, start_time, end_time, date),
```

```
290 FOREIGN KEY (mentor_id) REFERENCES mentors(user_id)
291 ON DELETE CASCADE
292 ON UPDATE CASCADE,
293 FOREIGN KEY (plan_id) REFERENCES plans(plan_id)
294 ON DELETE NO ACTION
295 ON UPDATE NO ACTION,
296 CHECK (end_time > start_time),
297 CHECK (DATEDIFF(MINUTE, start_time, end_time) BETWEEN 15 AND 120)
298 );
299
300 CREATE TABLE invoices(
301     invoice_id INT IDENTITY(1,1) PRIMARY KEY,
302     plan_registrations_id INT NOT NULL,
303     method NVARCHAR(50) NOT NULL,
304     paid_time DATETIME DEFAULT GETDATE(),
305     mentee_id INT NOT NULL,
306     stripe_session_id NVARCHAR(255) NULL,
307     stripe_customer_id NVARCHAR(255) NULL,
308     stripe_customer_email NVARCHAR(255) NULL,
309     stripe_payment_intent_id NVARCHAR(255) NULL,
310     stripe_charge_id NVARCHAR(255) NULL,
311     stripe_balance_transaction_id NVARCHAR(255) NULL,
312     stripe_receipt_url NVARCHAR(500) NULL,
313     payment_status NVARCHAR(50) NULL CHECK (payment_status IN (N'paid',
314     N'failed')),
315     currency NVARCHAR(10) NULL CHECK (currency IN (N'usd', N'vnd')),
316     amount_subtotal DECIMAL(10,2) NOT NULL,
317     amount_total DECIMAL(10,2) NOT NULL,
318     UNIQUE (invoice_id, plan_registrations_id),
319     FOREIGN KEY (plan_registrations_id) REFERENCES
320     plan_registrations(registration_id)
321     ON DELETE CASCADE
322     ON UPDATE CASCADE,
323     FOREIGN KEY (mentee_id) REFERENCES mentees(user_id)
324     ON DELETE CASCADE
325     ON UPDATE CASCADE
```

```
324 );
325
326 CREATE TABLE meetings(
327     meeting_id INT IDENTITY(1,1) PRIMARY KEY,
328     invoice_id INT NOT NULL,
329     plan_registrations_id INT NOT NULL,
330     status NVARCHAR(20) CHECK (status IN (N'Pending',N'Scheduled', N'
Completed', N'Cancelled')) DEFAULT N'Pending',
331     location NVARCHAR(255) NOT NULL,
332     review_link NVARCHAR(500) NULL CHECK
333     (review_link IS NULL OR review_link LIKE 'http://%' OR review_link
LIKE 'https://%'),
334     start_time DATETIME NOT NULL,
335     end_time DATETIME NOT NULL,
336     date DATE NOT NULL,
337     mentor_id INT NOT NULL,
338     UNIQUE (meeting_id, invoice_id, plan_registrations_id),
339     FOREIGN KEY (invoice_id, plan_registrations_id) REFERENCES
invoices(invoice_id, plan_registrations_id)
340         ON DELETE CASCADE
341         ON UPDATE CASCADE,
342     FOREIGN KEY (mentor_id, start_time, end_time, date) REFERENCES
slots(mentor_id, start_time, end_time, date)
343         ON DELETE NO ACTION
344         ON UPDATE NO ACTION
345 );
346
347 -- Triggers
348
349 ALTER TABLE mentors
350 ADD total_mentee INT DEFAULT 0, --DERIVED ATTRIBUTES FOR MENTORS
351     total_stars INT DEFAULT 0,
352     total_reviews INT DEFAULT 0,
353     rating DECIMAL(3,2) DEFAULT 0;
354 GO
355
```

```
356 -- an account must be email verified before it can be active; applies
    when an account is created and when it is modified
357
358 CREATE TRIGGER trg_user_email_verify
359 ON users
360 FOR INSERT, UPDATE
361 AS
362 BEGIN
363     IF EXISTS (
364         SELECT 1 FROM inserted
365         WHERE status = 'Active' AND is_email_verified = 0
366     )
367     BEGIN
368         RAISERROR(N'Email phải được xác minh trước khi kích hoạt tài
khoản.', 16, 1);
369         ROLLBACK TRAN;
370     END
371 END;
372 GO
373
374 -- a mentor must have a skill before a plan for them can be made;
    applies when a plan is created
375
376 CREATE TRIGGER trg_mentor_must_have_skill
377 ON plans
378 FOR INSERT
379 AS
380 BEGIN
381     IF EXISTS (
382         SELECT 1
383         FROM inserted i
384         WHERE NOT EXISTS (
385             SELECT 1 FROM set_skill s
386             WHERE s.mentor_id = i.mentor_id
387         )
388     )
```

```
389 BEGIN
390     RAISERROR(N'Mentor phải có ít nhất một kỹ năng trước khi mở buổi tư vấn.', 16, 1);
391     ROLLBACK TRAN;
392 END
393 END;
394 GO
395
396 -- mentee must complete one meeting of a mentor in order to give
    feedback to that mentor; applies when a feedback is created
397
398 CREATE TRIGGER trg_feedback_only_after_meeting
399 ON feedbacks
400 FOR INSERT
401 AS
402 BEGIN
403     IF EXISTS (
404         SELECT 1
405         FROM inserted f
406         WHERE NOT EXISTS (
407             SELECT 1
408             FROM meetings m
409             JOIN invoices i ON m.invoice_id = i.invoice_id
410             WHERE i.mentee_id = f.mentee_id
411                 AND m.mentor_id = f.mentor_id
412                 AND m.status = 'Completed'
413         )
414     )
415     BEGIN
416         RAISERROR(N'Chỉ mentee đã hoàn thành ít nhất 1 buổi tư vấn mới được gửi đánh giá.', 16, 1);
417         ROLLBACK TRAN;
418     END
419 END;
420 GO
421
```

```
422 -- booked slot is blocked for other mentee once used; applies when a
    meeting is created on that slot
423
424 CREATE TRIGGER trg_slot_one_booking
425 ON meetings
426 FOR INSERT
427 AS
428 BEGIN
429     -- check if slot is booked
430     IF EXISTS (
431         SELECT 1
432         FROM inserted i
433         JOIN slots s ON
434             s.mentor_id = i.mentor_id
435             AND s.start_time = i.start_time
436             AND s.end_time = i.end_time
437             AND s.date = i.date
438         WHERE s.status = 'Booked'
439     )
440     BEGIN
441         RAISERROR(N'Phiên làm việc này đã được đăng ký bởi mentee khác
442 .', 16, 1);
443         ROLLBACK TRAN;
444         RETURN;
445     END
446
447 -- otherwise set slot to booked (and ofc no rollback)
448 UPDATE s SET status = 'Booked'
449 FROM slots s
450 JOIN inserted i ON
451     s.mentor_id = i.mentor_id
452     AND s.start_time = i.start_time
453     AND s.end_time = i.end_time
454     AND s.date = i.date;
455 END;
GO
```

```
456
457 -- a message once sent (created) can not be modified; applies when
    message is changed
458
459 CREATE TRIGGER trg_messages_no_edit
460 ON messages
461 FOR UPDATE
462 AS
463 BEGIN
464     RAISERROR(N'Tin nhắn không thể chỉnh sửa sau khi gửi.', 16, 1);
465     ROLLBACK TRAN;
466 END;
467 GO
468
469 --Derived attribute triggers
470
471 CREATE TRIGGER trg_meeting_completed
472 ON meetings
473 AFTER UPDATE
474 AS
475 BEGIN
476     -- check for record changing to completed
477     IF EXISTS (
478         SELECT 1
479         FROM inserted i
480         WHERE i.status = 'Completed'
481     )
482     BEGIN
483         UPDATE mentors
484         SET total_mentee =
485         (
486             SELECT COUNT(DISTINCT Mentee_id)
487             FROM meetings m
488             JOIN invoices i ON m.invoice_id = i.invoice_id
489             WHERE m.mentor_id = inserted.mentor_id
490             AND m.status = 'Completed'
```

```
491         )
492     FROM inserted
493     WHERE mentors.user_id = inserted.mentor_id;
494 END
495 END;
496 GO
497
498 CREATE TRIGGER trg_meeting_insert
499 ON meetings
500 AFTER INSERT
501 AS
502 BEGIN
503     UPDATE mentors
504     SET total_mentee =
505     (
506         SELECT COUNT(DISTINCT Mentee_id)
507         FROM meetings m
508         JOIN invoices i ON m.invoice_id = i.invoice_id
509         WHERE m.mentor_id = inserted.mentor_id
510             AND m.status = 'Completed'
511     )
512     FROM inserted
513     WHERE mentors.user_id = inserted.mentor_id;
514 END;
515 GO
516
517 CREATE TRIGGER trg_feedback_insert
518 ON feedbacks
519 AFTER INSERT
520 AS
521 BEGIN
522     UPDATE mentors
523     SET
524         total_reviews =
525         (SELECT COUNT(*)
526          FROM feedbacks F
```



```
527         WHERE F.mentor_id = mentors.user_id),
528
529     total_stars =
530         (SELECT SUM(F.stars)
531          FROM feedbacks F
532          WHERE F.mentor_id = mentors.user_id),
533
534     rating =
535         CASE
536             WHEN (SELECT COUNT(*)
537                  FROM feedbacks F
538                  WHERE F.mentor_id = mentors.user_id) = 0
539             THEN 0
540             ELSE (
541                 SELECT SUM(F.stars) * 1.0 / COUNT(*)
542                 FROM feedbacks F
543                 WHERE F.mentor_id = mentors.user_id
544             )
545         END
546     FROM inserted i
547     WHERE mentors.user_id = i.mentor_id;
548 END;
549 GO
550
551 CREATE TRIGGER trg_feedback_update
552 ON feedbacks
553 AFTER UPDATE
554 AS
555 BEGIN
556     IF UPDATE(stars)
557     BEGIN
558         UPDATE mentors
559         SET
560             total_reviews =
561                 (SELECT COUNT(*)
562                 FROM feedbacks F
```

```
563         WHERE F.mentor_id = mentors.user_id),
564
565     total_stars =
566         (SELECT SUM(F.stars)
567          FROM feedbacks F
568          WHERE F.mentor_id = mentors.user_id),
569
570     rating =
571         CASE
572             WHEN (SELECT COUNT(*)
573                  FROM feedbacks F
574                  WHERE F.mentor_id = mentors.user_id) = 0
575             THEN 0
576             ELSE (
577                 SELECT SUM(F.stars) * 1.0 / COUNT(*)
578                 FROM feedbacks F
579                 WHERE F.mentor_id = mentors.user_id
580             )
581         END
582     FROM inserted i
583     WHERE mentors.user_id = i.mentor_id;
584 END
585 END;
586 GO
587
588 CREATE TRIGGER trg_feedback_delete
589 ON feedbacks
590 AFTER DELETE
591 AS
592 BEGIN
593     UPDATE mentors
594     SET
595         total_reviews =
596             (SELECT COUNT(*)
597              FROM feedbacks F
598              WHERE F.mentor_id = mentors.user_id),
```

```
599
600     total_stars =
601         (SELECT SUM(F.stars)
602          FROM feedbacks F
603          WHERE F.mentor_id = mentors.user_id),
604
605     rating =
606         CASE
607             WHEN (SELECT COUNT(*)
608                  FROM feedbacks F
609                  WHERE F.mentor_id = mentors.user_id) = 0
610             THEN 0
611             ELSE (
612                  SELECT SUM(F.stars) * 1.0 / COUNT(*)
613                  FROM feedbacks F
614                  WHERE F.mentor_id = mentors.user_id
615                 )
616         END
617 FROM deleted d
618 WHERE mentors.user_id = d.mentor_id;
619 END;
620 GO
```

1.2 Tạo dữ liệu mẫu

1.2.1 Các câu lệnh thêm dữ liệu

Sau khi hoàn tất quá trình tạo bảng, tiếp tục thực thi các câu lệnh chèn dữ liệu bên dưới. Để tránh làm báo cáo trở nên quá dài, nhóm chỉ trình bày 5 dòng dữ liệu mẫu cho mỗi bảng.

```
1 -- INSERT DATA FOR MENTORIA DATABASE
2 -- Password for all users: Password123!
3 -- Hashed password: $2b$10$UJAn0gdXdvLWzjkVVPCCsuUvXRKE0.
   NdAGVfEud0vUTEyZaMlapH2
4
```

```
5 -- Insert Users (5 Mentors + 5 Mentees + 1 Admin = 11 total)
6 INSERT INTO users (first_name, last_name, sex, email, password,
    avatar_url, country, role, timezone, status, is_email_verified,
    provider)
7 VALUES
8 (N'John', N'Doe', N'Male', N'john.doe@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=1', N'United States', N'Mentor', N'
    America/New_York', N'Active', 1, N'Local'),
9 (N'Sarah', N'Johnson', N'Female', N'sarah.johnson@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=5', N'United Kingdom', N'Mentor', N'
    Europe/London', N'Active', 1, N'Local'),
10 (N'Michael', N'Chen', N'Male', N'michael.chen@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=12', N'Singapore', N'Mentor', N'Asia/
    Singapore', N'Active', 1, N'Local'),
11 (N'Emily', N'Rodriguez', N'Female', N'emily.rodriguez@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=9', N'Spain', N'Mentor', N'Europe/
    Madrid', N'Active', 1, N'Local'),
12 (N'David', N'Kim', N'Male', N'david.kim@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=15', N'South Korea', N'Mentor', N'
    Asia/Seoul', N'Active', 1, N'Local'),
13 (N'Alice', N'Smith', N'Female', N'alice.smith@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=20', N'Canada', N'Mentee', N'America/
    Toronto', N'Active', 1, N'Local'),
14 (N'Bob', N'Wilson', N'Male', N'bob.wilson@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=33', N'Australia', N'Mentee', N'
    Australia/Sydney', N'Active', 1, N'Local'),
15 (N'Carol', N'Taylor', N'Female', N'carol.taylor@example.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=27', N'Ireland', N'Mentee', N'Europe/
```

```
Dublin', N'Active', 1, N'Local'),
16 (N'Daniel', N'Brown', N'Male', N'daniel.brown@example.com', N'
    $2b$10$UJAn0gdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=51', N'Germany', N'Mentee', N'Europe/
    Berlin', N'Active', 1, N'Local'),
17 (N'Emma', N'Davis', N'Female', N'emma.davis@example.com', N'
    $2b$10$UJAn0gdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=45', N'New Zealand', N'Mentee', N'
    Pacific/Auckland', N'Active', 1, N'Local'),
18 (N'Admin', N'User', N'Male', N'admin@mentoria.com', N'
    $2b$10$UJAn0gdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=68', N'United States', N'Admin', N'
    America/New_York', N'Active', 1, N'Local');
19
20 -- Insert User Social Links
21 INSERT INTO user_social_links (user_id, link, platform)
22 VALUES
23 (1, N'https://linkedin.com/in/johndoe', N'LinkedIn'),
24 (2, N'https://linkedin.com/in/sarahjohnson', N'LinkedIn'),
25 (3, N'https://linkedin.com/in/michaelchen', N'LinkedIn'),
26 (4, N'https://linkedin.com/in/emilyrodriguez', N'LinkedIn'),
27 (5, N'https://linkedin.com/in/davidkim', N'LinkedIn');
28
29 -- Insert Mentees
30 INSERT INTO mentees (user_id, goal)
31 VALUES
32 (6, N'Transition to a full-stack developer role within 6 months'),
33 (7, N'Learn mobile app development and launch my first app'),
34 (8, N'Break into the UI/UX design field and build a strong portfolio')
    ,
35 (9, N'Master data science and machine learning for career advancement'
    ),
36 (10, N'Develop leadership skills and prepare for management role');
37
38 -- Insert Mentors
```

```
39 INSERT INTO mentors (user_id, bio, headline, response_time, cv_url,
    bank_name, account_number, account_holder_name, bank_branch,
    swift_code)
40 VALUES
41 (1, N'Experienced full-stack developer with 10+ years in web
    development. Specialized in React, Node.js, and cloud architecture.
    ', N'Senior Full-Stack Developer & Tech Lead', N'Within 24 hours',
    N'https://example.com/cv/johndoe.pdf', N'Chase Bank', N'1234567890',
    N'John Doe', N'New York Main Branch', N'CHASUS33'),
42 (2, N'Product designer with expertise in creating user-centered
    digital experiences. Worked with startups and Fortune 500 companies
    .', N'Lead Product Designer', N'Within 12 hours', N'https://example
    .com/cv/sarahjohnson.pdf', N'Barclays Bank', N'2345678901', N'Sarah
    Johnson', N'London Central', N'BARCGB22'),
43 (3, N'Data scientist and ML engineer specializing in NLP and computer
    vision. Published researcher and conference speaker.', N'Senior
    Data Scientist & ML Engineer', N'Within 48 hours', N'https://
    example.com/cv/michaelchen.pdf', N'DBS Bank', N'3456789012', N'
    Michael Chen', N'Singapore Marina Bay', N'DBSSSGSG'),
44 (4, N'Digital marketing strategist with experience in growth hacking
    and content marketing. Helped 50+ startups scale.', N'Growth
    Marketing Lead', N'Within 24 hours', N'https://example.com/cv/
    emilyrodriguez.pdf', N'Santander Bank', N'4567890123', N'Emily
    Rodriguez', N'Madrid Centro', N'BSCHESMM'),
45 (5, N'Career coach and leadership consultant. Former tech executive
    with 15+ years of experience guiding professionals.', N'Career
    Coach & Leadership Consultant', N'Within 24 hours', N'https://
    example.com/cv/davidkim.pdf', N'Kookmin Bank', N'5678901234', N'
    David Kim', N'Seoul Gangnam', N'CZNBKRSE');
46
47 -- Insert Mentor Languages
48 INSERT INTO mentor_languages (mentor_id, mentor_language)
49 VALUES
50 (1, N'English'),
51 (2, N'English'),
52 (3, N'English'),
```

```
53 (4, N'Spanish'),
54 (5, N'Korean');
55
56 -- Insert Companies
57 INSERT INTO companies (cname)
58 VALUES
59 (N'Google'),
60 (N'Microsoft'),
61 (N'Amazon'),
62 (N'Meta'),
63 (N'Apple');
64
65 -- Insert Job Titles
66 INSERT INTO job_title (job_name)
67 VALUES
68 (N'Senior Software Engineer'),
69 (N'Tech Lead'),
70 (N'Lead Product Designer'),
71 (N'ML Engineer'),
72 (N'Growth Marketing Manager');
73
74 -- Insert Work Experience
75 INSERT INTO work_for (mentor_id, c_company_id, current_job_title_id)
76 VALUES
77 (1, 1, 1),
78 (2, 3, 3),
79 (3, 4, 4),
80 (4, 2, 5),
81 (5, 5, 2);
82
83 -- Insert Categories - Parent categories
84 INSERT INTO categories (category_name, super_category_id)
85 VALUES
86 (N'Engineering Mentors', NULL),
87 (N'Design Mentors', NULL),
88 (N'AI Mentors', NULL),
```

```
89 (N'Product Managers', NULL),
90 (N'Marketing Coaches', NULL);
91
92 -- Insert Skills
93 INSERT INTO skills (skill_name)
94 VALUES
95 (N'React'),
96 (N'Node.js'),
97 (N'Python'),
98 (N'Figma'),
99 (N'SEO');
100
101 -- Insert Own Skill
102 INSERT INTO own_skill (category_id, skill_id)
103 VALUES
104 (1, 1),
105 (1, 2),
106 (2, 4),
107 (3, 3),
108 (5, 5);
109
110 -- Insert Set Skill
111 INSERT INTO set_skill (mentor_id, skill_id)
112 VALUES
113 (1, 1),
114 (1, 2),
115 (2, 4),
116 (3, 3),
117 (4, 5);
118
119 -- Insert Plans
120 INSERT INTO plans (plan_description, plan_charge, plan_type, mentor_id
    )
121 VALUES
122 (N'30-minute focused mentorship session covering React, Node.js, and
    web development best practices', 50.00, N'Beginner', 1),
```



```
123 (N'Portfolio review and UX design consultation session', 60.00, N'  
    Beginner', 2),  
124 (N'ML/AI consultation session with hands-on guidance', 75.00, N'  
    Beginner', 3),  
125 (N'Marketing strategy consultation and growth tactics', 45.00, N'  
    Beginner', 4),  
126 (N'Career coaching session with resume review', 70.00, N'Beginner', 5)  
    ;  
127  
128 -- Insert Plan Sessions  
129 INSERT INTO plan_sessions (sessions_id, sessions_duration)  
130 VALUES  
131 (1, 30),  
132 (2, 60),  
133 (3, 75),  
134 (4, 45),  
135 (5, 70);  
136  
137 -- Insert Discounts  
138 INSERT INTO discounts (discount_name, discount_type, discount_value,  
    start_date, end_date, status, usage_limit, used_count)  
139 VALUES  
140 (N'WELCOME10', N'Percentage', 10.00, '2025-01-01', '2026-12-31', N'  
    Active', 1000, 15),  
141 (N'SAVE20', N'Percentage', 20.00, '2025-06-01', '2026-06-30', N'Active  
    ', 500, 42),  
142 (N'FIRST50', N'Fixed', 50.00, '2025-01-01', '2026-12-31', N'Active',  
    200, 67),  
143 (N'MENTOR25', N'Percentage', 25.00, '2025-09-01', '2025-03-31', N'  
    Inactive', 300, 89),  
144 (N'EARLYBIRD', N'Percentage', 15.00, '2025-01-01', '2026-12-31', N'  
    Inactive', 150, 150);  
145  
146 -- Insert Plan Registrations  
147 INSERT INTO plan_registrations (message, discount_id)  
148 VALUES
```

```
149 (N'Looking forward to learning web development with John!', 1),
150 (N'Excited to improve my design skills with Sarah', 2),
151 (N'Ready to dive into machine learning', 3),
152 (N'Need guidance on marketing strategies', 4),
153 (N'Want to advance my career to the next level', 1);
154
155 -- Insert Bookings
156 INSERT INTO bookings (mentee_id, plan_registrations_id, plan_id)
157 VALUES
158 (6, 1, 1),
159 (7, 2, 2),
160 (8, 3, 3),
161 (9, 4, 4),
162 (10, 5, 5);
163
164 -- Insert Slots
165 INSERT INTO slots (start_time, end_time, date, mentor_id, status,
166                    plan_id)
167 VALUES
168 ('2025-12-10 09:00:00', '2025-12-10 09:30:00', '2025-12-10', 1, N'
169     Available', 1),
170 ('2025-12-10 10:00:00', '2025-12-10 11:00:00', '2025-12-10', 2, N'
171     Available', 2),
172 ('2025-12-10 11:00:00', '2025-12-10 12:15:00', '2025-12-10', 3, N'
173     Available', 3),
174 ('2025-12-10 12:00:00', '2025-12-10 12:45:00', '2025-12-10', 4, N'
175     Available', 4),
176 ('2025-12-10 13:00:00', '2025-12-10 14:10:00', '2025-12-10', 5, N'
177     Available', 5);
178
179 -- Insert Invoices
180 INSERT INTO invoices (plan_registrations_id, method, paid_time,
181                      mentee_id, payment_status, currency, amount_subtotal, amount_total)
182 VALUES
183 (1, N'Credit Card', '2025-12-05 10:30:00', 6, N'paid', N'usd', 50.00,
184    45.00),
```

```
177 (2, N'PayPal', '2025-12-05 11:15:00', 7, N'paid', N'usd', 60.00,
    48.00),
178 (3, N'Credit Card', '2025-12-05 12:00:00', 8, N'paid', N'usd', 75.00,
    25.00),
179 (4, N'Debit Card', '2025-12-05 13:45:00', 9, N'paid', N'usd', 45.00,
    33.75),
180 (5, N'Credit Card', '2025-12-05 14:20:00', 10, N'paid', N'usd', 70.00,
    63.00);
181
182 -- Insert Meetings
183 INSERT INTO meetings (invoice_id, plan_registrations_id, status,
    location, start_time, end_time, date, mentor_id)
184 VALUES
185 (1, 1, N'Completed', N'https://zoom.us/j/123456789', '2025-12-10
    09:00:00', '2025-12-10 09:30:00', '2025-12-10', 1),
186 (2, 2, N'Completed', N'https://meet.google.com/abc-defg-hij', '
    2025-12-10 10:00:00', '2025-12-10 11:00:00', '2025-12-10', 2),
187 (3, 3, N'Completed', N'https://zoom.us/j/987654321', '2025-12-10
    11:00:00', '2025-12-10 12:15:00', '2025-12-10', 3),
188 (4, 4, N'Completed', N'https://meet.google.com/xyz-uvw-x-zya', '
    2025-12-10 12:00:00', '2025-12-10 12:45:00', '2025-12-10', 4),
189 (5, 5, N'Completed', N'https://zoom.us/j/555666777', '2025-12-10
    13:00:00', '2025-12-10 14:10:00', '2025-12-10', 5);
190
191 -- Insert Feedbacks
192 INSERT INTO feedbacks (mentee_id, mentor_id, stars, content, sent_time
    )
193 VALUES
194 (6, 1, 5, N'Excellent mentor! John provided great insights on React
    architecture.', '2025-12-11 16:30:00'),
195 (7, 2, 5, N'Sarah is amazing! Her design feedback was spot-on.', '
    2025-12-11 19:45:00'),
196 (8, 3, 4, N'Very knowledgeable about ML. Michael explained concepts
    clearly.', '2025-12-11 13:30:00'),
197 (9, 4, 5, N'Emily helped me create a comprehensive marketing strategy.
    ', '2025-12-11 14:00:00'),
```

```
198 (10, 5, 5, N'David provided excellent career guidance.', '2025-12-11
    15:30:00');
199
200 -- Insert Notifications
201 INSERT INTO notifications (title, content)
202 VALUES
203 (N>Welcome to Mentoria!', N>Thank you for joining Mentoria.'),
204 (N>New Mentor Available', N>Check out our latest mentors.'),
205 (N>Session Reminder', N>Your session is scheduled for tomorrow.'),
206 (N>Special Discount', N>Use code SAVE20 for 20% off!'),
207 (N>Profile Update', N>Complete your profile for better matching.');
```

```
208
209 -- Insert Sended
210 INSERT INTO sended (u_user_id, n_no_id, sent_time)
211 VALUES
212 (6, 1, '2025-12-01 10:00:00'),
213 (7, 2, '2025-12-01 10:30:00'),
214 (8, 3, '2025-12-01 11:00:00'),
215 (9, 4, '2025-12-01 11:30:00'),
216 (10, 5, '2025-12-01 12:00:00');
```

```
217
218 -- Insert Messages
219 INSERT INTO messages (content, sent_time, sender_id, receiver_id)
220 VALUES
221 (N>Hi John! Excited about our session on React.', '2024-12-08 10:00:00',
    6, 1),
222 (N>Hello Alice! Looking forward to it.', '2024-12-08 11:30:00', 1, 6),
223 (N>Hi Sarah, questions about my portfolio.', '2024-12-07 14:00:00', 7,
    2),
224 (N>Of course Bob! Send it over.', '2024-12-07 15:45:00', 2, 7),
225 (N>Michael, could we discuss NLP?', '2024-12-06 09:30:00', 8, 3);
```

1.2.2 Kết quả thu được

Sau khi thực thi các câu lệnh chèn dữ liệu, thu được các bảng kết quả như sau.

SQLQuery7.sq...ia (sa (70)) SQLQuery6.sql...oria (sa (88)) SQLQuery5.sql...oria (sa (86))

```
1 SELECT TOP (1000) [mentee_id]
2 , [plan_registrations_id]
3 , [plan_id]
4 FROM [test_mentoria].[dbo].[bookings]
5
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

	mentee_id	plan_registrations_id	plan_id
1	31	1	1
2	31	6	2
3	31	23	58
4	32	2	4
5	32	7	5
6	32	24	61
7	33	3	7
8	33	25	63
9	34	4	11
10	34	26	66
11	35	5	14
12	35	27	68
13	36	8	18
14	36	28	70
15	37	9	21
16	37	29	72
17	38	10	24

Query executed successfully. localhost (16.0 RTM) sa (70) test_mentoria 00:00:00 Row: 1, Col: 1 31 rows

Hình 1: Bảng bookings

SQLQuery6.sq...ia (sa (88)) SQLQuery5.sql...oria (sa (86))

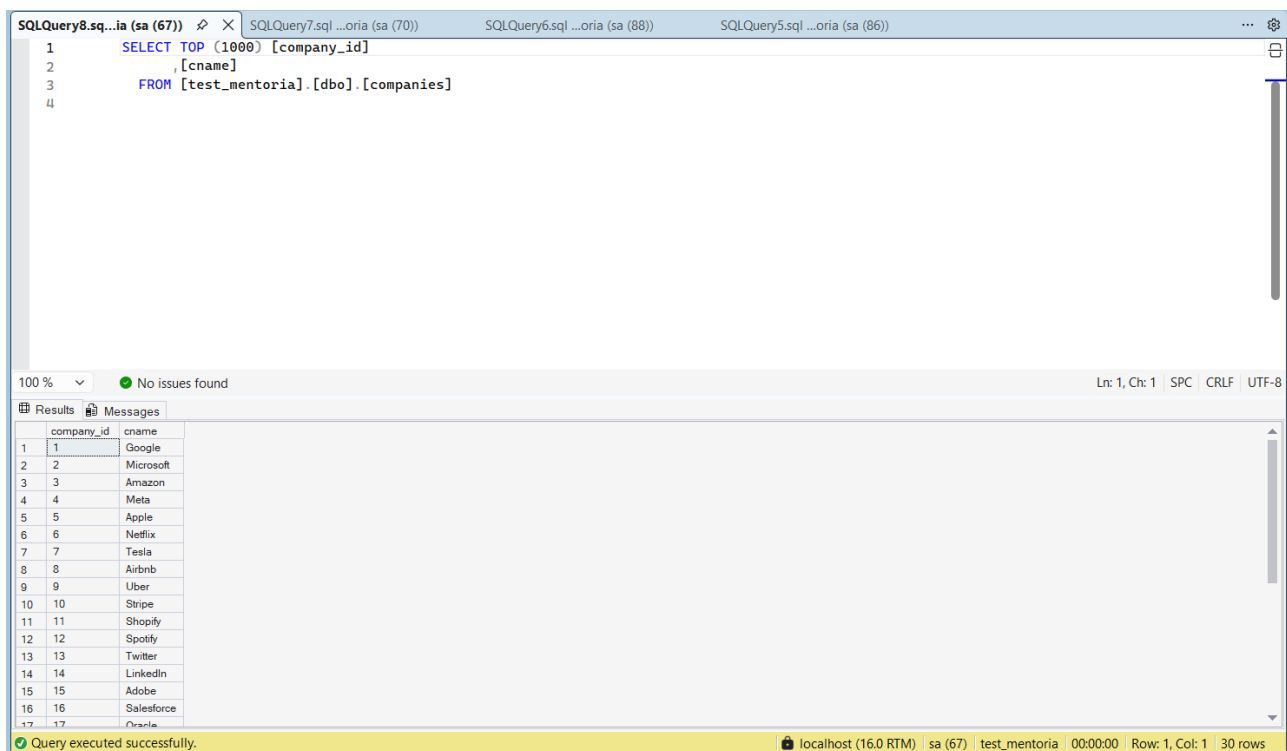
```
1 SELECT TOP (1000) [category_id]
2 , [category_name]
3 , [super_category_id]
4 FROM [test_mentoria].[dbo].[categories]
5
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

	category_id	category_name	super_category_id
1	1	Engineering Mentors	NULL
2	2	Design Mentors	NULL
3	3	Startup Mentors	NULL
4	4	AI Mentors	NULL
5	5	Product Managers	NULL
6	6	Marketing Coaches	NULL
7	7	Leadership Mentors	NULL
8	8	Web Development	1
9	9	Mobile Development	1
10	10	Cloud & DevOps	1
11	11	Cybersecurity	1
12	12	Backend Development	1
13	13	Frontend Development	1
14	14	Full Stack Development	1
15	15	Game Development	1
16	16	Embedded Systems	1
17	17	Blockchain Development	1
18	18	UI/UX Design	2
19	19	Product Design	2

Query executed successfully. localhost (16.0 RTM) sa (88) test_mentoria 00:00:00 Row: 1, Col: 1 48 rows

Hình 2: Bảng categories



SQLQuery8.sql...ia (sa (67))

```
1 SELECT TOP (1000) [company_id]
2     , [cname]
3 FROM [test_mentoria].[dbo].[companies]
4
```

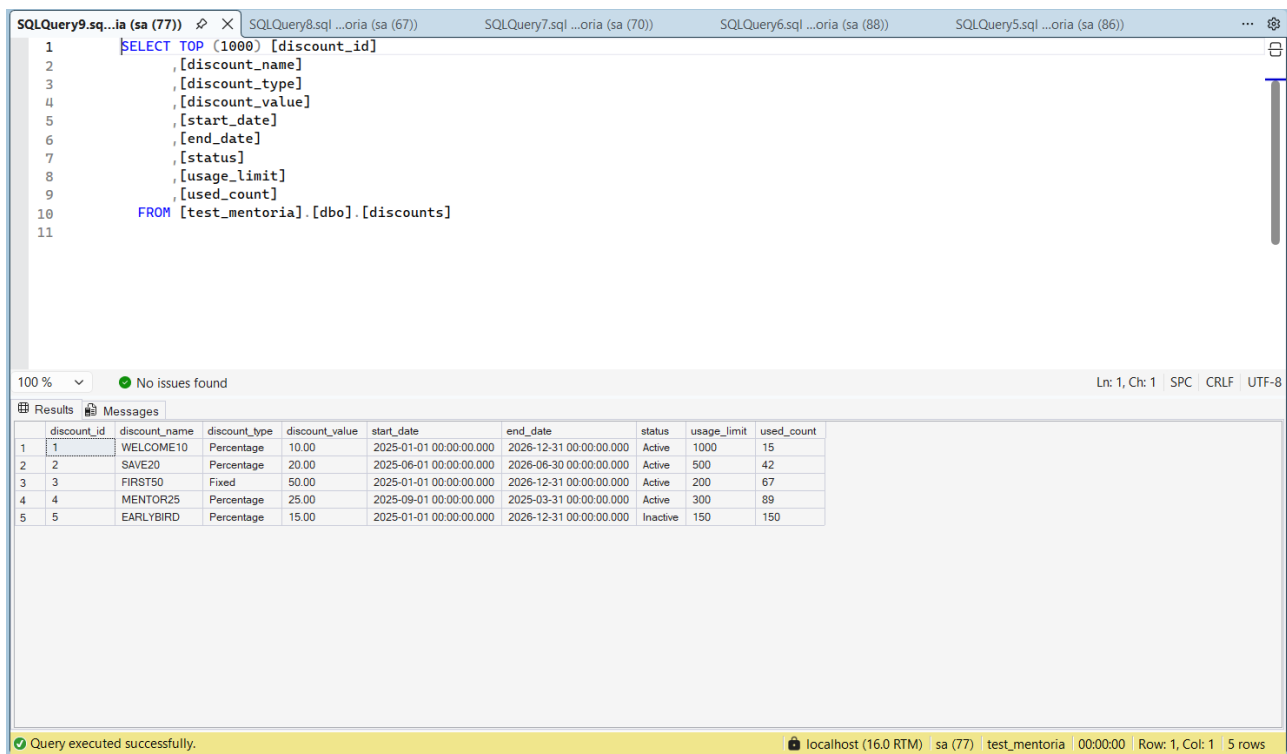
100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	company_id	cname
1	1	Google
2	2	Microsoft
3	3	Amazon
4	4	Meta
5	5	Apple
6	6	Netflix
7	7	Tesla
8	8	Airbnb
9	9	Uber
10	10	Stripe
11	11	Shopify
12	12	Spotify
13	13	Twitter
14	14	LinkedIn
15	15	Adobe
16	16	Salesforce
17	17	Oracle

Query executed successfully. localhost (16.0 RTM) sa (67) test_mentoria 00:00:00 Row: 1, Col: 1 30 rows

Hình 3: Bảng companies



SQLQuery9.sql...ia (sa (77))

```
1 SELECT TOP (1000) [discount_id]
2     , [discount_name]
3     , [discount_type]
4     , [discount_value]
5     , [start_date]
6     , [end_date]
7     , [status]
8     , [usage_limit]
9     , [used_count]
10 FROM [test_mentoria].[dbo].[discounts]
11
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	discount_id	discount_name	discount_type	discount_value	start_date	end_date	status	usage_limit	used_count
1	1	WELCOME10	Percentage	10.00	2025-01-01 00:00:00.000	2026-12-31 00:00:00.000	Active	1000	15
2	2	SAVE20	Percentage	20.00	2025-06-01 00:00:00.000	2026-06-30 00:00:00.000	Active	500	42
3	3	FIRST50	Fixed	50.00	2025-01-01 00:00:00.000	2026-12-31 00:00:00.000	Active	200	67
4	4	MENTOR25	Percentage	25.00	2025-09-01 00:00:00.000	2025-03-31 00:00:00.000	Active	300	89
5	5	EARLYBIRD	Percentage	15.00	2025-01-01 00:00:00.000	2026-12-31 00:00:00.000	Inactive	150	150

Query executed successfully. localhost (16.0 RTM) sa (77) test_mentoria 00:00:00 Row: 1, Col: 1 5 rows

Hình 4: Bảng discounts

SQLQuery11.s...ia (sa (70))

```
1 SELECT TOP (1000) [mentee_id]
2 , [mentor_id]
3 , [stars]
4 , [content]
5 , [sent_time]
6 FROM [test_mentoria].[dbo].[feedbacks]
7
```

100 % No issues found Lnr 7, Ctr 3 SPC CRLF UTF-8

Results Messages

	mentee_id	mentor_id	stars	content	sent_time
1	31	1	5	Excellent mentor! John provided great insights on Re...	2025-12-11 16:30:00.000
2	31	21	4	Patrick gave solid advice on engineering manageme...	2025-12-11 14:30:00.000
3	32	2	5	Sarah is amazing! Her design feedback was spot-on ...	2025-12-11 19:45:00.000
4	32	22	4	Isabella provided great social media strategy tips. My...	2025-12-11 15:45:00.000
5	33	3	4	Very knowledgeable about ML. Michael explained n...	2025-12-11 13:30:00.000
6	33	23	5	Henrik is a blockchain expert! His guidance on smart...	2025-12-11 11:45:00.000
7	34	4	5	Emily helped me create a comprehensive marketing ...	2025-12-11 14:00:00.000
8	34	24	4	Rachel's motion design tips transformed my animati...	2025-12-11 13:20:00.000
9	35	5	5	David provided excellent career guidance. His advice...	2025-12-11 15:30:00.000
10	35	25	4	Mohammed's IoT expertise is impressive. Learned a...	2025-12-11 12:30:00.000
11	36	6	5	Lisa is a cloud architecture guru! Her AWS guidance ...	2025-12-11 16:45:00.000
12	36	26	4	Yuki taught me how to write clear technical documen...	2025-12-11 15:00:00.000
13	37	7	5	James made mobile development fun! His practical ...	2025-12-11 11:30:00.000
14	37	27	4	Lucas helped me set up a comprehensive test autom...	2025-12-11 11:15:00.000
15	38	8	5	Anna has an amazing eye for design! She helped me...	2025-12-11 12:00:00.000
16	38	28	5	Chloe's e-commerce strategies increased my store c...	2025-12-11 13:20:00.000
17	39	9	5	Robert is a backend wizard! His microservices archit...	2025-12-11 13:30:00.000

Query executed successfully. localhost (16.0 RTM) sa (70) test_mentoria 00:00:00 Row: 1, Col: 1 29 rows

Hình 5: Bảng feedbacks

SQLQuery12.s...ia (sa (74)) SQLQuery11.sq...ria (sa (70))

```
1 SELECT TOP (1000) [invoice_id]
2 , [plan_registrations_id]
3 , [method]
4 , [paid_time]
5 , [mentee_id]
6 , [stripe_session_id]
7 , [stripe_customer_id]
8 , [stripe_customer_email]
9 , [stripe_payment_intent_id]
10 , [stripe_charge_id]
11 , [stripe_balance_transaction_id]
12 , [stripe_receipt_url]
13 , [payment_status]
14 , [currency]
15 , [amount_subtotal]
16 , [amount_total]
17 FROM [test_mentoria].[dbo].[invoices]
18
```

100 % No issues found Lnr 1, Ctr 1 SPC CRLF UTF-8

Results Messages

	invoice_id	plan_registrations_id	method	paid_time	mentee_id	stripe_session_id	stripe_customer_id	stripe_customer_email	stripe_payment_intent_id	stripe_charge_id	stripe_balance_transaction_id	stripe_receipt_url	payment_status	currency	amount_subtotal	amount_total
1	1	1	Credit Card	2025-12-05 10:30:00.000	31	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	50.00	45.00
2	2	2	PayPal	2025-12-05 11:15:00.000	32	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	60.00	48.00
3	3	3	Credit Card	2025-12-05 12:00:00.000	33	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	75.00	25.00
4	4	4	Debit Card	2025-12-05 13:45:00.000	34	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	45.00	33.75
5	5	5	Credit Card	2025-12-05 14:20:00.000	35	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	70.00	63.00
6	6	6	PayPal	2025-12-06 09:00:00.000	31	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	90.00	72.00
7	7	7	Credit Card	2025-12-06 10:30:00.000	32	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	190.00	149.00
8	8	8	Credit Card	2025-12-06 11:00:00.000	36	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	85.00	78.50
9	9	9	PayPal	2025-12-06 12:00:00.000	37	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	65.00	52.00
10	10	10	Debit Card	2025-12-06 13:00:00.000	38	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	55.00	5.00
11	11	11	Credit Card	2025-12-06 14:00:00.000	39	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	70.00	52.50
12	12	12	PayPal	2025-12-06 15:00:00.000	40	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	80.00	72.00
13	13	13	Credit Card	2025-12-07 09:00:00.000	41	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	90.00	81.00
14	14	14	Debit Card	2025-12-07 10:00:00.000	42	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	75.00	60.00
15	15	15	Credit Card	2025-12-07 11:00:00.000	43	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	95.00	45.00
16	16	16	PayPal	2025-12-07 12:00:00.000	44	NULL	NULL	NULL	NULL	NULL	NULL	NULL	paid	usd	85.00	48.00

Hình 6: Bảng invoices

SQLQuery32.s...ia (sa (77))

```

1 SELECT TOP (1000) [job_title_id]
2     ,[job_name]
3 FROM [test_mentoria].[dbo].[job_title]
4

```

100 % No issues found Ln: 4, Ch: 1 SPC CRLF UTF-8

job_title_id	job_name
1	Senior Software Engineer
2	Tech Lead
3	Lead Product Designer
4	Senior UX Designer
5	ML Engineer
6	Data Scientist
7	Growth Marketing Manager
8	Digital Marketing Lead
9	Engineering Manager
10	VP of Engineering
11	Product Manager
12	DevOps Engineer
13	Senior Backend Engineer
14	Senior Frontend Engineer
15	Full Stack Developer
16	Mobile Developer
17	UI Designer

Query executed successfully. localhost (16.0 RTM) sa (77) test_mentoria 00:00:00 Row: 1, Col: 1 30 rows

Hình 7: Bảng job_title

SQLQuery13.s...ia (sa (58))

```

1 SELECT TOP (1000) [meeting_id]
2     ,[invoice_id]
3     ,[plan_registrations_id]
4     ,[status]
5     ,[location]
6     ,[start_time]
7     ,[end_time]
8     ,[date]
9     ,[mentor_id]
10 FROM [test_mentoria].[dbo].[meetings]
11

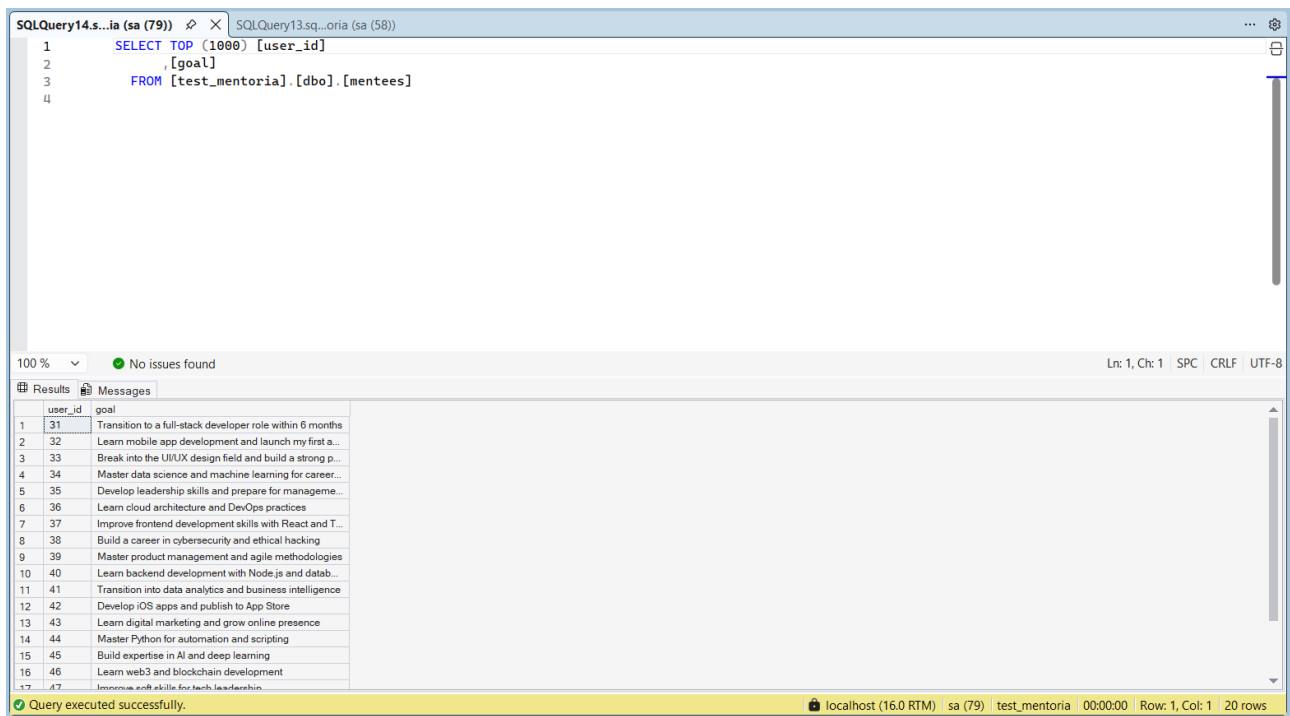
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

meeting_id	invoice_id	plan_registrations_id	status	location	start_time	end_time	date	mentor_id
1	1	1	Scheduled	https://zoom.us/j/123456789	2025-12-10 09:00:00.000	2025-12-10 09:30:00.000	2025-12-10	1
2	2	2	Scheduled	https://meet.google.com/abc-defg-hij	2025-12-10 10:00:00.000	2025-12-10 11:00:00.000	2025-12-10	2
3	3	3	Scheduled	https://zoom.us/j/987654321	2025-12-10 11:00:00.000	2025-12-10 12:15:00.000	2025-12-10	3
4	4	4	Scheduled	https://meet.google.com/xyz-uvw-xyz	2025-12-10 12:00:00.000	2025-12-10 12:45:00.000	2025-12-10	4
5	5	5	Scheduled	https://zoom.us/j/555666777	2025-12-10 13:00:00.000	2025-12-10 14:10:00.000	2025-12-10	5
6	6	6	Scheduled	https://zoom.us/j/111222333	2025-12-12 09:00:00.000	2025-12-12 10:00:00.000	2025-12-12	1
7	7	7	Pending	https://zoom.us/j/888999000	2025-12-14 14:00:00.000	2025-12-14 15:00:00.000	2025-12-14	2
8	8	8	Scheduled	https://zoom.us/j/888999000	2025-12-10 14:00:00.000	2025-12-10 15:25:00.000	2025-12-10	6
9	9	9	Scheduled	https://meet.google.com/ghi-jklm-nop	2025-12-10 09:00:00.000	2025-12-10 10:05:00.000	2025-12-10	7
10	10	10	Scheduled	https://zoom.us/j/444555666	2025-12-10 10:00:00.000	2025-12-10 10:55:00.000	2025-12-10	8
11	11	11	Scheduled	https://meet.google.com/qrs-tuvw-xyz	2025-12-10 11:00:00.000	2025-12-10 12:10:00.000	2025-12-10	9
12	12	12	Scheduled	https://zoom.us/j/777888999	2025-12-10 13:00:00.000	2025-12-10 14:20:00.000	2025-12-10	10
13	13	13	Scheduled	https://meet.google.com/abc-123-def	2025-12-10 09:00:00.000	2025-12-10 10:30:00.000	2025-12-10	11
14	14	14	Scheduled	https://zoom.us/j/333444555	2025-12-10 11:00:00.000	2025-12-10 12:15:00.000	2025-12-10	12
15	15	15	Scheduled	https://meet.google.com/mno-pqr-stu	2025-12-10 10:00:00.000	2025-12-10 11:35:00.000	2025-12-10	13
16	16	16	Scheduled	https://zoom.us/j/666777888	2025-12-10 13:00:00.000	2025-12-10 14:05:00.000	2025-12-10	14
17	17	17	Scheduled	https://meet.google.com/vwx-yz-abc	2025-12-10 09:00:00.000	2025-12-10 10:35:00.000	2025-12-10	15

Query executed successfully. localhost (16.0 RTM) sa (58) test_mentoria 00:00:00 Row: 1, Col: 1 31 rows

Hình 8: Bảng meetings



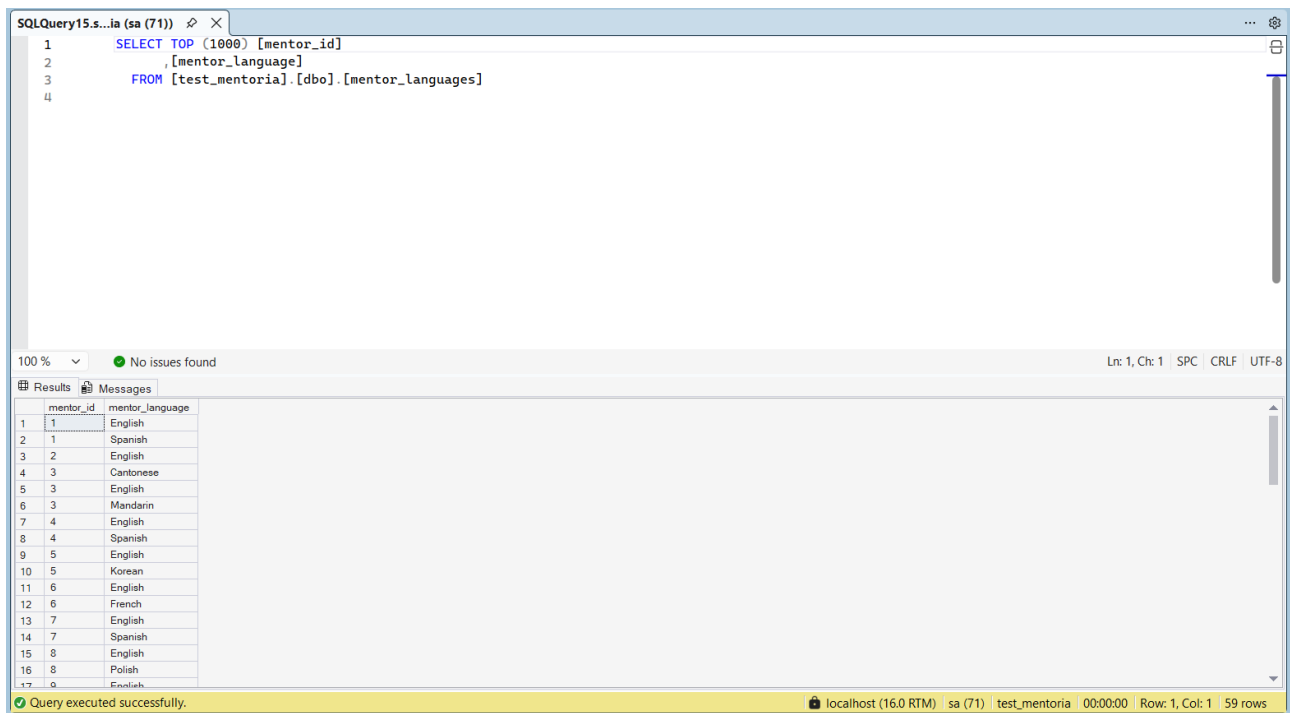
The screenshot shows a SQL query window with the following query:

```
1 SELECT TOP (1000) [user_id]
2     ,[goal]
3 FROM [test_mentoria].[dbo].[mentees]
4
```

The results pane displays 20 rows of data. The status bar indicates 'Query executed successfully' and 'Row: 1, Col: 1 20 rows'.

user_id	goal
31	Transition to a full-stack developer role within 6 months
32	Learn mobile app development and launch my first a...
33	Break into the UI/UX design field and build a strong p...
34	Master data science and machine learning for career...
35	Develop leadership skills and prepare for manage...
36	Learn cloud architecture and DevOps practices
37	Improve frontend development skills with React and T...
38	Build a career in cybersecurity and ethical hacking
39	Master product management and agile methodologies
40	Learn backend development with Node.js and datab...
41	Transition into data analytics and business intelligence
42	Develop iOS apps and publish to App Store
43	Learn digital marketing and grow online presence
44	Master Python for automation and scripting
45	Build expertise in AI and deep learning
46	Learn web3 and blockchain development
47	Improve soft skills for tech leadership

Hình 9: Bảng mentees



The screenshot shows a SQL query window with the following query:

```
1 SELECT TOP (1000) [mentor_id]
2     ,[mentor_language]
3 FROM [test_mentoria].[dbo].[mentor_languages]
4
```

The results pane displays 59 rows of data. The status bar indicates 'Query executed successfully' and 'Row: 1, Col: 1 59 rows'.

mentor_id	mentor_language
1	English
1	Spanish
2	English
3	Cantonese
3	English
3	Mandarin
4	English
4	Spanish
5	English
5	Korean
6	English
6	French
7	English
7	Spanish
8	English
8	Polish
9	English

Hình 10: Bảng mentor_languages

SQLQuery16.s...la (sa (73)) X SQLQuery15.sq...oria (sa (71))

```
1 SELECT TOP (1000) [user_id]
2 , [bio]
3 , [headline]
4 , [response_time]
5 , [cv_url]
6 , [bank_name]
7 , [account_number]
8 , [account_holder_name]
9 , [bank_branch]
10 , [swift_code]
11 FROM [test_mentoria].[dbo].[mentors]
```

100 % No issues found Lnc: 1, Ch: 1 SPC CRLF UTF-8

user_id	bio	headline	response_time	cv_url	bank_name	account_number	account_holder_name	bank_branch	swift_code
1	Experienced full-stack developer with 10+ years in...	Senior Full-Stack Developer & Tech Lead	Within 24 hours	https://example.com/cv/johndoe.pdf	Chase Bank	1234567890	John Doe	New York Main Branch	CHASUS33
2	Product designer with expertise in creating user-ce...	Lead Product Designer	Within 12 hours	https://example.com/cv/sarahjohnson.pdf	Barclays Bank	2345678901	Sarah Johnson	London Central	BARCGB22
3	Data scientist and ML engineer specializing in NL...	Senior Data Scientist & ML Engineer	Within 48 hours	https://example.com/cv/michaelchen.pdf	DBS Bank	3456789012	Michael Chen	Singapore Marina Bay	DBSSSGSG
4	Digital marketing strategist with experience in gro...	Growth Marketing Lead	Within 24 hours	https://example.com/cv/emilyrodriguez.pdf	Santander Bank	4567890123	Emily Rodriguez	Madrid Centro	BSCHEMM
5	Career coach and leadership consultant. Former te...	Career Coach & Leadership Consultant	Within 24 hours	https://example.com/cv/davidkim.pdf	Kookmin Bank	5678901234	David Kim	Seoul Gangnam	CZNBKRSE
6	Cloud architect and DevOps specialist. Expert in A...	Senior Cloud Architect	Within 24 hours	https://example.com/cv/isaanderson.pdf	Royal Bank of Canada	6789012345	Lisa Anderson	Toronto Downtown	ROYCAT22
7	Mobile developer with expertise in iOS and Androi...	Senior Mobile Developer	Within 36 hours	https://example.com/cv/jamesmartinez.pdf	BBVA Bancomer	7890123456	James Martinez	Mexico City Reforma	BONRMXMM
8	UI/UX designer focused on creating beautiful and i...	Senior UI/UX Designer	Within 12 hours	https://example.com/cv/annakowalski.pdf	PKO Bank Polski	8901234567	Anna Kowalski	Warsaw Centrum	BPKOPLPW
9	Backend engineer specializing in scalable system...	Senior Backend Engineer	Within 24 hours	https://example.com/cv/robertthompson.pdf	Commonwealth Bank	9012345678	Robert Thompson	Sydney CBD	CTBAU225
10	Digital transformation consultant. Helping busines...	Digital Transformation Consultant	Within 48 hours	https://example.com/cv/mariagarcia.pdf	Banco Galicia	0123456789	Maria Garcia	Buenos Aires Centro	GABSAFAR
11	Software architect with deep knowledge in system...	Principal Software Architect	Within 24 hours	https://example.com/cv/thomaslee.pdf	MUFG Bank	1122334455	Thomas Lee	Tokyo Marunouchi	BOTKJPJT
12	Creative director and brand strategist. Worked with...	Creative Director & Brand Strategist	Within 24 hours	https://example.com/cv/sophiedubois.pdf	BNP Paribas	2233445566	Sophie Dubois	Paris Opera	BNPFRPPP
13	Cybersecurity expert with certifications in ethical ha...	Senior Cybersecurity Consultant	Within 24 hours	https://example.com/cv/kevinwalsh.pdf	AIB Bank	3344556677	Kevin Walsh	Dublin O'Connell Street	AIBKIE2D
14	Product manager with strong technical background...	Senior Product Manager	Within 12 hours	https://example.com/cv/aurasmuller.pdf	Deutsche Bank	4455667788	Laura Muller	Berlin Alexanderplatz	DEUTDE33
15	AI researcher and engineer. Published papers on...	AI Research Engineer	Within 48 hours	https://example.com/cv/danielsantos.pdf	Banco do Brasil	5566778899	Daniel Santos	Sao Paulo Paulista	BRASBRRL
16	Graphic designer and illustrator. Specialized in bra...	Senior Graphic Designer	Within 24 hours	https://example.com/cv/emmawilson.pdf	ANZ Bank	6677889900	Emma Wilson	Auckland Queen Street	ANZBNZ22
17	Full-stack developer with expertise in cloud archite...	Lead Full-Stack Developer	Within 24 hours	https://example.com/cv/alexanderjohnson.pdf	HSBC Bank	7788990011	Alexander Johnson	London Victoria	HSBCGB22

Query executed successfully. localhost (16.0 RTM) sa (73) test_mentoria 00:00:00 Row: 1, Col: 1 30 rows

Hình 11: Bảng mentors

SQLQuery17.sq...oria (sa (63)) X

```
1 SELECT TOP (1000) [mentorships_id]
2 , [benefit_description]
3 FROM [test_mentoria].[dbo].[mentorships_benefits]
```

100 % No issues found Lnc: 1, Ch: 1 SPC CRLF UTF-8

mentorships_id	benefit_description
1	2 calls per month (60min/call)
2	Access to design resources
3	Email support
4	Portfolio review and redesign guidance
5	4 calls per month (60min/call)
6	Access to exclusive design resources
7	Design system consultation
8	Job application review
9	Portfolio review and redesign guidance
10	Unlimited chat support
11	4 calls per month (60min/call)
12	Code review and optimization
13	Email support
14	Real-world ML project guidance
15	Research paper discussions
16	3 calls per month (60min/call)
17	Career guidance in AI/ML field

Query executed successfully. localhost (16.0 RTM) sa (63) test_mentoria 00:00:00 Row: 1, Col: 1 210 rows

Hình 12: Bảng mentorships_benefits

SQLQuery18.s...ia (sa (73)) X SQLQuery17.sq...oria (sa (63))

```
1 SELECT TOP (1000) [message_id]
2     , [content]
3     , [sent_time]
4     , [sender_id]
5     , [receiver_id]
6 FROM [test_mentoria].[dbo].[messages]
7
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	message_id	content	sent_time	sender_id	receiver_id
1	1	Hi John! I'm excited about our upcoming session on ...	2024-12-08 10:00:00.000	31	1
2	2	Hello Alice! I'm looking forward to it too. Do you have a...	2024-12-08 11:30:00.000	1	31
3	3	Hi Sarah, I have some questions about my portfolio de...	2024-12-07 14:00:00.000	32	2
4	4	Of course Bob! Feel free to send it over and I'll take a l...	2024-12-07 15:45:00.000	2	32
5	5	Michael, could we discuss NLP in our next session?	2024-12-06 09:30:00.000	33	3
6	6	Absolutely Carol! NLP is a fascinating topic. We can g...	2024-12-06 13:20:00.000	3	33
7	7	Emily, thank you for the marketing tips from last sessio...	2024-12-05 16:00:00.000	34	4
8	8	You're welcome Daniel! Let me know how the imple...	2024-12-05 17:30:00.000	4	34
9	9	David, I've updated my resume as you suggested.	2024-12-04 11:00:00.000	35	5
10	10	Great Emma! Send it over and I'll review the changes.	2024-12-04 14:15:00.000	5	35

Query executed successfully. localhost (16.0 RTM) sa (73) test_mentoria 00:00:00 Row: 1, Col: 1 10 rows

Hình 13: Bảng messages

SQLQuery19.s...ia (sa (63)) X

```
1 SELECT TOP (1000) [no_id]
2     , [title]
3     , [content]
4 FROM [test_mentoria].[dbo].[notifications]
5
```

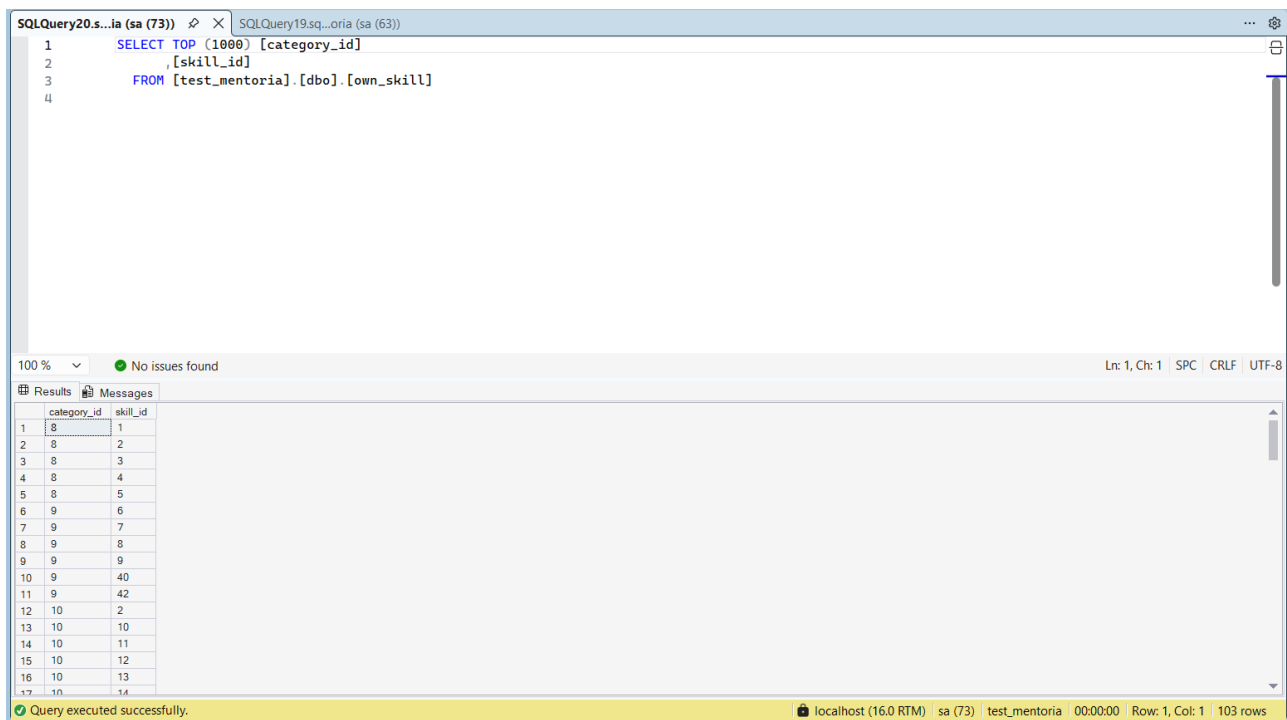
100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	no_id	title	content
1	1	Welcome to Mentor!	Thank you for joining Mentor. Start your journey b...
2	2	New Mentor Available	Check out our latest mentors specializing in AI, Ma...
3	3	Session Reminder	Your session is scheduled for tomorrow. Please ch...
4	4	Special Discount Available	Use code SAVE20 to get 20% off your next mentors...
5	5	Profile Update	Don't forget to complete your profile to help mentor...

Query executed successfully. localhost (16.0 RTM) sa (63) test_mentoria 00:00:00 Row: 1, Col: 1 5 rows

Hình 14: Bảng notifications



The screenshot shows a SQL query window with the following query:

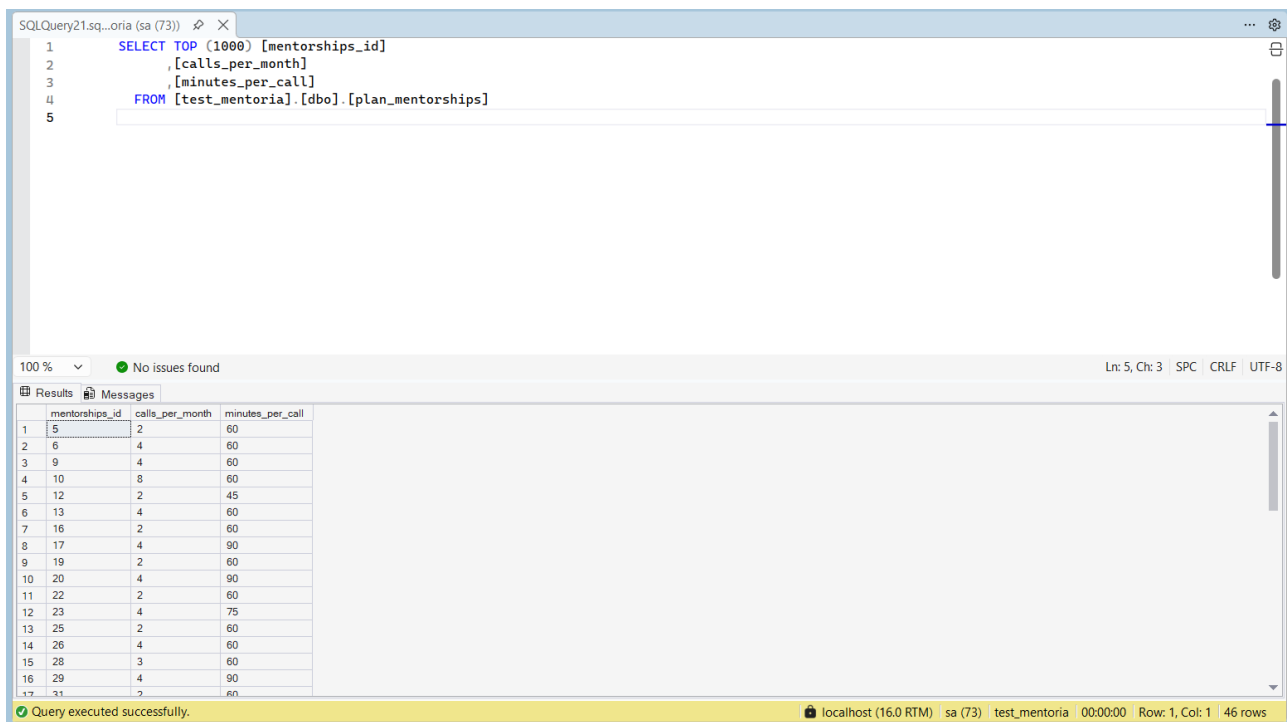
```
1 SELECT TOP (1000) [category_id]
2     ,[skill_id]
3 FROM [test_mentoria].[dbo].[own_skill]
4
```

The results pane displays a table with two columns: category_id and skill_id. The data is as follows:

category_id	skill_id
8	1
8	2
8	3
8	4
8	5
9	6
9	7
9	8
9	9
9	40
9	42
10	2
10	10
10	11
10	12
10	13
10	14

The status bar at the bottom indicates: Query executed successfully. localhost (16.0 RTM) sa (73) test_mentoria 00:00:00 Row: 1, Col: 1 103 rows

Hình 15: Bảng own_skill



The screenshot shows a SQL query window with the following query:

```
1 SELECT TOP (1000) [mentorships_id]
2     ,[calls_per_month]
3     ,[minutes_per_call]
4 FROM [test_mentoria].[dbo].[plan_mentorships]
5
```

The results pane displays a table with three columns: mentorships_id, calls_per_month, and minutes_per_call. The data is as follows:

mentorships_id	calls_per_month	minutes_per_call
5	2	60
6	4	60
9	4	60
10	8	60
12	2	45
13	4	60
16	2	60
17	4	90
19	2	60
20	4	90
22	2	60
23	4	75
25	2	60
26	4	60
28	3	60
29	4	90
31	2	60

The status bar at the bottom indicates: Query executed successfully. localhost (16.0 RTM) sa (73) test_mentoria 00:00:00 Row: 1, Col: 1 46 rows

Hình 16: Bảng plan_mentorships

SQLQuery22.s...ia (sa (69)) X SQLQuery21.sq...oria (sa (73))

```

1 SELECT TOP (1000) [registration_id]
2     , [message]
3     , [discount_id]
4 FROM [test_mentoria].[dbo].[plan_registrations]
5

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

registration_id	message	discount_id
1	ward to learning web development with...	1
2	...improve my design skills with Sarah	2
3	Ready to dive into machine learning	3
4	Need guidance on marketing strategies	4
5	Want to advance my career to the next level	1
6	Interested in full-stack development mentorship	2
7	Hope to build better UI/UX for my projects	3
8	Seeking mentorship for cloud architecture	1
9	Want to learn mobile app development	2
10	Looking to improve backend development skills	3
11	Interested in digital transformation consulting	4
12	Need help with system design and architecture	1
13	Want to learn brand strategy	2
14	Looking for cybersecurity guidance	3
15	Interested in product management mentorship	4
16	Ready to dive into AI and deep learning	1
17	Want to improve my machine learning skills	2

Query executed successfully. localhost (16.0 RTM) sa (69) test_mentoria 00:00:00 Row: 1, Col: 1 31 rows

Hình 17: Bảng plan_registrations

SQLQuery23.s...ia (sa (74)) X SQLQuery22.sq...oria (sa (69)) SQLQuery21.sq...oria (sa (73))

```

1 SELECT TOP (1000) [sessions_id]
2     , [sessions_duration]
3 FROM [test_mentoria].[dbo].[plan_sessions]
4

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

sessions_id	sessions_duration
1	30
2	60
3	120
4	60
5	75
6	120
7	45
8	70
9	120
10	85
11	65
12	55
13	70
14	80
15	90
16	75
17	95

Query executed successfully. localhost (16.0 RTM) sa (74) test_mentoria 00:00:00 Row: 1, Col: 1 34 rows

Hình 18: Bảng plan_sessions

SQLQuery24.s...ia (sa (73))

```
1 SELECT TOP (1000) [plan_id]
2 , [plan_description]
3 , [plan_charge]
4 , [plan_type]
5 , [mentor_id]
6 FROM [test_mentoria].[dbo].[plans]
7
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	plan_id	plan_description	plan_charge	plan_type	mentor_id
1	1	30-minute focused mentorship session covering Re...	50.00	Beginner	1
2	2	1-hour deep-dive session with code review and archi...	90.00	Introductory	1
3	3	2-hour comprehensive technical consultation with sy...	150.00	Introductory	1
4	4	Portfolio review and UX design consultation session	60.00	Beginner	2
5	5	Monthly mentorship program for aspiring designers ...	199.00	Lite	2
6	6	Comprehensive design mentorship - 4 calls per mon...	349.00	Standard	2
7	7	ML/AI consultation session with hands-on guidance	75.00	Beginner	3
8	8	Advanced machine learning project review and opti...	130.00	Introductory	3
9	9	ML mentorship with weekly sessions - 4 calls per m...	399.00	Standard	3
10	10	Premium ML mentorship with intensive support - 8 c...	699.00	Premium	3
11	11	Marketing strategy consultation and growth tactics	45.00	Beginner	4
12	12	Digital marketing mentorship - 2 calls per month (45...	199.00	Lite	4
13	13	Comprehensive marketing mentorship - 4 calls per ...	349.00	Standard	4
14	14	Career coaching session with resume review	70.00	Beginner	5
15	15	Leadership development and career growth session	120.00	Introductory	5
16	16	Executive leadership mentorship - 2 calls per month ...	399.00	Lite	5
17	17	Premium leadership coaching - 4 calls per month (8...	700.00	Premium	5

Query executed successfully. localhost (16.0 RTM) sa (73) test_mentoria 00:00:00 Row: 1, Col: 1 80 rows

Hình 19: Bảng plans

SQLQuery25.s...ia (sa (75)) SQLQuery24.sq...oria (sa (73))

```
1 SELECT TOP (1000) [u_user_id]
2 , [n_no_id]
3 , [sent_time]
4 FROM [test_mentoria].[dbo].[sended]
5
```

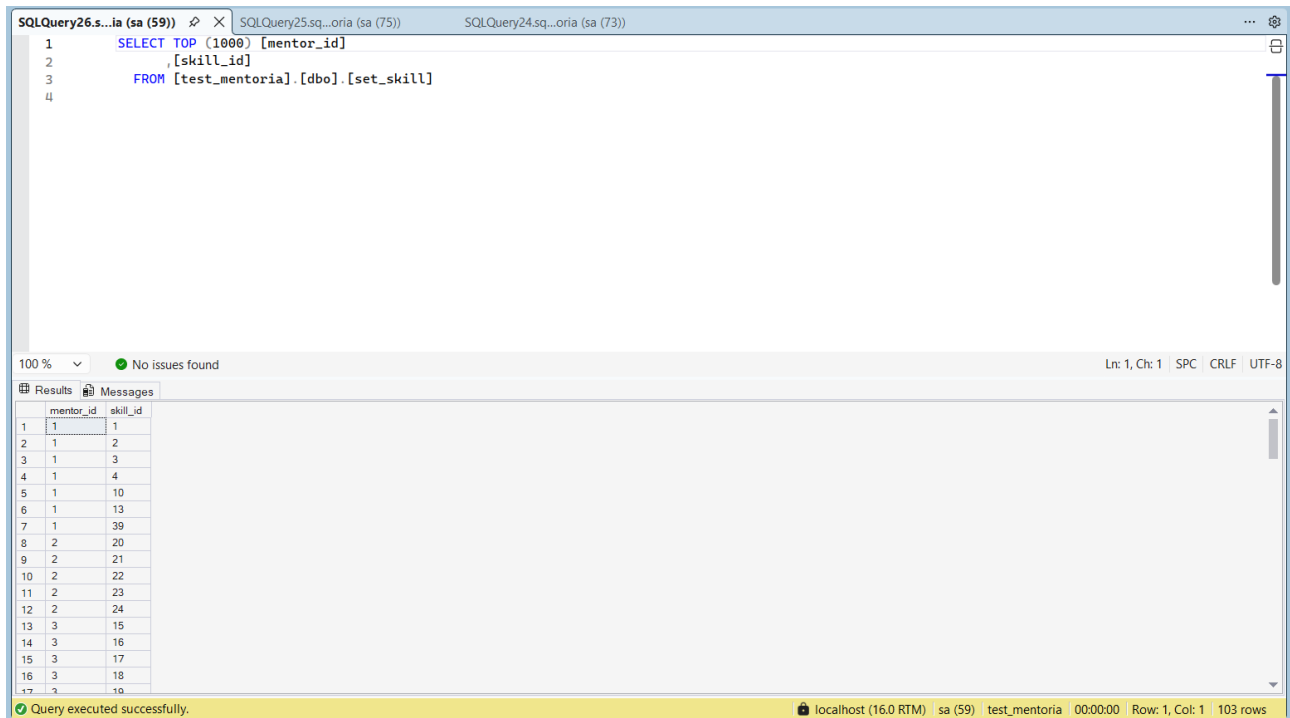
100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	u_user_id	n_no_id	sent_time
1	31	1	2025-12-01 10:00:00.000
2	31	3	2025-12-02 08:00:00.000
3	32	1	2025-12-01 10:30:00.000
4	32	3	2025-12-02 08:30:00.000
5	33	1	2025-12-01 11:00:00.000
6	33	2	2025-12-02 14:00:00.000
7	34	1	2025-12-01 11:30:00.000
8	34	4	2025-12-02 09:00:00.000
9	35	1	2025-12-01 12:00:00.000
10	35	5	2025-12-02 15:00:00.000
11	36	1	2025-12-01 12:30:00.000
12	37	1	2025-12-01 13:00:00.000
13	38	1	2025-12-01 13:30:00.000
14	39	1	2025-12-01 14:00:00.000
15	40	1	2025-12-01 14:30:00.000
16	41	1	2025-12-01 15:00:00.000
17	42	1	2025-12-01 15:30:00.000

Query executed successfully. localhost (16.0 RTM) sa (75) test_mentoria 00:00:00 Row: 1, Col: 1 25 rows

Hình 20: Bảng sended



The screenshot shows a SQL query window with the following query:

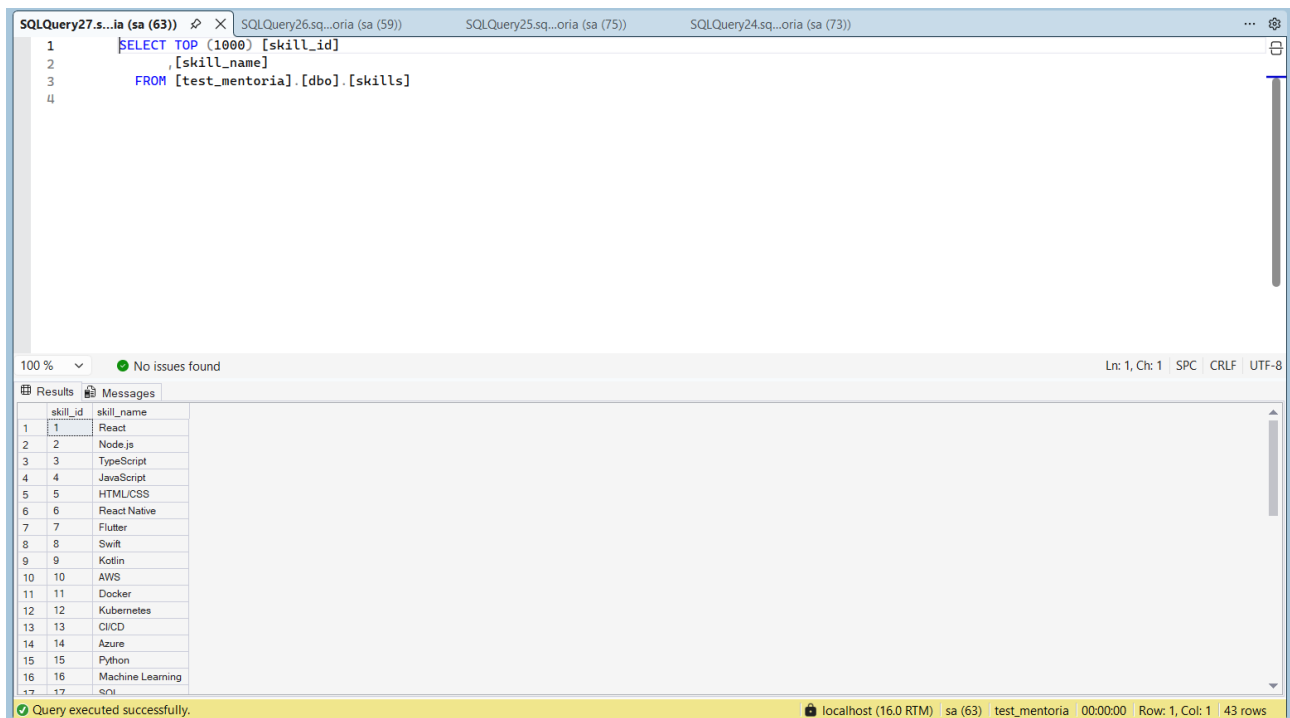
```
1 SELECT TOP (1000) [mentor_id]
2     ,[skill_id]
3 FROM [test_mentoria].[dbo].[set_skill]
4
```

The results pane displays the following data:

mentor_id	skill_id
1	1
2	2
3	3
4	4
5	10
6	13
7	39
8	20
9	21
10	22
11	23
12	24
13	15
14	16
15	17
16	18
17	19

Query executed successfully. localhost (16.0 RTM) sa (59) test_mentoria 00:00:00 Row: 1, Col: 1 103 rows

Hình 21: Bảng set_skill



The screenshot shows a SQL query window with the following query:

```
1 SELECT TOP (1000) [skill_id]
2     ,[skill_name]
3 FROM [test_mentoria].[dbo].[skills]
4
```

The results pane displays the following data:

skill_id	skill_name
1	React
2	Node.js
3	TypeScript
4	JavaScript
5	HTML/CSS
6	React Native
7	Flutter
8	Swift
9	Kotlin
10	AWS
11	Docker
12	Kubernetes
13	CI/CD
14	Azure
15	Python
16	Machine Learning
17	SQL

Query executed successfully. localhost (16.0 RTM) sa (63) test_mentoria 00:00:00 Row: 1, Col: 1 43 rows

Hình 22: Bảng skills

SQLQuery28.sq...oria (sa (70)) SQLQuery27.sq...oria (sa (63)) SQLQuery26.sq...oria (sa (59)) SQLQuery25.sq...oria (sa (75)) SQLQuery24.sq...oria (sa (73))

```

1 SELECT TOP (1000) [start_time]
2     ,[end_time]
3     ,[date]
4     ,[mentor_id]
5     ,[status]
6     ,[plan_id]
7 FROM [test_mentoria].[dbo].[slots]
8

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

	start_time	end_time	date	mentor_id	status	plan_id
1	2025-12-10 09:00:00.000	2025-12-10 09:30:00.000	2025-12-10	1	Booked	1
2	2025-12-10 10:00:00.000	2025-12-10 11:00:00.000	2025-12-10	1	Available	2
3	2025-12-11 14:00:00.000	2025-12-11 16:00:00.000	2025-12-11	1	Available	3
4	2025-12-12 09:00:00.000	2025-12-12 10:00:00.000	2025-12-12	1	Booked	2
5	2025-12-13 15:00:00.000	2025-12-13 15:30:00.000	2025-12-13	1	Available	1
6	2025-12-14 11:00:00.000	2025-12-14 13:00:00.000	2025-12-14	1	Available	3
7	2025-12-15 09:30:00.000	2025-12-15 10:00:00.000	2025-12-15	1	Available	1
8	2025-12-10 10:00:00.000	2025-12-10 11:00:00.000	2025-12-10	2	Booked	4
9	2025-12-11 13:00:00.000	2025-12-11 14:00:00.000	2025-12-11	2	Available	5
10	2025-12-12 11:00:00.000	2025-12-12 12:00:00.000	2025-12-12	2	Available	6
11	2025-12-13 09:00:00.000	2025-12-13 10:00:00.000	2025-12-13	2	Available	4
12	2025-12-14 14:00:00.000	2025-12-14 15:00:00.000	2025-12-14	2	Booked	5
13	2025-12-15 10:00:00.000	2025-12-15 11:00:00.000	2025-12-15	2	Available	6
14	2025-12-10 11:00:00.000	2025-12-10 12:15:00.000	2025-12-10	3	Booked	7
15	2025-12-11 15:00:00.000	2025-12-11 17:00:00.000	2025-12-11	3	Available	8
16	2025-12-12 10:00:00.000	2025-12-12 11:00:00.000	2025-12-12	3	Available	9
17	2025-12-13 13:00:00.000	2025-12-13 14:00:00.000	2025-12-13	3	Available	10

Query executed successfully. localhost (16.0 RTM) sa (70) test_mentoria 00:00:00 Row: 1, Col: 1 156 rows

Hình 23: Bảng slots

SQLQuery29.s...ia (sa (78)) SQLQuery28.sq...oria (sa (70)) SQLQuery27.sq...oria (sa (63)) SQLQuery26.sq...oria (sa (59)) SQLQuery25.sq...oria (sa (75))

```

1 SELECT TOP (1000) [user_id]
2     ,[link]
3     ,[platform]
4 FROM [test_mentoria].[dbo].[user_social_links]
5

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

	user_id	link	platform
1	1	https://github.com/johndoe	GitHub
2	1	https://linkedin.com/in/johndoe	LinkedIn
3	1	https://twitter.com/johndoe	Twitter
4	2	https://behance.net/sarahjohnson	Behance
5	2	https://dribbble.com/sarahjohnson	Dribbble
6	2	https://linkedin.com/in/sarahjohnson	LinkedIn
7	3	https://github.com/michaelchen	GitHub
8	3	https://kaggle.com/michaelchen	Kaggle
9	3	https://linkedin.com/in/michaelchen	LinkedIn
10	4	https://linkedin.com/in/emilyrodriguez	LinkedIn
11	4	https://twitter.com/emilyrodriguez	Twitter
12	5	https://linkedin.com/in/davidkim	LinkedIn
13	5	https://twitter.com/davidkim	Twitter
14	6	https://github.com/lisaanderson	GitHub
15	6	https://linkedin.com/in/lisaanderson	LinkedIn
16	7	https://github.com/jamesmartinez	GitHub
17	7	https://linkedin.com/in/jamesmartinez	LinkedIn

Query executed successfully. localhost (16.0 RTM) sa (78) test_mentoria 00:00:00 Row: 1, Col: 1 89 rows

Hình 24: Bảng user_social_links

SQLQuery30.s...ia (sa (65)) X SQLQuery29.sq...oria (sa (78)) SQLQuery28.sq...oria (sa (70)) SQLQuery27.sq...oria (sa (63)) SQLQuery26.sq...oria (sa (59))

```

1 SELECT TOP (1000) [user_id]
2   ,[first_name]
3   ,[last_name]
4   ,[sex]
5   ,[created_at]
6   ,[updated_at]
7   ,[email]
8   ,[password]
9   ,[avatar_url]
10  ,[country]
11  ,[role]
12  ,[timezone]
13  ,[status]
14  ,[is_email_verified]
15  ,[otp]
16  ,[otp_expiration]
17  ,[reset_password_token]
18  ,[reset_password_token_expiration]

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

	user_id	first_name	last_name	sex	created_at	updated_at	email	password	avatar_url	country	role	timezone
1	1	John	Doe	Male	2025-12-09 16:52:28.380	NULL	john.doe@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=1	United States	Mentor	America/New_York
2	2	Sarah	Johnson	Female	2025-12-09 16:52:28.380	NULL	sarah.johnson@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=5	United Kingdom	Mentor	Europe/London
3	3	Michael	Chen	Male	2025-12-09 16:52:28.380	NULL	michael.chen@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=12	Singapore	Mentor	Asia/Singapore
4	4	Emily	Rodriguez	Female	2025-12-09 16:52:28.380	NULL	emily.rodriguez@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=9	Spain	Mentor	Europe/Madrid
5	5	David	Kim	Male	2025-12-09 16:52:28.380	NULL	david.kim@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=15	South Korea	Mentor	Asia/Seoul
6	6	Lisa	Anderson	Female	2025-12-09 16:52:28.380	NULL	lisa.anderson@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=10	Canada	Mentor	America/Toronto
7	7	James	Marinez	Male	2025-12-09 16:52:28.380	NULL	james.marinex@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=13	Mexico	Mentor	America/Mexico_City
8	8	Anna	Kowalski	Female	2025-12-09 16:52:28.380	NULL	anna.kowalski@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=16	Poland	Mentor	Europe/Warsaw
9	9	Robert	Thompson	Male	2025-12-09 16:52:28.380	NULL	robert.thompson@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=17	Australia	Mentor	Australia/Sydney
10	10	Maria	Garcia	Female	2025-12-09 16:52:28.380	NULL	maria.garcia@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=23	Argentina	Mentor	America/Argentina/Buenos
11	11	Thomas	Lee	Male	2025-12-09 16:52:28.380	NULL	thomas.lee@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=18	Japan	Mentor	Asia/Tokyo
12	12	Sophie	Dubois	Female	2025-12-09 16:52:28.380	NULL	sophie.dubois@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=24	France	Mentor	Europe/Paris
13	13	Kevin	Walsh	Male	2025-12-09 16:52:28.380	NULL	kevin.walsh@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=25	Ireland	Mentor	Europe/Dublin
14	14	Laura	Muller	Female	2025-12-09 16:52:28.380	NULL	laura.muller@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=29	Germany	Mentor	Europe/Berlin
15	15	Daniel	Santos	Male	2025-12-09 16:52:28.380	NULL	daniel.santos@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=26	Brazil	Mentor	America/Sao_Paulo
16	16	Emma	Wilson	Female	2025-12-09 16:52:28.380	NULL	emma.wilson@example.com	\$2b\$10\$UjAn0gdXkLWqkVPCsuUjXrKE0NdAGVEudOv...	https://pravatar.cc/150?img=27	New Zealand	Mentor	Pacific/Auckland

Query executed successfully. localhost (16.0 RTM) sa (65) test_mentoria 00:00:00 Row: 1, Col: 1 51 rows

Hình 25: Bảng users

SQLQuery31.s...ia (sa (80)) X SQLQuery30.sq...oria (sa (65)) SQLQuery29.sq...oria (sa (78)) SQLQuery28.sq...oria (sa (70)) SQLQuery27.sq...oria (sa (63))

```

1 SELECT TOP (1000) [mentor_id]
2   ,[c_company_id]
3   ,[current_job_title_id]
4 FROM [test_mentoria].[dbo].[work_for]
5

```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

	mentor_id	c_company_id	current_job_title_id
1	1	1	1
2	1	2	2
3	2	3	3
4	2	8	4
5	3	4	5
6	3	6	6
7	4	9	8
8	4	11	7
9	5	5	9
10	5	10	10
11	6	2	23
12	6	3	11
13	7	1	16
14	7	5	16
15	8	15	17
16	9	3	13
17	9	6	13

Query executed successfully. localhost (16.0 RTM) sa (80) test_mentoria 00:00:00 Row: 1, Col: 1 57 rows

Hình 26: Bảng work_for

2 Các thủ tục, trigger và hàm

2.1 Thủ tục INSERT / UPDATE / DELETE dữ liệu vào 1 bảng

Bảng dữ liệu được lựa chọn để minh họa trong phần này là bảng **slots**.

2.1.1 Thủ tục INSERT

Mô tả thủ tục: Thủ tục `Proc_Insert_Slot` được sử dụng để tạo mới một Slot (lịch rảnh của mentor). Thủ tục được kích hoạt trong trường hợp mentor muốn tạo lịch rảnh để mentee có thể thấy và đặt lịch học.

Usecase:

- Mentor tạo thêm slot mới cho các gói mentoring.

Thủ tục bao gồm các thao tác chính:

1. Kiểm tra sự tồn tại của mentor.
2. Xác nhận rằng gói Plan thuộc về mentor.
3. Kiểm tra tính hợp lệ về thời gian (không trong quá khứ, cùng ngày, độ dài không dưới 15 và không trên 120 phút).
4. Kiểm tra xem slot có bị trùng thời gian với slot đã tồn tại hay không.
5. Thực hiện chèn slot mới vào bảng.

Input:

- `@MentorID (INT)`: ID của mentor.
- `@StartTime (DATETIME)`: Thời gian bắt đầu slot.
- `@EndTime (DATETIME)`: Thời gian kết thúc slot.
- `@Date (DATE)`: Ngày tạo slot.
- `@PlanID (INT)`: Mã gói mentoring của mentor.

Output:

- Chuỗi thông báo thành công: "Slot created successfully."
- Báo lỗi bằng RAISERROR trong các trường hợp không hợp lệ.

Câu lệnh tạo thủ tục:

```
1 CREATE OR ALTER PROCEDURE Proc_Insert_Slot
2     @MentorID INT,
3     @StartTime DATETIME,
4     @EndTime DATETIME,
5     @Date DATE,
6     @PlanID INT
7 AS
8 BEGIN
9     SET NOCOUNT ON;
10
11     IF NOT EXISTS (SELECT 1 FROM mentors WHERE user_id = @MentorID)
12     BEGIN
13         RAISERROR(N'Mentor does not exist!', 16, 1);
14         RETURN;
15     END
16
17     IF NOT EXISTS (
18         SELECT 1 FROM plans
19         WHERE plan_id = @PlanID AND mentor_id = @MentorID
20     )
21     BEGIN
22         RAISERROR(N'Invalid Plan: This plan does not belong to this
23 mentor!', 16, 1);
24         RETURN;
25     END
26
27     IF @StartTime < GETDATE()
28     BEGIN
29         RAISERROR(N'Start time must not be in the past!', 16, 1);
30         RETURN;
31     END
```

```
31
32 IF CAST(@StartTime AS DATE) <> @Date
33 BEGIN
34     RAISERROR(N'StartTime must fall on the same day as Date!', 16,
35     1);
36     RETURN;
37 END
38
39 DECLARE @Duration INT = DATEDIFF(MINUTE, @StartTime, @EndTime);
40
41 IF @Duration < 15
42 BEGIN
43     RAISERROR(N'Slot duration must be at least 15 minutes!', 16,
44     1);
45     RETURN;
46 END
47
48 IF @Duration > 120
49 BEGIN
50     RAISERROR(N'Slot duration must not exceed 120 minutes!', 16,
51     1);
52     RETURN;
53 END
54
55 IF EXISTS (
56     SELECT 1 FROM slots
57     WHERE mentor_id = @MentorID
58           AND date = @Date
59           AND (@StartTime < end_time AND @EndTime > start_time)
60 )
61 BEGIN
62     RAISERROR(N'Time slot overlaps with an existing slot!', 16, 1)
63 ;
64     RETURN;
65 END
```

```
63      INSERT INTO slots (start_time, end_time, date, mentor_id, plan_id)
64      VALUES (@StartTime, @EndTime, @Date, @MentorID, @PlanID);
65
66      SELECT 'Slot created successfully.' AS Message;
67 END;
68 GO
```

Nhóm câu lệnh thực thi thủ tục mẫu:

```
1 PRINT N'===== TESTCASE INSERT
   ====='
2 PRINT N'Slot created successfully.'
3 EXEC Proc_Insert_Slot
4     @MentorID = 1,
5     @StartTime = '2025-12-12 10:00',
6     @EndTime   = '2025-12-12 11:00',
7     @Date      = '2025-12-12',
8     @PlanID    = 1;
9
10 PRINT N'Mentor does not exist!'
11 EXEC Proc_Insert_Slot
12     @MentorID = 999,
13     @StartTime = '2025-12-12 10:00',
14     @EndTime   = '2025-12-12 11:00',
15     @Date      = '2025-12-12',
16     @PlanID    = 1;
17
18 PRINT N'Invalid Plan'
19 EXEC Proc_Insert_Slot
20     @MentorID = 1,
21     @StartTime = '2025-12-12 10:00',
22     @EndTime   = '2025-12-12 11:00',
23     @Date      = '2025-12-12',
24     @PlanID    = 999;
25
26 PRINT N'Start time must not be in the past!'
27 EXEC Proc_Insert_Slot
```

```
28      @MentorID = 1,
29      @StartTime = '2020-01-01 10:00',
30      @EndTime   = '2020-01-01 11:00',
31      @Date       = '2020-01-01',
32      @PlanID     = 1;
33
34 PRINT N'StartTime must fall on the same day as Date!'
35 EXEC Proc_Insert_Slot
36      @MentorID = 1,
37      @StartTime = '2025-12-12 10:00',
38      @EndTime   = '2025-12-12 11:00',
39      @Date       = '2025-12-13',
40      @PlanID     = 1;
41
42 PRINT N'Slot duration must be at least 15 minutes'
43 EXEC Proc_Insert_Slot
44      @MentorID = 1,
45      @StartTime = '2025-12-12 10:00',
46      @EndTime   = '2025-12-12 10:10',
47      @Date       = '2025-12-12',
48      @PlanID     = 1;
49
50 PRINT N'Slot duration must not exceed 120 minutes'
51 EXEC Proc_Insert_Slot
52      @MentorID = 1,
53      @StartTime = '2025-12-12 10:00',
54      @EndTime   = '2025-12-12 13:30',
55      @Date       = '2025-12-12',
56      @PlanID     = 1;
57
58 PRINT N'Time slot overlaps with existing slot'
59 EXEC Proc_Insert_Slot
60      @MentorID = 1,
61      @StartTime = '2025-12-10 08:30',
62      @EndTime   = '2025-12-10 09:30',
63      @Date       = '2025-12-10',
```

```
@PlanID      = 1;
```

Kết quả minh họa:

```
===== TESTCASE INSERT =====  
Slot created successfully.  
Mentor does not exist!  
Msg 50000, Level 16, State 1, Procedure Proc_Insert_Slot, Line 14 [Batch Start Line 237]  
Mentor does not exist!  
Invalid Plan  
Msg 50000, Level 16, State 1, Procedure Proc_Insert_Slot, Line 23 [Batch Start Line 237]  
Invalid Plan: This plan does not belong to this mentor!  
Start time must not be in the past!  
Msg 50000, Level 16, State 1, Procedure Proc_Insert_Slot, Line 29 [Batch Start Line 237]  
Start time must not be in the past!  
StartTime must fall on the same day as Date!  
Msg 50000, Level 16, State 1, Procedure Proc_Insert_Slot, Line 35 [Batch Start Line 237]  
StartTime must fall on the same day as Date!  
Slot duration must be at least 15 minutes  
Msg 50000, Level 16, State 1, Procedure Proc_Insert_Slot, Line 43 [Batch Start Line 237]  
Slot duration must be at least 15 minutes!  
Slot duration must not exceed 120 minutes  
Msg 50000, Level 16, State 1, Procedure Proc_Insert_Slot, Line 49 [Batch Start Line 237]  
Slot duration must not exceed 120 minutes!  
Time slot overlaps with existing slot  
Msg 50000, Level 16, State 1, Procedure Proc_Insert_Slot, Line 60 [Batch Start Line 237]  
Time slot overlaps with an existing slot!
```

Hình 27: Kết quả thực thi thủ tục INSERT

2.1.2 Thủ tục UPDATE

Mô tả thủ tục: Thủ tục Proc_Update_Slot dùng để cập nhật slot của một mentor.

Usecase:

- Cập nhật thời gian slot. (khi @Status = NULL)
- Cập nhật trạng thái slot thành Booked. (khi @Status = 'Booked')

Thủ tục bao gồm các thao tác chính:

1. Slot cũ phải tồn tại.
2. Không được cập nhật nếu slot đang ở trạng thái Booked.
3. Nếu cập nhật thời gian: phải kiểm tra hợp lệ về ngày, thời lượng và trùng lặp.

Input:

- @MentorID (INT): ID mentor.
- @OldStart (DATETIME): Thời gian bắt đầu cũ của slot.
- @OldEnd (DATETIME): Thời gian kết thúc cũ của slot.
- @OldDate (DATE): Ngày cũ của slot.
- @Status (NVARCHAR(20)): Trạng thái slot (NULL hoặc 'Booked').
- @NewStart (DATETIME): Thời gian bắt đầu mới (nếu cập nhật).
- @NewEnd (DATETIME): Thời gian kết thúc mới (nếu cập nhật).
- @NewDate (DATE): Ngày mới của slot (nếu cập nhật).

Output:

- Chuỗi thông báo thành công "Slot time updated successfully."
- Chuỗi thông báo thành công "Slot booked successfully."
- Báo lỗi với RAISERROR trong các trường hợp không hợp lệ.

Câu lệnh tạo thủ tục:

```
1 CREATE OR ALTER PROCEDURE Proc_Update_Slot
2     @MentorID INT,
3     @OldStart DATETIME,
4     @OldEnd DATETIME,
5     @OldDate DATE,
6     @Status NVARCHAR(20) = NULL,
7     @NewStart DATETIME = NULL,
8     @NewEnd DATETIME = NULL,
9     @NewDate DATE = NULL
10 AS
11 BEGIN
12     SET NOCOUNT ON;
13
14     IF NOT EXISTS (
15         SELECT 1 FROM slots
16         WHERE mentor_id = @MentorID
```



```
17         AND start_time = @OldStart
18         AND end_time = @OldEnd
19         AND date = @OldDate
20     )
21 BEGIN
22     RAISERROR(N'Original slot does not exist!', 16, 1);
23     RETURN;
24 END
25
26 DECLARE @CurrentStatus NVARCHAR(20);
27 SELECT @CurrentStatus = status
28 FROM slots
29 WHERE mentor_id = @MentorID
30        AND start_time = @OldStart
31        AND end_time = @OldEnd
32        AND date = @OldDate;
33
34 IF @CurrentStatus = N'Booked'
35 BEGIN
36     RAISERROR(N'Cannot update: Slot is currently booked!', 16, 1);
37     RETURN;
38 END
39
40 IF @Status IS NULL
41 BEGIN
42     IF @NewStart IS NULL OR @NewEnd IS NULL OR @NewDate IS NULL
43     BEGIN
44         RAISERROR(N'Updating time requires NewStart, NewEnd, and
45 NewDate!', 16, 1);
46         RETURN;
47     END
48
49     IF DATEDIFF(MINUTE, @NewStart, @NewEnd) NOT BETWEEN 15 AND 120
50     BEGIN
51         RAISERROR(N'Slot duration must be between 15 and 120
52 minutes!', 16, 1);
```

```
51         RETURN;  
52     END  
53  
54     IF CAST(@NewStart AS DATE) <> @NewDate  
55     BEGIN  
56         RAISERROR(N'NewStart must fall on NewDate!', 16, 1);  
57         RETURN;  
58     END  
59  
60     IF @NewStart <= GETDATE()  
61     BEGIN  
62         RAISERROR(N'NewStart cannot be in the past!', 16, 1);  
63         RETURN;  
64     END  
65  
66     IF EXISTS (  
67         SELECT 1 FROM slots  
68         WHERE mentor_id = @MentorID  
69             AND date = @NewDate  
70             AND NOT (start_time = @OldStart AND end_time = @OldEnd  
71 AND date = @OldDate)  
72             AND @NewStart < end_time  
73             AND @NewEnd > start_time  
74     )  
75     BEGIN  
76         RAISERROR(N'New time range overlaps with another slot!',  
77 16, 1);  
78         RETURN;  
79     END  
80  
81     UPDATE slots  
82     SET start_time = @NewStart,  
83         end_time = @NewEnd,  
84         date = @NewDate
```

```
85         WHERE mentor_id = @MentorID
86         AND start_time BETWEEN DATEADD(SECOND, -1, @OldStart) AND
DATEADD(SECOND, 1, @OldStart)
87         AND end_time BETWEEN DATEADD(SECOND, -1, @OldEnd) AND
DATEADD(SECOND, 1, @OldEnd)
88         AND date = @OldDate
89
90         AND date = @OldDate
91
92     IF @@ROWCOUNT = 0
93     BEGIN
94         RAISERROR(N'No slot matched to update!',16,1);
95         RETURN;
96     END
97
98     SELECT 'Slot updated successfully.' AS Message;
99     RETURN;
100 END
101
102 IF @Status = N'Booked'
103 BEGIN
104     IF @NewStart IS NOT NULL OR @NewEnd IS NOT NULL OR @NewDate IS
NOT NULL
105     BEGIN
106         RAISERROR(N'Cannot change time when setting status to
Booked!', 16, 1);
107         RETURN;
108     END
109
110     UPDATE slots
111     SET status = N'Booked'
112     WHERE mentor_id = @MentorID
113         AND start_time >= @OldStart AND start_time < DATEADD(SECOND
,1,@OldStart)
114         AND end_time >= @OldEnd AND end_time < DATEADD(SECOND
,1,@OldEnd)
```

```
115         AND date = @OldDate;
116
117     SELECT 'Slot booked successfully.' AS Message;
118     RETURN;
119 END
120 RAISERROR(N'Invalid status! Must be NULL (time update) or Booked.'
121 , 16, 1);
122 END
123 GO
```

Nhóm câu lệnh thực thi thủ tục mẫu:

```
1 PRINT N'===== TESTCASE UPDATE
   ====='
2 INSERT INTO slots (start_time, end_time, date, mentor_id, status,
   plan_id)
3 VALUES ('2025-12-12 09:30', '2025-12-12 10:30', '2025-12-12', 1, '
   Available', 1);
4
5 PRINT N'New time overlaps with another slot'
6 EXEC Proc_Update_Slot
7     @MentorID = 1,
8     @OldStart = '2025-12-10 08:00',
9     @OldEnd   = '2025-12-10 09:00',
10    @OldDate  = '2025-12-10',
11    @Status   = NULL,
12    @NewStart = '2025-12-12 10:00',
13    @NewEnd   = '2025-12-12 11:00',
14    @NewDate  = '2025-12-12';
15
16 PRINT N'Slot booked successfully.'
17 EXEC Proc_Update_Slot
18     @MentorID = 1,
19     @OldStart = '2025-12-10 08:00',
20     @OldEnd   = '2025-12-10 09:00',
21     @OldDate  = '2025-12-10',
22     @Status   = N'Booked';
```

Kết quả minh họa:

```
===== TESTCASE UPDATE =====  
  
(1 row affected)  
overlaps with another slot  
Msg 50000, Level 16, State 1, Procedure Proc_Update_Slot, Line 76 [Batch Start Line 237]  
New time range overlaps with another slot!  
Slot booked successfully.
```

Hình 28: Kết quả thực thi thủ tục UPDATE

2.1.3 Thủ tục DELETE

Mô tả thủ tục: Thủ tục Proc_Delete_Slot dùng để xóa một slot của mentor.

Usecase:

- Mentor muốn xóa một slot mà họ đã tạo trước đó.

Thủ tục bao gồm các kiểm tra:

- Slot phải tồn tại.
- Không được xóa nếu slot đang ở trạng thái Booked.

Input:

- @MentorID (INT): ID của mentor.
- @StartTime (DATETIME): Thời gian bắt đầu của slot.
- @EndTime (DATETIME): Thời gian kết thúc của slot.
- @Date (DATE): Ngày của slot.

Output:

- Chuỗi thông báo thành công "Slot deleted successfully."
- Báo lỗi với RAISERROR trong các trường hợp không hợp lệ.

Câu lệnh tạo thủ tục:

```
1 CREATE OR ALTER PROCEDURE Proc_Delete_Slot  
2     @MentorID INT,  
3     @StartTime DATETIME,  
4     @EndTime DATETIME,  
5     @Date DATE
```

```
6 AS
7 BEGIN
8     SET NOCOUNT ON;
9
10    IF NOT EXISTS (
11        SELECT 1 FROM slots
12        WHERE mentor_id = @MentorID
13            AND start_time = @StartTime
14            AND end_time    = @EndTime
15            AND date = @Date
16    )
17    BEGIN
18        RAISERROR(N'Slot not found.', 16, 1);
19        RETURN;
20    END
21
22    IF EXISTS (
23        SELECT 1 FROM slots
24        WHERE mentor_id = @MentorID
25            AND start_time >= @StartTime
26            AND end_time    >= @EndTime
27            AND date = @Date
28            AND status = N'Booked'
29    )
30    BEGIN
31        RAISERROR(N'Cannot delete: Slot is currently booked!', 16, 1);
32        RETURN;
33    END
34
35    DELETE FROM slots
36    WHERE mentor_id = @MentorID
37        AND start_time >= @StartTime
38        AND end_time    >= @EndTime
39        AND date = @Date;
40
41    SELECT 'Slot deleted successfully.' AS Message;
```

42 END
43 GO

Nhóm câu lệnh thực thi mẫu:

```
1 PRINT N'===== TESTCASE DELETE
   ====='
2 INSERT INTO slots (start_time, end_time, date, mentor_id, status,
   plan_id)
3 VALUES ('2025-12-12 12:30', '2025-12-12 13:30', '2025-12-12', 1, '
   Available', 1);
4
5 PRINT N'Slot deleted successfully.'
6 EXEC Proc_Delete_Slot
7     @MentorID = 1,
8     @StartTime = '2025-12-12 12:30',
9     @EndTime   = '2025-12-12 13:30',
10    @Date       = '2025-12-12';
11
12 PRINT N'Slot not found'
13 EXEC Proc_Delete_Slot
14     @MentorID = 1,
15     @StartTime = '2000-01-01 00:00',
16     @EndTime   = '2000-01-01 01:00',
17     @Date       = '2000-01-01';
18
19 UPDATE slots SET status = 'Booked'
20 WHERE mentor_id = 1
21    AND start_time = '2025-12-10 08:00';
22
23 PRINT N'Cannot delete: Slot is currently booked!'
24 EXEC Proc_Delete_Slot
25     @MentorID = 1,
26     @StartTime = '2025-12-10 08:00',
27     @EndTime   = '2025-12-10 09:00',
28     @Date       = '2025-12-10';
```

Kết quả minh họa:

```
===== TESTCASE DELETE =====

(1 row affected)
Slot deleted successfully.
Slot not found
Msg 50000, Level 16, State 1, Procedure Proc_Delete_Slot, Line 20 [Batch Start Line 237]
Slot not found.

(1 row affected)
Cannot delete: Slot is currently booked!
Msg 50000, Level 16, State 1, Procedure Proc_Delete_Slot, Line 33 [Batch Start Line 237]
Cannot delete: Slot is currently booked!
```

Hình 29: Kết quả thực thi thủ tục DELETE

2.2 Các trigger

2.2.1 Trigger kiểm tra ràng buộc ngữ nghĩa

Mô tả trigger: Trigger này được sử dụng để kiểm tra ràng buộc nghiệp vụ yêu cầu một tài khoản chỉ được kích hoạt (status = 'Active') khi email đã được xác minh. Thủ tục được sử dụng trong các trường hợp có thao tác INSERT hoặc UPDATE lên bảng users, nhằm phát hiện những bản ghi vi phạm quy tắc: status = 'Active' nhưng is_email_verified = 0.

Usecase:

- Khi hệ thống thêm mới người dùng (INSERT) và cố gắng kích hoạt tài khoản trước khi xác minh email.
- Khi hệ thống cập nhật người dùng (UPDATE) và thay đổi trạng thái tài khoản thành Active nhưng trường xác minh email vẫn là 0.

Trigger bao gồm các thao tác chính:

1. Kiểm tra các bản ghi trong bảng ảo inserted.
2. Xác định xem có bản ghi nào thỏa điều kiện vi phạm hay không.
3. Nếu có, trigger thông báo lỗi (RAISERROR) và rollback transaction để ngăn ghi dữ liệu sai.

Câu lệnh tạo trigger:

```
1 CREATE TRIGGER trg_user_email_verify
2 ON users
3 FOR INSERT, UPDATE
4 AS
5 BEGIN
6     IF EXISTS (
7         SELECT 1 FROM inserted
8         WHERE status = 'Active' AND is_email_verified = 0
9     )
10    BEGIN
11        RAISERROR(N'Email phải được xác minh trước khi kích hoạt tài
12        khoản.', 16, 1);
13        ROLLBACK TRAN;
14    END
15 END;
16 GO
```

Nhóm câu lệnh thực thi mẫu:

```
1 --user email verification trigger test
2 INSERT INTO users (first_name, last_name, sex, email, password,
3     avatar_url, country, role, timezone, status, is_email_verified,
4     provider)
5 VALUES
6 (N'John', N'Doe', N'Male', N'john.doe.test@example.com', N'
7     $2b$10$UJAnOgdXdvLWzjkVVPcCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
8     https://i.pravatar.cc/150?img=1', N'United States', N'Mentor', N'
9     America/New_York', N'Active', 0, N'Local');
10 GO --will not work
11 INSERT INTO users (first_name, last_name, sex, email, password,
12     avatar_url, country, role, timezone, status, is_email_verified,
13     provider)
14 VALUES
15 (N'John', N'Doe', N'Male', N'john.doe.test@example.com', N'
16     $2b$10$UJAnOgdXdvLWzjkVVPcCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
17     https://i.pravatar.cc/150?img=1', N'United States', N'Mentor', N'
```

```
America/New_York', N'Active', 1, N'Local');  
9 GO --will work  
10 UPDATE users SET is_email_verified = 0;  
11 GO --will not work
```

Kết quả minh họa:

```
Msg 50000, Level 16, State 1, Procedure trg_user_email_verify, Line 31 [Batch Start Line 0]  
Email phải được xác minh trước khi kích hoạt tài khoản.  
Msg 3609, Level 16, State 1, Line 2  
The transaction ended in the trigger. The batch has been aborted.  
  
(1 row affected)  
Msg 50000, Level 16, State 1, Procedure trg_user_email_verify, Line 31 [Batch Start Line 9]  
Email phải được xác minh trước khi kích hoạt tài khoản.  
Msg 3609, Level 16, State 1, Line 10  
The transaction ended in the trigger. The batch has been aborted.
```

Hình 30: Kết quả kiểm tra trigger xác thực email

2.2.2 Sử dụng trigger để tính toán, cập nhật thuộc tính dẫn xuất

Mô tả trigger: Nhóm trigger này được dùng để tự động tính toán và cập nhật các thuộc tính dẫn xuất của mentor: `total_mentee`, `total_reviews`, `total_stars` và `rating`. Các thuộc tính này phản ánh số lượng mentee theo học, tổng số lượt đánh giá, tổng số sao nhận được và điểm đánh giá trung bình của mentor. Thủ tục được kích hoạt khi có thay đổi dữ liệu đánh giá (feedbacks) hoặc buổi học (meeting).

Usecase:

- Nhóm trigger trên bảng feedbacks (`trg_feedback_insert`, `trg_feedback_update`, `trg_feedback_delete`):
 - Khi mentee gửi đánh giá mới cho mentor (INSERT).
 - Khi mentee cập nhật số sao đã đánh giá (UPDATE cột stars).
 - Khi một feedback bị xóa (DELETE).
- Nhóm trigger trên bảng meetings (`trg_meeting_insert`, `trg_meeting_completed`):
 - Khi tạo mới một buổi hẹn giữa mentee và mentor (INSERT vào meetings).

- Khi trạng thái buổi hẹn được cập nhật sang Completed (UPDATE status).

Các thao tác chính:

1. Xác định mentor bị ảnh hưởng, lấy danh sách mentor_id từ bảng ảo inserted (đối với INSERT/UPDATE) hoặc deleted (đối với DELETE).
2. Với trigger trên bảng feedbacks:
 - Đếm lại tổng số feedback của mentor → cập nhật total_reviews.
 - Tính tổng số sao từ tất cả các feedback của mentor → cập nhật total_stars.
 - Tính rating:
 - Nếu không có feedback nào → gán rating = 0.
 - Nếu có feedback → $\text{rating} = \text{SUM}(\text{stars}) * 1.0 / \text{COUNT}(*).$
3. Với trigger trên bảng meetings:
 - Lọc các record meetings của mentor có status = 'Completed'.
 - Kết hợp với invoices để xác định mentee_id tương ứng.
 - Đếm số mentee khác nhau ($\text{COUNT}(\text{DISTINCT mentee_id})$) đã có ít nhất một buổi gặp Completed với mentor → cập nhật total_mentee.
4. Ghi lại các giá trị mới (total_mentee, total_reviews, total_stars, rating) cho đúng user_id của mentor liên quan.

Câu lệnh tạo các thuộc tính dẫn xuất:

```
1 ALTER TABLE mentors
2 ADD total_mentee INT DEFAULT 0, --DERIVED ATTRIBUTES FOR MENTORS
3    total_stars INT DEFAULT 0,
4    total_reviews INT DEFAULT 0,
5    rating DECIMAL(3,2) DEFAULT 0;
6 GO
```

Câu lệnh tạo các trigger tính các thuộc tính dẫn xuất:

```
1 CREATE TRIGGER trg_meeting_completed
2 ON meetings
3 AFTER UPDATE
4 AS
5 BEGIN
6 -- check for record changing to completed
7 IF EXISTS (
8     SELECT 1
9     FROM inserted i
10    WHERE i.status = 'Completed'
11 )
12 BEGIN
13     UPDATE mentors
14     SET total_mentee =
15     (
16         SELECT COUNT(DISTINCT Mentee_id)
17         FROM meetings m
18         JOIN invoices i ON m.invoice_id = i.invoice_id
19         WHERE m.mentor_id = inserted.mentor_id
20         AND m.status = 'Completed'
21     )
22     FROM inserted
23     WHERE mentors.user_id = inserted.mentor_id;
24 END
25 END;
26 GO
27
28 CREATE TRIGGER trg_meeting_insert
29 ON meetings
30 AFTER INSERT
31 AS
32 BEGIN
33     UPDATE mentors
34     SET total_mentee =
35     (
```

```
36     SELECT COUNT(DISTINCT Mentee_id)
37     FROM meetings m
38     JOIN invoices i ON m.invoice_id = i.invoice_id
39     WHERE m.mentor_id = inserted.mentor_id
40     AND m.status = 'Completed'
41 )
42 FROM inserted
43 WHERE mentors.user_id = inserted.mentor_id;
44 END;
45 GO
46
47 CREATE TRIGGER trg_feedback_insert
48 ON feedbacks
49 AFTER INSERT
50 AS
51 BEGIN
52     UPDATE mentors
53     SET
54         total_reviews =
55             (SELECT COUNT(*)
56              FROM feedbacks F
57              WHERE F.mentor_id = mentors.user_id),
58
59         total_stars =
60             (SELECT SUM(F.stars)
61              FROM feedbacks F
62              WHERE F.mentor_id = mentors.user_id),
63
64         rating =
65             CASE
66                 WHEN (SELECT COUNT(*)
67                      FROM feedbacks F
68                      WHERE F.mentor_id = mentors.user_id) = 0
69                 THEN 0
70                 ELSE (
71                     SELECT SUM(F.stars) * 1.0 / COUNT(*)
```

```
72         FROM feedbacks F
73         WHERE F.mentor_id = mentors.user_id
74     )
75     END
76     FROM inserted i
77     WHERE mentors.user_id = i.mentor_id;
78 END;
79 GO
80
81 CREATE TRIGGER trg_feedback_update
82 ON feedbacks
83 AFTER UPDATE
84 AS
85 BEGIN
86     IF UPDATE(stars)
87     BEGIN
88         UPDATE mentors
89         SET
90             total_reviews =
91                 (SELECT COUNT(*)
92                  FROM feedbacks F
93                  WHERE F.mentor_id = mentors.user_id),
94
95             total_stars =
96                 (SELECT SUM(F.stars)
97                  FROM feedbacks F
98                  WHERE F.mentor_id = mentors.user_id),
99
100             rating =
101                 CASE
102                     WHEN (SELECT COUNT(*)
103                          FROM feedbacks F
104                          WHERE F.mentor_id = mentors.user_id) = 0
105                     THEN 0
106                     ELSE (
107                         SELECT SUM(F.stars) * 1.0 / COUNT(*)
```

```
108         FROM feedbacks F
109         WHERE F.mentor_id = mentors.user_id
110     )
111     END
112     FROM inserted i
113     WHERE mentors.user_id = i.mentor_id;
114 END
115 END;
116 GO
117
118 CREATE TRIGGER trg_feedback_delete
119 ON feedbacks
120 AFTER DELETE
121 AS
122 BEGIN
123     UPDATE mentors
124     SET
125         total_reviews =
126             (SELECT COUNT(*)
127              FROM feedbacks F
128              WHERE F.mentor_id = mentors.user_id),
129
130         total_stars =
131             (SELECT SUM(F.stars)
132              FROM feedbacks F
133              WHERE F.mentor_id = mentors.user_id),
134
135         rating =
136             CASE
137                 WHEN (SELECT COUNT(*)
138                      FROM feedbacks F
139                      WHERE F.mentor_id = mentors.user_id) = 0
140                 THEN 0
141                 ELSE (
142                     SELECT SUM(F.stars) * 1.0 / COUNT(*)
143                     FROM feedbacks F
```

```
144         WHERE F.mentor_id = mentors.user_id
145     )
146     END
147 FROM deleted d
148 WHERE mentors.user_id = d.mentor_id;
149 END;
150 GO
```

Nhóm câu lệnh thực thi mẫu:

```
1 -- Insert Mentor: William Chen
2 INSERT INTO users (first_name, last_name, sex, email, password,
3     avatar_url, country, role, timezone, status, is_email_verified,
4     provider)
5 VALUES
6 (N'William', N'Chen', N'Male', N'william.chen.test@mentoria.com', N'
7     $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
8     https://i.pravatar.cc/150?img=70', N'United States', N'Mentor', N'
9     America/Chicago', N'Active', 1, N'Local');
10 GO
11
12 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
13     = N'william.chen.test@mentoria.com');
14
15 INSERT INTO mentors (user_id, bio, headline, response_time, cv_url,
16     bank_name, account_number, account_holder_name, bank_branch,
17     swift_code)
18 VALUES
19 (@mentor_id, N'Test mentor profile for database testing. Specializes
20     in Python, Django, and Machine Learning. 8+ years of teaching
21     experience.', N'Python & ML Specialist', N'Within 12 hours', N'
22     https://test.example.com/cv/williamchen.pdf', N'Test Bank', N'
23     9999888877', N'William Chen', N'Chicago Test Branch', N'TESTUS99');
24 GO
25
26 -- Insert test skill
27 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
```



```
    = N'william.chen.test@mentoria.com');
16 DECLARE @skill_id INT;
17
18 -- Check if skill exists, if not create it
19 IF NOT EXISTS (SELECT 1 FROM skills WHERE skill_name = N'Python
    Testing')
20 BEGIN
21     INSERT INTO skills (skill_name) VALUES (N'Python Testing');
22 END
23 SET @skill_id = (SELECT skill_id FROM skills WHERE skill_name = N'
    Python Testing');
24
25 INSERT INTO set_skill (mentor_id, skill_id) VALUES (@mentor_id,
    @skill_id);
26 GO
27
28 -- Insert plans for William Chen
29 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
30
31 INSERT INTO plans (plan_description, plan_charge, plan_type, mentor_id
    )
32 VALUES
33 (N'Test plan: 45-minute Python and ML fundamentals session', 75.00, N'
    Beginner', @mentor_id),
34 (N'Test plan: 1.5-hour Python deep-dive with code review', 120.00, N'
    Introductory', @mentor_id),
35 (N'Test plan: 2-hour ML project consultation', 180.00, N'Advanced',
    @mentor_id);
36 GO
37
38 -- Insert Mentee 1: Jessica Wong
39 INSERT INTO users (first_name, last_name, sex, email, password,
    avatar_url, country, role, timezone, status, is_email_verified,
    provider)
40 VALUES
```

```
41 (N'Jessica', N'Wong', N'Female', N'jessica.wong.test@mentoria.com', N'  
    $2b$10$UJAn0gdXdvLWzjkVVPcCsUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'  
    https://i.pravatar.cc/150?img=71', N'Canada', N'Mentee', N'America/  
    Vancouver', N'Active', 1, N'Local');  
42 GO  
43  
44 DECLARE @mentee1_id INT = (SELECT TOP 1 user_id FROM users WHERE email  
    = N'jessica.wong.test@mentoria.com');  
45  
46 INSERT INTO mentees (user_id, goal)  
47 VALUES  
48 (@mentee1_id, N'Learn Python for data analysis and automation');  
49 GO  
50  
51 -- Insert discount  
52 IF NOT EXISTS (SELECT 1 FROM discounts WHERE discount_name = N'  
    TESTCODE50')  
53 BEGIN  
54     INSERT INTO discounts (discount_name, discount_type,  
        discount_value, start_date, end_date, status, usage_limit,  
        used_count)  
55     VALUES (N'TESTCODE50', N'Percentage', 15.00, '2025-01-01', '  
        2026-12-31', N'Active', 500, 5);  
56 END  
57 GO  
58  
59 -- Insert plan registration for Jessica  
60 DECLARE @discount_id INT = (SELECT discount_id FROM discounts WHERE  
    discount_name = N'TESTCODE50');  
61  
62 INSERT INTO plan_registrations (message, discount_id)  
63 VALUES (N'Test registration: Looking forward to learning Python with  
    William!', @discount_id);  
64 GO  
65  
66 -- Get IDs for invoice and meeting
```

```
67 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
68 DECLARE @mentee1_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'jessica.wong.test@mentoria.com');
69 DECLARE @plan1_id INT = (SELECT TOP 1 plan_id FROM plans WHERE
    mentor_id = @mentor_id ORDER BY plan_id);
70 DECLARE @plan_reg_id INT = (SELECT TOP 1 registration_id FROM
    plan_registrations ORDER BY registration_id DESC);
71
72 -- Insert slots for William Chen
73 INSERT INTO slots (start_time, end_time, date, mentor_id, status,
    plan_id)
74 VALUES
75 ('2025-12-16 10:00:00', '2025-12-16 10:45:00', '2025-12-16',
    @mentor_id, N'Available', @plan1_id),
76 ('2025-12-21 10:30:00', '2025-12-21 11:15:00', '2025-12-21',
    @mentor_id, N'Available', @plan1_id);
77
78 -- Insert invoice for Jessica
79 INSERT INTO invoices (plan_registrations_id, method, paid_time,
    mentee_id, payment_status, currency, amount_subtotal, amount_total)
80 VALUES (@plan_reg_id, N'PayPal', '2025-12-08 14:20:00', @mentee1_id, N
    'paid', N'usd', 75.00, 63.75);
81 GO
82
83 -- Insert meeting for Jessica
84 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
85 DECLARE @mentee1_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'jessica.wong.test@mentoria.com');
86 DECLARE @invoice1_id INT = (SELECT TOP 1 invoice_id FROM invoices
    WHERE mentee_id = @mentee1_id ORDER BY invoice_id DESC);
87 DECLARE @plan_reg_id INT = (SELECT TOP 1 registration_id FROM
    plan_registrations ORDER BY registration_id DESC);
88
89 INSERT INTO meetings (invoice_id, plan_registrations_id, status,
```

```
location, start_time, end_time, date, mentor_id)
90 VALUES
91 (@invoice1_id, @plan_reg_id, N'Completed', N'https://meet.google.com/
    test-xyz-abc', '2025-12-16 10:00:00', '2025-12-16 10:45:00', '
    2025-12-16', @mentor_id);
92 GO
93
94 -- Insert Mentee 2: Marcus Taylor
95 INSERT INTO users (first_name, last_name, sex, email, password,
    avatar_url, country, role, timezone, status, is_email_verified,
    provider)
96 VALUES
97 (N'Marcus', N'Taylor', N'Male', N'marcus.taylor.test@mentoria.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=72', N'United Kingdom', N'Mentee', N'
    Europe/London', N'Active', 1, N'Local');
98 GO
99
100 DECLARE @mentee2_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'marcus.taylor.test@mentoria.com');
101
102 INSERT INTO mentees (user_id, goal)
103 VALUES (@mentee2_id, N'Build machine learning models for business
    applications');
104 GO
105
106 -- Insert invoice and meeting for Marcus
107 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
108 DECLARE @mentee2_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'marcus.taylor.test@mentoria.com');
109 DECLARE @plan_reg_id INT = (SELECT TOP 1 registration_id FROM
    plan_registrations ORDER BY registration_id DESC);
110
111 INSERT INTO invoices (plan_registrations_id, method, paid_time,
    mentee_id, payment_status, currency, amount_subtotal, amount_total)
```

```
112 VALUES (@plan_reg_id, N'Credit Card', '2025-12-09 09:15:00',
    @mentee2_id, N'paid', N'usd', 75.00, 63.75);
113
114 DECLARE @invoice2_id INT = SCOPE_IDENTITY();
115
116 INSERT INTO meetings (invoice_id, plan_registrations_id, status,
    location, start_time, end_time, date, mentor_id)
117 VALUES
118 (@invoice2_id, @plan_reg_id, N'Completed', N'https://meet.google.com/
    test-def-ghi', '2025-12-21 10:30:00', '2025-12-21 11:15:00', '
    2025-12-21', @mentor_id);
119 GO
120
121 -- =====
122 -- TEST CASE 1: trg_feedback_insert
123 -- =====
124 PRINT '=====';
125 PRINT 'TEST CASE 1: trg_feedback_insert';
126 PRINT '=====';
127 PRINT 'Insert 2 feedbacks with stars: 5 and 1';
128 PRINT 'Expected: total_reviews=2, total_stars=6, rating=3.00';
129 PRINT '';
130
131 -- Insert feedbacks (triggers will calculate derived attributes)
132 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
133 DECLARE @mentee1_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'jessica.wong.test@mentoria.com');
134 DECLARE @mentee2_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'marcus.taylor.test@mentoria.com');
135
136 INSERT INTO feedbacks (mentee_id, mentor_id, stars, content, sent_time
    )
137 VALUES
138 (@mentee1_id, @mentor_id, 5, N'Outstanding mentor! William explained
    Python concepts very clearly and provided practical examples. His
```

```
teaching style is perfect for beginners.', '2025-12-16 12:00:00'),
139 (@mentee2_id, @mentor_id, 1, N'Not satisfied with the session quality.
    Expected more depth in ML topics.', '2025-12-21 13:30:00');
140 GO
141
142 -- Check results after INSERT
143 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
144
145 SELECT
146     m.user_id,
147     CONCAT(u.first_name, ' ', u.last_name) as mentor_name,
148     m.total_mentee,
149     m.total_stars,
150     m.total_reviews,
151     m.rating,
152     CASE
153         WHEN m.total_mentee = 2 AND m.total_stars = 6 AND m.
total_reviews = 2 AND m.rating = 3.00
154         THEN 'PASS'
155         ELSE 'FAIL'
156     END as test_status
157 FROM mentors m
158 JOIN users u ON m.user_id = u.user_id
159 WHERE m.user_id = @mentor_id;
160 PRINT '';
161 GO
162
163 -- =====
164 -- TEST CASE 2: trg_feedback_update
165 -- =====
166 PRINT '=====';
167 PRINT 'TEST CASE 2: trg_feedback_update';
168 PRINT '=====';
169 PRINT 'Update Marcus feedback from 1 star to 4 stars';
170 PRINT 'Expected: total_reviews=2, total_stars=9, rating=4.50';
```

```
171 PRINT '';
172
173 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
174 DECLARE @mentee2_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'marcus.taylor.test@mentoria.com');
175
176 UPDATE feedbacks
177 SET stars = 4,
178     content = N'After reconsideration, the session was actually quite
    helpful. William provided good insights on ML fundamentals.'
179 WHERE mentee_id = @mentee2_id AND mentor_id = @mentor_id;
180 GO
181
182 -- Check results after UPDATE
183 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
184
185 SELECT
186     m.user_id,
187     CONCAT(u.first_name, ' ', u.last_name) as mentor_name,
188     m.total_mentee,
189     m.total_stars,
190     m.total_reviews,
191     m.rating,
192     CASE
193         WHEN m.total_stars = 9 AND m.total_reviews = 2 AND m.rating =
194             4.50
195         THEN 'PASS'
196         ELSE 'FAIL'
197     END as test_status
198 FROM mentors m
199 JOIN users u ON m.user_id = u.user_id
200 WHERE m.user_id = @mentor_id;
201 PRINT '';
202 GO
```

```
202
203 -- =====
204 -- TEST CASE 3: trg_feedback_delete
205 -- =====
206 PRINT '=====';
207 PRINT 'TEST CASE 3: trg_feedback_delete';
208 PRINT '=====';
209 PRINT 'Delete Marcus feedback (4 stars)';
210 PRINT 'Expected: total_reviews=1, total_stars=5, rating=5.00';
211 PRINT '';
212
213 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
214 DECLARE @mentee2_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'marcus.taylor.test@mentoria.com');
215
216 DELETE FROM feedbacks
217 WHERE mentee_id = @mentee2_id AND mentor_id = @mentor_id;
218 GO
219
220 -- Check results after DELETE
221 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
222
223 SELECT
224     m.user_id,
225     CONCAT(u.first_name, ' ', u.last_name) as mentor_name,
226     m.total_mentee,
227     m.total_stars,
228     m.total_reviews,
229     m.rating,
230     CASE
231         WHEN m.total_stars = 5 AND m.total_reviews = 1 AND m.rating =
232             5.00
233         THEN 'PASS'
234         ELSE 'FAIL'
```



```
234     END as test_status
235 FROM mentors m
236 JOIN users u ON m.user_id = u.user_id
237 WHERE m.user_id = @mentor_id;
238 PRINT '';
239 GO
240
241 -- =====
242 -- TEST CASE 4: trg_meeting_completed
243 -- =====
244 PRINT '=====';
245 PRINT 'TEST CASE 4: trg_meeting_completed';
246 PRINT '=====';
247 PRINT 'Create a new meeting with status Scheduled, then update to
    Completed';
248 PRINT 'Expected: total_mentee increases from 2 to 3';
249 PRINT '';
250
251 -- Insert Mentee 3: Sophia Lee
252 INSERT INTO users (first_name, last_name, sex, email, password,
    avatar_url, country, role, timezone, status, is_email_verified,
    provider)
253 VALUES
254 (N'Sophia', N'Lee', N'Female', N'sophia.lee.test@mentoria.com', N'
    $2b$10$UJAnOgdXdvLWzjkVVPCCsuUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'
    https://i.pravatar.cc/150?img=73', N'Australia', N'Mentee', N'
    Australia/Sydney', N'Active', 1, N'Local');
255 GO
256
257 DECLARE @mentee3_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'sophia.lee.test@mentoria.com');
258
259 INSERT INTO mentees (user_id, goal)
260 VALUES (@mentee3_id, N'Learn advanced Python programming and data
    structures');
261 GO
```

```
262
263 -- Insert plan registration for Sophia
264 DECLARE @discount_id INT = (SELECT discount_id FROM discounts WHERE
    discount_name = N'TESTCODE50');
265
266 INSERT INTO plan_registrations (message, discount_id)
267 VALUES (N'Test registration: Excited to learn Python with William!',
    @discount_id);
268 GO
269
270 -- Get IDs and create meeting with Scheduled status
271 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
272 DECLARE @mentee3_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'sophia.lee.test@mentoria.com');
273 DECLARE @plan1_id INT = (SELECT TOP 1 plan_id FROM plans WHERE
    mentor_id = @mentor_id ORDER BY plan_id);
274 DECLARE @plan_reg_id INT = (SELECT TOP 1 registration_id FROM
    plan_registrations ORDER BY registration_id DESC);
275
276 -- Insert slot for Sophia's meeting
277 INSERT INTO slots (start_time, end_time, date, mentor_id, status,
    plan_id)
278 VALUES
279 ('2025-12-22 14:00:00', '2025-12-22 14:45:00', '2025-12-22',
    @mentor_id, N'Available', @plan1_id);
280
281 INSERT INTO invoices (plan_registrations_id, method, paid_time,
    mentee_id, payment_status, currency, amount_subtotal, amount_total)
282 VALUES (@plan_reg_id, N'Bank Transfer', '2025-12-10 11:00:00',
    @mentee3_id, N'paid', N'usd', 75.00, 63.75);
283
284 DECLARE @invoice3_id INT = SCOPE_IDENTITY();
285
286 INSERT INTO meetings (invoice_id, plan_registrations_id, status,
    location, start_time, end_time, date, mentor_id)
```

```
287 VALUES
288 (@invoice3_id, @plan_reg_id, N'Scheduled', N'https://meet.google.com/
    test-jkl-mno', '2025-12-22 14:00:00', '2025-12-22 14:45:00', '
    2025-12-22', @mentor_id);
289 GO
290
291 -- Now UPDATE the meeting status to Completed (this triggers
    trg_meeting_completed)
292 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
293 DECLARE @mentee3_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'sophia.lee.test@mentoria.com');
294
295 UPDATE meetings
296 SET status = 'Completed'
297 WHERE mentor_id = @mentor_id
298     AND status = 'Scheduled'
299     AND date = '2025-12-22';
300 GO
301
302 -- Check results after UPDATE
303 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
304
305 SELECT
306     m.user_id,
307     CONCAT(u.first_name, ' ', u.last_name) as mentor_name,
308     m.total_mentee,
309     CASE
310         WHEN m.total_mentee = 3
311         THEN 'PASS'
312         ELSE 'FAIL'
313     END as test_status
314 FROM mentors m
315 JOIN users u ON m.user_id = u.user_id
316 WHERE m.user_id = @mentor_id;
```

```
317 PRINT '';  
318 GO  
319  
320 -- =====  
321 -- TEST CASE 5: trg_meeting_insert  
322 -- =====  
323 PRINT '=====';  
324 PRINT 'TEST CASE 5: trg_meeting_insert';  
325 PRINT '=====';  
326 PRINT 'Insert a new meeting directly with Completed status';  
327 PRINT 'Expected: total_mentee increases from 3 to 4';  
328 PRINT '';  
329  
330 -- Insert Mentee 4: Daniel Kim  
331 INSERT INTO users (first_name, last_name, sex, email, password,  
    avatar_url, country, role, timezone, status, is_email_verified,  
    provider)  
332 VALUES  
333 (N'Daniel', N'Kim', N'Male', N'daniel.kim.test@mentoria.com', N'  
    $2b$10$UJAnOgdXdvLWzjkVVPcCsUvXRKE0.NdAGVfEud0vUTEyZaMlapH2', N'  
    https://i.pravatar.cc/150?img=74', N'South Korea', N'Mentee', N'  
    Asia/Seoul', N'Active', 1, N'Local');  
334 GO  
335  
336 DECLARE @mentee4_id INT = (SELECT TOP 1 user_id FROM users WHERE email  
    = N'daniel.kim.test@mentoria.com');  
337  
338 INSERT INTO mentees (user_id, goal)  
339 VALUES (@mentee4_id, N'Master machine learning algorithms and model  
    deployment');  
340 GO  
341  
342 -- Insert plan registration for Daniel  
343 DECLARE @discount_id INT = (SELECT discount_id FROM discounts WHERE  
    discount_name = N'TESTCODE50');  
344
```

```
345 INSERT INTO plan_registrations (message, discount_id)
346 VALUES (N'Test registration: Ready to dive deep into ML with William!'
    , @discount_id);
347 GO
348
349 -- Get IDs and create meeting directly with Completed status (this
    triggers trg_meeting_insert)
350 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
351 DECLARE @mentee4_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'daniel.kim.test@mentoria.com');
352 DECLARE @plan1_id INT = (SELECT TOP 1 plan_id FROM plans WHERE
    mentor_id = @mentor_id ORDER BY plan_id);
353 DECLARE @plan_reg_id INT = (SELECT TOP 1 registration_id FROM
    plan_registrations ORDER BY registration_id DESC);
354
355 -- Insert slot for Daniel's meeting
356 INSERT INTO slots (start_time, end_time, date, mentor_id, status,
    plan_id)
357 VALUES
358 ('2025-12-23 09:30:00', '2025-12-23 10:15:00', '2025-12-23',
    @mentor_id, N'Available', @plan1_id);
359
360 INSERT INTO invoices (plan_registrations_id, method, paid_time,
    mentee_id, payment_status, currency, amount_subtotal, amount_total)
361 VALUES (@plan_reg_id, N'PayPal', '2025-12-11 16:30:00', @mentee4_id, N
    'paid', N'usd', 75.00, 63.75);
362
363 DECLARE @invoice4_id INT = SCOPE_IDENTITY();
364
365 -- This INSERT will trigger trg_meeting_insert
366 INSERT INTO meetings (invoice_id, plan_registrations_id, status,
    location, start_time, end_time, date, mentor_id)
367 VALUES
368 (@invoice4_id, @plan_reg_id, N'Completed', N'https://meet.google.com/
    test-pqr-stu', '2025-12-23 09:30:00', '2025-12-23 10:15:00', '
```

```

2025-12-23', @mentor_id);
369 GO
370
371 -- Check results after INSERT
372 DECLARE @mentor_id INT = (SELECT TOP 1 user_id FROM users WHERE email
    = N'william.chen.test@mentoria.com');
373
374 SELECT
375     m.user_id,
376     CONCAT(u.first_name, ' ', u.last_name) as mentor_name,
377     m.total_mentee,
378     CASE
379         WHEN m.total_mentee = 4
380         THEN 'PASS'
381         ELSE 'FAIL'
382     END as test_status
383 FROM mentors m
384 JOIN users u ON m.user_id = u.user_id
385 WHERE m.user_id = @mentor_id;
386 PRINT '';
387 GO

```

Kết quả minh họa:



	user_id	mentor_name	total_mentee	total_stars	total_reviews	rating	test_status
1	52	William Chen	2	6	2	3.00	PASS
1	52	William Chen	2	9	2	4.50	PASS
1	52	William Chen	2	5	1	5.00	PASS
1	52	William Chen	3				PASS
1	52	William Chen	4				PASS

Query executed successfully. localhost (16.0 RTM) : sa (80) : test_mentoria : 00:00:00 : Row: 1, Col: 1 : 5 rows

Hình 31: Kết quả kiểm tra trigger tính thuộc tính dẫn xuất

2.3 Thủ tục truy vấn

2.3.1 Tìm kiếm/lọc nâng cao Mentor

Mô tả thủ tục: Thủ tục `sp_SearchMentors` được sử dụng để tìm kiếm và lọc danh sách các Mentor theo nhiều tiêu chí khác nhau. Phục vụ cho chức năng tìm kiếm nâng cao trên giao diện người dùng, giúp Mentee có thể tìm kiếm Mentor phù hợp với nhu cầu cá nhân.

Usecase:

- Người dùng (Mentee) muốn tìm Mentor theo tên, công ty, kỹ năng, danh mục.
- Lọc Mentor theo quốc gia, ngôn ngữ, đánh giá, mức giá.
- Sắp xếp kết quả theo các tiêu chí: Đánh giá (rating), số lượng review, giá plan.
- Phân trang kết quả tìm kiếm.

Các thao tác chính:

1. Lọc Mentor theo nhiều điều kiện: tên, họ, công ty, job title, kỹ năng, danh mục, quốc gia, ngôn ngữ, trạng thái.
2. Lọc theo đánh giá tối thiểu và khoảng giá plan.
3. Tính toán và hiển thị thông tin tổng hợp: rating trung bình, tổng review, tổng plans, giá thấp nhất/cao nhất.
4. Sắp xếp kết quả theo nhiều cột khác nhau.
5. Phân trang động với tính toán tổng số trang.

Input:

- `@SearchName(NVARCHAR(50))`: Họ tên Mentor cần tìm.
- `@FirstName(NVARCHAR(50))`: Tên Mentor cần tìm.
- `@LastName(NVARCHAR(50))`: Họ Mentor cần tìm.
- `@CompanyName(NVARCHAR(255))`: Tên công ty Mentor đang/đã làm việc.

- @JobTitleName(NVARCHAR(255)): Chức danh công việc hiện tại.
- @SkillName(NVARCHAR(100)): Tên kỹ năng Mentor sở hữu.
- @CategoryName(NVARCHAR(100)): Tên danh mục kỹ năng.
- @Country(NVARCHAR(100)): Quốc gia của Mentor.
- @Language(NVARCHAR(100)): Ngôn ngữ Mentor sử dụng.
- @Status(NVARCHAR(20)): Trạng thái tài khoản (Active, Inactive).
- @MinRating(DECIMAL(3,2)): Đánh giá (Số sao) tối thiểu.
- @MaxPrice(DECIMAL(10,2)): Giá tối đa của plan.
- @MinPrice(DECIMAL(10,2)): Giá tối thiểu của plan.
- @SortColumn(NVARCHAR(50)): Cột dùng để sắp xếp (mặc định là user_id).
- @SortDirection(NVARCHAR(4)): Hướng sắp xếp ASC/DESC (mặc định là ASC).
- @Page(INT): Số trang hiện tại (mặc định là 1).
- @Limit(INT): Số bản ghi mỗi trang (mặc định là 10, tối đa là 100).

Output:

Trả về danh sách các Mentor thỏa mãn điều kiện tìm kiếm, bao gồm các thông tin:

- Thông tin cá nhân: user_id, first_name, last_name, email, avatar_url, country, status, timezone.
- Thông tin Mentor: bio, headline, response_time, cv_url.
- Thông tin công việc: company_names, job_title.
- Thông tin đánh giá: average_rating, total_reviews.
- Thông tin plan: total_plans, lowest_plan_price, highest_plan_price, average_plan_price.
- Thông tin kỹ năng: skills, languages, categories.
- Thông tin phân trang: total_items, current_page, items_per_page, total_pages.

Câu lệnh tạo thủ tục:

```
1 IF OBJECT_ID('dbo.sp_SearchMentors', 'P') IS NOT NULL
2     DROP PROCEDURE dbo.sp_SearchMentors;
3 GO
4
5 CREATE PROCEDURE dbo.sp_SearchMentors
6     @SearchName NVARCHAR(100) = NULL,
7     @FirstName NVARCHAR(50) = NULL,
8     @LastName NVARCHAR(50) = NULL,
9     @CompanyName NVARCHAR(255) = NULL,
10    @JobTitleName NVARCHAR(255) = NULL,
11    @SkillName NVARCHAR(100) = NULL,
12    @CategoryName NVARCHAR(100) = NULL,
13    @Country NVARCHAR(100) = NULL,
14    @Language NVARCHAR(100) = NULL,
15    @Status NVARCHAR(20) = NULL,
16    @MinRating DECIMAL(3,2) = NULL,
17    @MaxPrice DECIMAL(10,2) = NULL,
18    @MinPrice DECIMAL(10,2) = NULL,
19    @SortColumn NVARCHAR(50) = N'user_id',
20    @SortDirection NVARCHAR(4) = N'ASC',
21    @Page INT = 1,
22    @Limit INT = 10
23 AS
24 BEGIN
25     SET NOCOUNT ON;
26
27     -- Kiem tra SortColumn tranh sql injection
28     IF @SortColumn NOT IN (
29         N'user_id', N'first_name', N'last_name', N'email', N'country',
30         N'company_name', N'job_title', N'average_rating', N'
total_reviews',
31         N'total_plans', N'lowest_plan_price'
32     )
33         SET @SortColumn = N'user_id';
34
```

```
35 IF @SortDirection NOT IN (N'ASC', N'DESC')
36     SET @SortDirection = N'ASC';
37
38 IF @Page < 1 SET @Page = 1;
39 IF @Limit < 1 OR @Limit > 100 SET @Limit = 10;
40
41 DECLARE @Offset INT = (@Page - 1) * @Limit;
42
43 -- CTE để xử lý các filter conditions cho mentor (tránh duplicate
44 rows)
45 WITH FilteredMentors AS (
46     SELECT DISTINCT m.user_id
47     FROM mentors m
48     INNER JOIN users u ON m.user_id = u.user_id
49     WHERE
50         u.role = N'Mentor'
51         AND (@FirstName IS NULL OR u.first_name LIKE N'%' +
52 @FirstName + N'%'')
53         AND (@LastName IS NULL OR u.last_name LIKE N'%' +
54 @LastName + N'%'')
55         AND (@SearchName IS NULL OR
56             u.first_name LIKE N'%' + @SearchName + N'%' OR
57             u.last_name LIKE N'%' + @SearchName + N'%' OR
58             (u.first_name + N' ' + u.last_name) LIKE N'%' +
59 @SearchName + N'%'')
60     )
61     AND (@Country IS NULL OR u.country = @Country)
62     AND (@Status IS NULL OR u.status = @Status)
63     -- Loc theo company
64     AND (@CompanyName IS NULL OR EXISTS (
65         SELECT 1
66         FROM work_for wf
67         INNER JOIN companies c ON wf.c_company_id = c.
68 company_id
69         WHERE wf.mentor_id = m.user_id
70         AND c.cname LIKE N'%' + @CompanyName + N'%'')
```

```
66         ))
67         -- Loc theo job title
68         AND (@JobTitleName IS NULL OR EXISTS (
69             SELECT 1
70             FROM work_for wf
71             INNER JOIN job_title jt ON wf.current_job_title_id =
jt.job_title_id
72             WHERE wf.mentor_id = m.user_id
73             AND jt.job_name LIKE N'%' + @JobTitleName + N'%'
74         ))
75         -- Loc theo skill
76         AND (@SkillName IS NULL OR EXISTS (
77             SELECT 1
78             FROM set_skill ss
79             INNER JOIN skills s ON ss.skill_id = s.skill_id
80             WHERE ss.mentor_id = m.user_id
81             AND s.skill_name LIKE N'%' + @SkillName + N'%'
82         ))
83         -- Loc theo category
84         AND (@CategoryName IS NULL OR EXISTS (
85             SELECT 1
86             FROM set_skill ss
87             INNER JOIN skills s ON ss.skill_id = s.skill_id
88             INNER JOIN own_skill os ON s.skill_id = os.skill_id
89             INNER JOIN categories cat ON os.category_id = cat.
category_id
90             WHERE ss.mentor_id = m.user_id
91             AND cat.category_name LIKE N'%' + @CategoryName + N'
%'
92         ))
93         -- Loc theo language
94         AND (@Language IS NULL OR EXISTS (
95             SELECT 1
96             FROM mentor_languages ml
97             CROSS APPLY STRING_SPLIT(@Language, ',') AS lang
98             WHERE ml.mentor_id = m.user_id
```

```
99         AND LTRIM(RTRIM(ml.mentor_language)) = LTRIM(RTRIM(
100 lang.value))
101     ))
102 ),
103 -- CTE de lay thong tin cong ty va job title
104 CompanyInfo AS (
105     SELECT
106         wf.mentor_id,
107         STUFF((
108             SELECT DISTINCT N', ' + c2.cname
109             FROM work_for wf2
110             INNER JOIN companies c2 ON c2.company_id = wf2.
111 c_company_id
112             WHERE wf2.mentor_id = wf.mentor_id
113             FOR XML PATH(''), TYPE
114             ).value('',' NVARCHAR(MAX)'), 1, 2, '') AS company_names,
115 -- Lay job title dau tien
116 (SELECT TOP 1 jt2.job_name
117 FROM work_for wf3
118 INNER JOIN job_title jt2 ON wf3.current_job_title_id =
119 jt2.job_title_id
120 WHERE wf3.mentor_id = wf.mentor_id
121 ) AS job_title
122 FROM work_for wf
123 WHERE wf.mentor_id IN (SELECT user_id FROM FilteredMentors)
124 GROUP BY wf.mentor_id
125 ),
126 -- CTE lay thong tin plans va gia
127 PlanInfo AS (
128     SELECT
129         p.mentor_id,
130         COUNT(DISTINCT p.plan_id) AS total_plans,
131         MIN(p.plan_charge) AS lowest_plan_price,
```

```
132         WHERE p.mentor_id IN (SELECT user_id FROM FilteredMentors)
133         GROUP BY p.mentor_id
134     ),
135     -- CTE lay skills
136     SkillInfo AS (
137         SELECT
138             ss.mentor_id,
139             STUFF((
140                 SELECT DISTINCT N', ' + s2.skill_name
141                 FROM set_skill ss2
142                 INNER JOIN skills s2 ON ss2.skill_id = s2.skill_id
143                 WHERE ss2.mentor_id = ss.mentor_id
144                 FOR XML PATH(''), TYPE
145             ).value('.', 'NVARCHAR(MAX)'), 1, 2, '') AS skills
146         FROM set_skill ss
147         WHERE ss.mentor_id IN (SELECT user_id FROM FilteredMentors)
148         GROUP BY ss.mentor_id
149     ),
150     -- CTE lay languages
151     LanguageInfo AS (
152         SELECT
153             ml.mentor_id,
154             STUFF((
155                 SELECT DISTINCT N', ' + ml2.mentor_language
156                 FROM mentor_languages ml2
157                 WHERE ml2.mentor_id = ml.mentor_id
158                 FOR XML PATH(''), TYPE
159             ).value('.', 'NVARCHAR(MAX)'), 1, 2, '') AS languages
160         FROM mentor_languages ml
161         WHERE ml.mentor_id IN (SELECT user_id FROM FilteredMentors)
162         GROUP BY ml.mentor_id
163     ),
164     -- CTE lay categories
165     CategoryInfo AS (
166         SELECT DISTINCT
167             ss.mentor_id,
```

```
168         STUFF((
169             SELECT DISTINCT N', ' + cat2.category_name
170             FROM set_skill ss2
171             INNER JOIN own_skill os2 ON ss2.skill_id = os2.
skill_id
172             INNER JOIN categories cat2 ON os2.category_id = cat2.
category_id
173             WHERE ss2.mentor_id = ss.mentor_id
174             FOR XML PATH(''), TYPE
175             ).value('','NVARCHAR(MAX)'), 1, 2, '') AS categories
176 FROM set_skill ss
177 INNER JOIN own_skill os ON ss.skill_id = os.skill_id
178 INNER JOIN categories cat ON os.category_id = cat.category_id
179 WHERE ss.mentor_id IN (SELECT user_id FROM FilteredMentors)
180 GROUP BY ss.mentor_id
181 ),
182 -- CTE tat ca filter
183 FinalData AS (
184     SELECT
185         m.user_id,
186         u.first_name,
187         u.last_name,
188         u.email,
189         u.avatar_url,
190         u.country,
191         u.status,
192         u.timezone,
193         m.bio,
194         m.headline,
195         dbo.CalculateMentorResponseTime(m.user_id) AS
response_time,
196         m.cv_url,
197         -- su dung derived attr tu bang mentors
198         ISNULL(m.rating, 0) AS average_rating,
199         ISNULL(m.total_reviews, 0) AS total_reviews,
200         m.total_mentee,
```

```
201         m.total_stars ,
202         ISNULL(ci.company_names, N'') AS company_names ,
203         ISNULL(ci.job_title, N'') AS job_title ,
204         ISNULL(pi.total_plans, 0) AS total_plans ,
205         ISNULL(pi.lowest_plan_price, 0) AS lowest_plan_price ,
206         ISNULL(pi.highest_plan_price, 0) AS highest_plan_price ,
207         ISNULL(pi.average_plan_price, 0) AS average_plan_price ,
208         ISNULL(si.skills, N'') AS skills ,
209         ISNULL(li.languages, N'') AS languages ,
210         ISNULL(cati.categories, N'') AS categories
211     FROM mentors m
212     INNER JOIN users u ON m.user_id = u.user_id
213     INNER JOIN FilteredMentors fm ON m.user_id = fm.user_id
214     LEFT JOIN CompanyInfo ci ON m.user_id = ci.mentor_id
215     LEFT JOIN PlanInfo pi ON m.user_id = pi.mentor_id
216     LEFT JOIN SkillInfo si ON m.user_id = si.mentor_id
217     LEFT JOIN LanguageInfo li ON m.user_id = li.mentor_id
218     LEFT JOIN CategoryInfo cati ON m.user_id = cati.mentor_id
219     WHERE (@MinRating IS NULL OR ISNULL(m.rating, 0) >= @MinRating
220 )
221     AND (@MaxPrice IS NULL OR ISNULL(pi.lowest_plan_price,
222 999999) <= @MaxPrice)
223     AND (@MinPrice IS NULL OR ISNULL(pi.lowest_plan_price, 0) >=
224 @MinPrice)
225 ),
226 -- Count tong cho trang
227 TotalCount AS (
228     SELECT COUNT(*) AS total FROM FinalData
229 )
230 SELECT
231     fd.*,
232     tc.total AS total_items ,
233     @Page AS current_page ,
234     @Limit AS items_per_page ,
235     CEILING(CAST(tc.total AS FLOAT) / @Limit) AS total_pages
236 FROM FinalData fd
```

```
234     CROSS JOIN TotalCount tc
235     ORDER BY
236         CASE WHEN @SortColumn = N'first_name' AND @SortDirection = N'
ASC' THEN fd.first_name END ASC,
237         CASE WHEN @SortColumn = N'first_name' AND @SortDirection = N'
DESC' THEN fd.first_name END DESC,
238         CASE WHEN @SortColumn = N'last_name' AND @SortDirection = N'
ASC' THEN fd.last_name END ASC,
239         CASE WHEN @SortColumn = N'last_name' AND @SortDirection = N'
DESC' THEN fd.last_name END DESC,
240         CASE WHEN @SortColumn = N'email' AND @SortDirection = N'ASC'
THEN fd.email END ASC,
241         CASE WHEN @SortColumn = N'email' AND @SortDirection = N'DESC'
THEN fd.email END DESC,
242         CASE WHEN @SortColumn = N'country' AND @SortDirection = N'ASC'
THEN fd.country END ASC,
243         CASE WHEN @SortColumn = N'country' AND @SortDirection = N'DESC
' THEN fd.country END DESC,
244         CASE WHEN @SortColumn = N'company_name' AND @SortDirection = N
'ASC' THEN fd.company_names END ASC,
245         CASE WHEN @SortColumn = N'company_name' AND @SortDirection = N
'DESC' THEN fd.company_names END DESC,
246         CASE WHEN @SortColumn = N'job_title' AND @SortDirection = N'
ASC' THEN fd.job_title END ASC,
247         CASE WHEN @SortColumn = N'job_title' AND @SortDirection = N'
DESC' THEN fd.job_title END DESC,
248         CASE WHEN @SortColumn = N'average_rating' AND @SortDirection =
N'ASC' THEN fd.average_rating END ASC,
249         CASE WHEN @SortColumn = N'average_rating' AND @SortDirection =
N'DESC' THEN fd.average_rating END DESC,
250         CASE WHEN @SortColumn = N'total_reviews' AND @SortDirection =
N'ASC' THEN fd.total_reviews END ASC,
251         CASE WHEN @SortColumn = N'total_reviews' AND @SortDirection =
N'DESC' THEN fd.total_reviews END DESC,
252         CASE WHEN @SortColumn = N'total_plans' AND @SortDirection = N'
ASC' THEN fd.total_plans END ASC,
```



```
253         CASE WHEN @SortColumn = N'total_plans' AND @SortDirection = N'  
DESC' THEN fd.total_plans END DESC,  
254         CASE WHEN @SortColumn = N'lowest_plan_price' AND  
@SortDirection = N'ASC' THEN fd.lowest_plan_price END ASC,  
255         CASE WHEN @SortColumn = N'lowest_plan_price' AND  
@SortDirection = N'DESC' THEN fd.lowest_plan_price END DESC,  
256         CASE WHEN @SortColumn = N'user_id' AND @SortDirection = N'ASC'  
THEN fd.user_id END ASC,  
257         CASE WHEN @SortColumn = N'user_id' AND @SortDirection = N'DESC  
' THEN fd.user_id END DESC,  
258         fd.user_id ASC  
259     OFFSET @Offset ROWS  
260     FETCH NEXT @Limit ROWS ONLY;  
261 END;  
262 GO
```

Nhóm câu lệnh thực thi thủ tục mẫu:

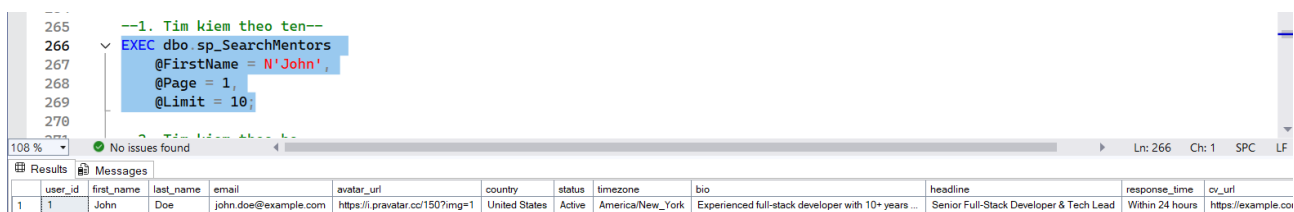
```
1  --1. Tim kiem theo ten--  
2  EXEC dbo.sp_SearchMentors  
3      @FirstName = N'John',  
4      @Page = 1,  
5      @Limit = 10;  
6  
7  --2. Tim kiem theo ky nang React--  
8  EXEC dbo.sp_SearchMentors  
9      @SkillName = N'React',  
10     @SortColumn = N'average_rating',  
11     @SortDirection = N'DESC',  
12     @Page = 1,  
13     @Limit = 10;  
14  
15 --3. Tim kiem phuc tap: Python + English + Rating >= 4--  
16 EXEC dbo.sp_SearchMentors  
17     @SkillName = N'Python',  
18     @Language = N'English',  
19     @MinRating = 4.0,
```

```

20     @SortColumn = N'total_plans',
21     @SortDirection = N'DESC',
22     @Page = 1,
23     @Limit = 10;
24
25 --4. Sắp xếp theo tên A-Z--
26 EXEC dbo.sp_SearchMentors
27     @SortColumn = N'first_name',
28     @SortDirection = N'ASC',
29     @Page = 1,
30     @Limit = 10;
31
32 --5. Tìm kiếm lỗi, không có kết quả--
33 EXEC dbo.sp_SearchMentors
34     @SkillName = N'SkillKhongTonTai123',
35     @Page = 1,
36     @Limit = 10;

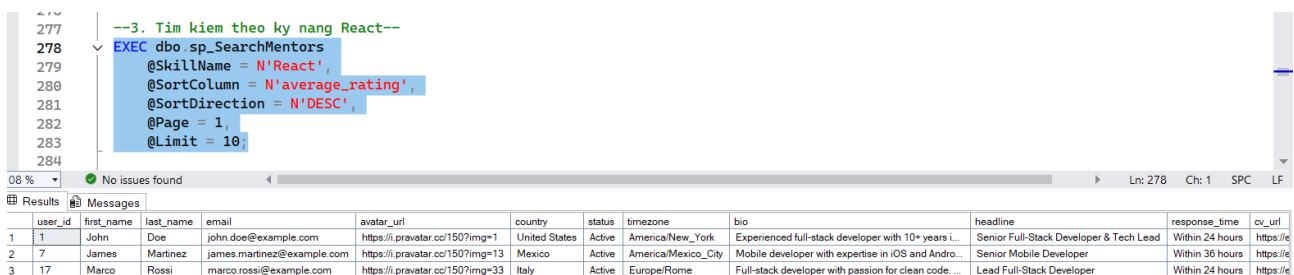
```

Kết quả các màn hình hiển thị từ DBMS minh họa cho việc thực hiện thủ tục:



user_id	first_name	last_name	email	avatar_url	country	status	timezone	bio	headline	response_time	cv_url
1	John	Doe	john.doe@example.com	https://i.pravatar.co/150?img=1	United States	Active	America/New_York	Experienced full-stack developer with 10+ years ...	Senior Full-Stack Developer & Tech Lead	Within 24 hours	https://example.co

Hình 32: Kết quả thực thi thủ tục mẫu 1



user_id	first_name	last_name	email	avatar_url	country	status	timezone	bio	headline	response_time	cv_url
1	John	Doe	john.doe@example.com	https://i.pravatar.co/150?img=1	United States	Active	America/New_York	Experienced full-stack developer with 10+ years i...	Senior Full-Stack Developer & Tech Lead	Within 24 hours	https://e
7	James	Martinez	james.martinez@example.com	https://i.pravatar.co/150?img=13	Mexico	Active	America/Mexico_City	Mobile developer with expertise in iOS and Andro...	Senior Mobile Developer	Within 36 hours	https://e
17	Marco	Rossi	marco.rossi@example.com	https://i.pravatar.co/150?img=33	Italy	Active	Europe/Rome	Full-stack developer with passion for clean code. ...	Lead Full-Stack Developer	Within 24 hours	https://e

Hình 33: Kết quả thực thi thủ tục mẫu 2

367

--16. Tìm kiếm phức tạp: Python + English + Rating >= 4--

368

EXEC dbo.sp_SearchMentors

369

@SkillName = N'Python',

370

@Language = N'English',

371

@MinRating = 4.0,

372

@SortColumn = N'total_plans',

373

@SortDirection = N'DESC',

374

@Page = 1,

375

@Limit = 10;

376

108 %

No issues found

Ln: 375

Ch: 17

SPC

LF

Results

Messages

	user_id	first_name	last_name	email	avatar_url	country	status	timezone	bio	headline	response_time	cv_url
1	3	Michael	Chen	michael.chen@example.com	https://i.pravatar.co/1507/img=12	Singapore	Active	Asia/Singapore	Data scientist and ML engineer specializing in NLP...	Senior Data Scientist & ML Engineer	Within 48 hours	https://
2	11	Thomas	Lee	thomas.lee@example.com	https://i.pravatar.co/1507/img=18	Japan	Active	Asia/Tokyo	Software architect with deep knowledge in system d...	Principal Software Architect	Within 24 hours	https://
3	15	Daniel	Santos	daniel.santos@example.com	https://i.pravatar.co/1507/img=26	Brazil	Active	America/Sao_Paulo	AI researcher and engineer. Published papers on de...	AI Research Engineer	Within 48 hours	https://
4	19	Alex	Petrov	alex.petrov@example.com	https://i.pravatar.co/1507/img=36	Russia	Active	Europe/Moscow	Game developer with 12 years experience. Worked ...	Senior Game Developer	Within 36 hours	https://
5	23	Henrik	Andersson	henrik.andersson@example.com	https://i.pravatar.co/1507/img=43	Sweden	Active	Europe/Stockholm	Blockchain developer and Web3 enthusiast. Built De...	Senior Blockchain Developer	Within 48 hours	https://
6	25	Mohammed	Ahmed	mohammed.ahmed@example.com	https://i.pravatar.co/1507/img=46	UAE	Active	Asia/Dubai	IoT engineer and embedded systems developer. Bui...	Senior IoT Engineer	Within 36 hours	https://

Hình 34: Kết quả thực thi thủ tục mẫu 3

430

--24. Sắp xếp theo tên A-Z--

431

EXEC dbo.sp_SearchMentors

432

@SortColumn = N'first_name',

433

@SortDirection = N'ASC',

434

@Page = 1,

435

@Limit = 10;

436

100%

100 issues found

Ln: 431

Ch: 1

SPC

LF

Results

Messages

	user_id	first_name	last_name	email	avatar_url	country	status	timezone	bio	headline	response_time	cv_url
1	19	Alex	Petrov	alex.petrov@example.com	https://i.pravatar.co/1507/img=36	Russia	Active	Europe/Moscow	Game developer with 12 years experience. Worked...	Senior Game Developer	Within 36 hours	https://
2	8	Anna	Kowalski	anna.kowalski@example.com	https://i.pravatar.co/1507/img=16	Poland	Active	Europe/Warsaw	UI/UX designer focused on creating beautiful and int...	Senior UI/UX Designer	Within 12 hours	https://
3	28	Chloe	Martin	chloe.martin@example.com	https://i.pravatar.co/1507/img=49	Belgium	Active	Europe/Brussels	E-commerce specialist and conversion optimization...	E-commerce Strategy Lead	Within 24 hours	https://
4	15	Daniel	Santos	daniel.santos@example.com	https://i.pravatar.co/1507/img=26	Brazil	Active	America/Sao_Paulo	AI researcher and engineer. Published papers on de...	AI Research Engineer	Within 48 hours	https://
5	5	David	Kim	david.kim@example.com	https://i.pravatar.co/1507/img=15	South Korea	Active	Asia/Seoul	Career coach and leadership consultant. Former tec...	Career Coach & Leadership Consultant	Within 24 hours	https://
6	4	Emily	Rodriguez	emily.rodriguez@example.com	https://i.pravatar.co/1507/img=9	Spain	Active	Europe/Madrid	Digital marketing strategist with experience in growt...	Growth Marketing Lead	Within 24 hours	https://
7	16	Emma	Wilson	emma.wilson@example.com	https://i.pravatar.co/1507/img=32	New Zealand	Active	Pacific/Auckland	Graphic designer and illustrator. Specialized in bran...	Senior Graphic Designer	Within 24 hours	https://
8	29	Erik	Hansen	erik.hansen@example.com	https://i.pravatar.co/1507/img=52	Denmark	Active	Europe/Copenhagen	Site reliability engineer. Expert in monitoring, observ...	Senior SRE	Within 24 hours	https://
9	23	Henrik	Andersson	henrik.andersson@example.com	https://i.pravatar.co/1507/img=43	Sweden	Active	Europe/Stockholm	Blockchain developer and Web3 enthusiast. Built De...	Senior Blockchain Developer	Within 48 hours	https://
10	22	Isabella	Lopez	isabella.lopez@example.com	https://i.pravatar.co/1507/img=42	Colombia	Active	America/Bogota	Social media strategist and influencer marketing exp...	Social Media Strategy Lead	Within 12 hours	https://

Hình 35: Kết quả thực thi thủ tục mẫu 4

```

437 --25. Tìm kiếm lỗi, không có kết quả--
438 EXEC dbo.sp_SearchMentors
439 @SkillName = N'SkillKhongTonTai123',
440 @Page = 1,
441 @Limit = 10;

```

user_id	first_name	last_name	email	avatar_url	country	status	timezone	bio	headline	response_time	cv_url	company_names	job_title	average_rating	total_reviews	total_plans	lowest_plan_price	highest_plan_price	average_plan_price
---------	------------	-----------	-------	------------	---------	--------	----------	-----	----------	---------------	--------	---------------	-----------	----------------	---------------	-------------	-------------------	--------------------	--------------------

Hình 36: Kết quả thực thi thủ tục mẫu 5

2.3.2 Thống kê Dashboard

Mô tả thủ tục: Thủ tục `sp_DashboardStatistics` được sử dụng để thống kê và tổng hợp dữ liệu nghiệp vụ cho giao diện Dashboard quản trị. Thủ tục này hỗ trợ thống kê theo nhiều nhóm khác nhau (Mentor, tháng, danh mục) và cung cấp các số liệu quan trọng về doanh thu, booking, đánh giá.

Usecase:

- Admin muốn xem thống kê tổng quan về hoạt động của hệ thống

- Thống kê doanh thu và hiệu suất của từng Mentor.
- Phân tích xu hướng doanh thu theo thời gian (theo tháng).
- Đánh giá hiệu quả của từng danh mục kỹ năng.
- So sánh các chỉ số giữa các Mentor, tháng, danh mục.

Các thao tác chính:

1. Thống kê theo Mentor: tổng doanh thu, số lượng booking, đánh giá trung bình, số meeting hoàn thành.
2. Thống kê theo tháng: doanh thu theo tháng, số lượng Mentee, Mentor, booking.
3. Thống kê theo danh mục: số Mentor, số booking, doanh thu của từng category.
4. Lọc theo khoảng thời gian, công ty, danh mục.
5. Sắp xếp kết quả theo nhiều tiêu chí.
6. Sử dụng các hàm Aggregate (SUM, COUNT, AVG) kết hợp với CTE để tính toán doanh thu, số lượng booking, rating trung bình và xử lý chuỗi dữ liệu (tên công ty, kỹ năng).

Input:

- @GroupBy(NVARCHAR(50)): Nhóm thống kê - 'mentor', 'month', hoặc 'category' (mặc định là 'mentor')
- @StartDate(DATETIME): Ngày bắt đầu khoảng thời gian thống kê (mặc định là 12 tháng trước).
- @EndDate(DATETIME): Ngày kết thúc khoảng thời gian thống kê (mặc định là hôm nay).
- @CompanyId(INT): Lọc Mentor theo công ty.
- @CategoryId(INT): Lọc Mentor theo danh mục kỹ năng.

- @MinRevenue(DECIMAL(10,2)): Doanh thu tối thiểu để hiển thị.
- @MinBookingCount(INT): Số booking tối thiểu để hiển thị.
- @SortBy(NVARCHAR(50)): Cột sắp xếp (mặc định là 'total_revenue').
- @SortOrder(NVARCHAR(4)): Hướng sắp xếp ASC/DESC (mặc định là 'DESC').

Output:

Khi @GroupBy= 'mentor':

- mentor_id, mentor_name, mentor_email, mentor_avatar_url, mentor_country, mentor_status, mentor_headline.
- company_names, job_titles, skills.
- total_bookings, total_invoices, total_revenue, average_invoice_amount.
- completed_meetings, cancelled_meetings, upcoming_meetings.
- average_rating, total_reviews, total_plans.

Khi @GroupBy= 'month':

- year, month, month_name.
- total_invoices, total_mentees, total_mentors.
- total_revenue, average_invoice_amount, total_bookings.

Khi @GroupBy= 'category':

- category_id, category_name.
- total_mentors_with_skill, total_bookings, total_revenue.

Câu lệnh tạo thủ tục:

```
1 IF OBJECT_ID('dbo.sp_DashboardStatistics', 'P') IS NOT NULL
2     DROP PROCEDURE dbo.sp_DashboardStatistics;
3 GO
4
5 CREATE PROCEDURE dbo.sp_DashboardStatistics
6     @GroupBy NVARCHAR(50) = N'mentor',
7     @StartDate DATETIME = NULL,
8     @EndDate DATETIME = NULL,
```

```
9      @CompanyId INT = NULL ,
10     @CategoryId INT = NULL ,
11     @MinRevenue DECIMAL(10,2) = NULL ,
12     @MinBookingCount INT = NULL ,
13     @SortBy NVARCHAR(50) = N'total_revenue' ,
14     @SortOrder NVARCHAR(4) = N'DESC'
15 AS
16 BEGIN
17     SET NOCOUNT ON;
18
19     -- Xu ly mac dinh
20     IF @SortOrder NOT IN (N'ASC', N'DESC') SET @SortOrder = N'DESC';
21     IF @StartDate IS NULL SET @StartDate = DATEADD(MONTH, -12, GETDATE
22     ());
23     IF @EndDate IS NULL SET @EndDate = GETDATE();
24
25     -- =====
26     -- GROUP BY MENTOR
27     -- =====
28     IF @GroupBy = N'mentor'
29     BEGIN
30         IF @SortBy NOT IN (N'total_revenue', N'total_bookings', N'
31         average_rating',
32                             N'completed_meetings', N'mentor_name', N'
33         average_invoice_amount')
34             SET @SortBy = N'total_revenue';
35
36     -- CTE Loc Mentor co ban (Theo Company, Category)
37     WITH BaseMentors AS (
38         SELECT DISTINCT m.user_id
39         FROM mentors m
40         INNER JOIN users u ON m.user_id = u.user_id
41         WHERE (@CompanyId IS NULL OR EXISTS (
42             SELECT 1 FROM work_for wf
43             WHERE wf.mentor_id = m.user_id AND wf.c_company_id =
44             @CompanyId
```

```
41         ))
42         AND (@CategoryId IS NULL OR EXISTS (
43             SELECT 1 FROM set_skill ss
44             INNER JOIN own_skill os ON ss.skill_id = os.skill_id
45             WHERE ss.mentor_id = m.user_id AND os.category_id =
@CategoryId
46         ))
47     ),
48     -- CTE Tính Doanh thu & Booking
49     RevenueStats AS (
50         SELECT
51             p.mentor_id,
52             COUNT(DISTINCT b.plan_registrations_id) AS
total_bookings,
53             COUNT(DISTINCT inv.invoice_id) AS total_invoices,
54             SUM(inv.amount_total) AS total_revenue,
55             AVG(inv.amount_total) AS average_invoice_amount
56         FROM plans p
57         INNER JOIN bookings b ON p.plan_id = b.plan_id
58         INNER JOIN plan_registrations pr ON b.
plan_registrations_id = pr.registration_id
59         INNER JOIN invoices inv ON pr.registration_id = inv.
plan_registrations_id
60         WHERE inv.paid_time BETWEEN @StartDate AND @EndDate
61             AND inv.payment_status = N'paid'
62             AND p.mentor_id IN (SELECT user_id FROM BaseMentors)
63         GROUP BY p.mentor_id
64     ),
65
66     -- CTE Tính Meeting (Tinh rieng)
67     MeetingStats AS (
68         SELECT
69             mentor_id,
70             COUNT(CASE WHEN status = N'Completed' THEN 1 END) AS
completed_meetings,
71             COUNT(CASE WHEN status = N'Cancelled' THEN 1 END) AS
```

```
cancelled_meetings ,
72         COUNT(CASE WHEN status IN (N'Pending', N'Scheduled')
THEN 1 END) AS upcoming_meetings
73     FROM meetings
74     WHERE mentor_id IN (SELECT user_id FROM BaseMentors)
75     GROUP BY mentor_id
76 ),
77 -- CTE Gop thông tin chuỗi (Cty, Skill...)
78 StringAggStats AS (
79     SELECT
80         bm.user_id,
81         -- Danh sách cty
82         STUFF((SELECT DISTINCT N', ' + c.cname
83             FROM work_for wf JOIN companies c ON wf.
c_company_id = c.company_id
84             WHERE wf.mentor_id = bm.user_id FOR XML PATH(''
), TYPE).value('',' NVARCHAR(MAX)'), 1, 2, '') AS company_names,
85         -- Danh sách chức danh
86         STUFF((SELECT DISTINCT N', ' + jt.job_name
87             FROM work_for wf JOIN job_title jt ON wf.
current_job_title_id = jt.job_title_id
88             WHERE wf.mentor_id = bm.user_id FOR XML PATH(''
), TYPE).value('',' NVARCHAR(MAX)'), 1, 2, '') AS job_titles,
89         -- Danh sách kỹ năng
90         STUFF((SELECT DISTINCT N', ' + s.skill_name
91             FROM set_skill ss JOIN skills s ON ss.skill_id
= s.skill_id
92             WHERE ss.mentor_id = bm.user_id FOR XML PATH(''
), TYPE).value('',' NVARCHAR(MAX)'), 1, 2, '') AS skills
93     FROM BaseMentors bm
94 )
95 -- JOIN TẤT CẢ LẠI
96 SELECT
97     m.user_id AS mentor_id,
98     u.first_name + N' ' + u.last_name AS mentor_name,
99     u.email AS mentor_email,
```



```
100         u.avatar_url AS mentor_avatar_url ,
101         u.country AS mentor_country ,
102         u.status AS mentor_status ,
103         m.headline AS mentor_headline ,
104         ISNULL(sa.company_names, '') AS company_names ,
105         ISNULL(sa.job_titles, '') AS job_titles ,
106         ISNULL(sa.skills, '') AS skills ,
107         ISNULL(rs.total_bookings, 0) AS total_bookings ,
108         ISNULL(rs.total_invoices, 0) AS total_invoices ,
109         ISNULL(rs.total_revenue, 0) AS total_revenue ,
110         ISNULL(rs.average_invoice_amount, 0) AS
average_invoice_amount ,
111         ISNULL(ms.completed_meetings, 0) AS completed_meetings ,
112         ISNULL(ms.cancelled_meetings, 0) AS cancelled_meetings ,
113         ISNULL(ms.upcoming_meetings, 0) AS upcoming_meetings ,
114         -- sdung derived attr tu bang mentors
115         ISNULL(m.rating, 0) AS average_rating ,
116         ISNULL(m.total_reviews, 0) AS total_reviews ,
117         m.total_mentee ,
118         m.total_stars ,
119         (SELECT COUNT(*) FROM plans WHERE mentor_id = m.user_id)
AS total_plans
120     FROM mentors m
121     INNER JOIN users u ON m.user_id = u.user_id
122     INNER JOIN BaseMentors bm ON m.user_id = bm.user_id
123     LEFT JOIN RevenueStats rs ON m.user_id = rs.mentor_id
124     LEFT JOIN MeetingStats ms ON m.user_id = ms.mentor_id
125     LEFT JOIN StringAggStats sa ON m.user_id = sa.user_id
126     WHERE
127         (@MinRevenue IS NULL OR ISNULL(rs.total_revenue, 0) >=
@MinRevenue)
128         AND (@MinBookingCount IS NULL OR ISNULL(rs.total_bookings,
0) >= @MinBookingCount)
129     ORDER BY
130         CASE WHEN @SortBy = N'total_revenue' AND @SortOrder = N'
DESC' THEN rs.total_revenue END DESC ,
```

```
131         CASE WHEN @SortBy = N'total_revenue' AND @SortOrder = N'  
ASC' THEN rs.total_revenue END ASC,  
132         CASE WHEN @SortBy = N'total_bookings' AND @SortOrder = N'  
DESC' THEN rs.total_bookings END DESC,  
133         CASE WHEN @SortBy = N'total_bookings' AND @SortOrder = N'  
ASC' THEN rs.total_bookings END ASC,  
134         CASE WHEN @SortBy = N'average_rating' AND @SortOrder = N'  
DESC' THEN m.rating END DESC,  
135         CASE WHEN @SortBy = N'average_rating' AND @SortOrder = N'  
ASC' THEN m.rating END ASC,  
136         CASE WHEN @SortBy = N'completed_meetings' AND @SortOrder =  
N'DESC' THEN ms.completed_meetings END DESC,  
137         CASE WHEN @SortBy = N'completed_meetings' AND @SortOrder =  
N'ASC' THEN ms.completed_meetings END ASC,  
138         CASE WHEN @SortBy = N'average_invoice_amount' AND  
@SortOrder = N'DESC' THEN rs.average_invoice_amount END DESC,  
139         CASE WHEN @SortBy = N'average_invoice_amount' AND  
@SortOrder = N'ASC' THEN rs.average_invoice_amount END ASC,  
140         CASE WHEN @SortBy = N'mentor_name' AND @SortOrder = N'DESC  
' THEN u.first_name END DESC,  
141         CASE WHEN @SortBy = N'mentor_name' AND @SortOrder = N'ASC'  
THEN u.first_name END ASC;  
142     END  
143  
144     -- =====  
145     -- GROUP BY MONTH  
146     -- =====  
147     ELSE IF @GroupBy = N'month'  
148     BEGIN  
149         SELECT  
150             YEAR(inv.paid_time) AS year,  
151             MONTH(inv.paid_time) AS month,  
152             DATENAME(MONTH, inv.paid_time) AS month_name,  
153             COUNT(DISTINCT inv.invoice_id) AS total_invoices,  
154             COUNT(DISTINCT b.mentee_id) AS total_mentees,  
155             COUNT(DISTINCT p.mentor_id) AS total_mentors,
```

```
156         SUM(inv.amount_total) AS total_revenue ,
157         AVG(inv.amount_total) AS average_invoice_amount ,
158         COUNT(DISTINCT b.plan_registrations_id) AS total_bookings
159     FROM invoices inv
160     INNER JOIN plan_registrations pr ON inv.
plan_registrations_id = pr.registration_id
161     INNER JOIN bookings b ON pr.registration_id = b.
plan_registrations_id
162     INNER JOIN plans p ON b.plan_id = p.plan_id
163     WHERE inv.paid_time BETWEEN @StartDate AND @EndDate
164         AND inv.payment_status = N'paid'
165         AND (@CompanyId IS NULL OR EXISTS (SELECT 1 FROM work_for wf
WHERE wf.mentor_id = p.mentor_id AND wf.c_company_id = @CompanyId)
)
166         AND (@CategoryId IS NULL OR EXISTS (SELECT 1 FROM set_skill
ss JOIN own_skill os ON ss.skill_id = os.skill_id WHERE ss.
mentor_id = p.mentor_id AND os.category_id = @CategoryId))
167     GROUP BY YEAR(inv.paid_time), MONTH(inv.paid_time), DATENAME(
MONTH, inv.paid_time)
168     HAVING (@MinRevenue IS NULL OR SUM(inv.amount_total) >=
@MinRevenue)
169     AND (@MinBookingCount IS NULL OR COUNT(DISTINCT b.
plan_registrations_id) >= @MinBookingCount)
170     ORDER BY year DESC, month DESC;
171 END
172
173 -- =====
174 --GROUP BY CATEGORY
175 -- =====
176 ELSE IF @GroupBy = N'category'
177 BEGIN
178     -- Tính riêng doanh thu
179     WITH CategoryRevenue AS (
180         SELECT
181             os.category_id ,
182             SUM(inv.amount_total) AS total_revenue ,
```

```
183         COUNT(DISTINCT b.plan_registrations_id) AS
total_bookings
184     FROM set_skill ss
185     JOIN own_skill os ON ss.skill_id = os.skill_id
186     JOIN plans p ON ss.mentor_id = p.mentor_id
187     JOIN bookings b ON p.plan_id = b.plan_id
188     JOIN plan_registrations pr ON b.plan_registrations_id =
pr.registration_id
189     JOIN invoices inv ON pr.registration_id = inv.
plan_registrations_id
190     WHERE inv.paid_time BETWEEN @StartDate AND @EndDate AND
inv.payment_status = N'paid'
191     GROUP BY os.category_id
192 )
193 SELECT
194     cat.category_id,
195     cat.category_name,
196     COUNT(DISTINCT ss.mentor_id) AS total_mentors_with_skill,
197     ISNULL(cr.total_bookings, 0) AS total_bookings,
198     ISNULL(cr.total_revenue, 0) AS total_revenue
199 FROM categories cat
200 INNER JOIN own_skill os ON cat.category_id = os.category_id
201 INNER JOIN set_skill ss ON os.skill_id = ss.skill_id
202 LEFT JOIN CategoryRevenue cr ON cat.category_id = cr.
category_id
203 WHERE (@CategoryId IS NULL OR cat.category_id = @CategoryId)
204 GROUP BY cat.category_id, cat.category_name, cr.total_bookings
, cr.total_revenue
205 HAVING (@MinRevenue IS NULL OR ISNULL(cr.total_revenue, 0) >=
@MinRevenue)
206 ORDER BY total_revenue DESC;
207 END
208 ELSE
209 BEGIN
210     RAISERROR('Invalid GroupBy param. Supported: mentor, month,
category', 16, 1);
```

```
211     END
212 END;
213 GO
```

Nhóm câu lệnh thực thi thủ tục mẫu:

```
1  --1. Mac dinh (Group by Mentor, 12 thang gan nhat)--
2  EXEC dbo.sp_DashboardStatistics;
3
4  --2. Thong ke Mentor theo doanh thu cao nhat--
5  EXEC dbo.sp_DashboardStatistics
6      @GroupBy = N'mentor',
7      @SortBy = N'total_revenue',
8      @SortOrder = N'DESC';
9
10 --3. Thong ke theo danh muc (Web Development, UI/UX Design,...)--
11 EXEC dbo.sp_DashboardStatistics
12     @GroupBy = N'category',
13     @SortBy = N'total_revenue',
14     @SortOrder = N'DESC';
15
16 --4. Loc Mentor co doanh thu >= 50$ va co it nhat 1 booking--
17 EXEC dbo.sp_DashboardStatistics
18     @GroupBy = N'mentor',
19     @MinRevenue = 50.00,
20     @MinBookingCount = 1,
21     @SortBy = N'total_revenue',
22     @SortOrder = N'DESC';
23
24 --5. Sap xep theo gia tri hoa don trung binh--
25 EXEC dbo.sp_DashboardStatistics
26     @GroupBy = N'mentor',
27     @SortBy = N'average_invoice_amount',
28     @SortOrder = N'DESC';
29
30 --6. Truong hop loi - GroupBy khong hop le--
31 EXEC dbo.sp_DashboardStatistics @GroupBy = N'invalid_param';
```

Kết quả các màn hình hiển thị từ DBMS minh họa cho việc thực hiện thủ tục:

215

--1. Mac định (Group by Mentor, 12 tháng gần nhất)--

216

EXEC dbo.sp_DashboardStatistics;

118%

6

↑

↓

Ln: 216

Ch: 1

SPC

LF

Results

Messages

	mentor_id	mentor_name	mentor_email	mentor_avatar_url	mentor_country	mentor_status	mentor_headline	company_names	job_titles	skills
7	10	Maria Garcia	maria.garcia@example.com	https://i.pravatar.cc/1507/img=23	Argentina	Active	Digital Transformation Consultant	Oracle, Salesforce	Business Analyst	Communication, Syst
8	29	Erik Hansen	erik.hansen@example.com	https://i.pravatar.cc/1507/img=52	Denmark	Active	Senior SRE	Dropbox, Google	Site Reliability Engineer	CI/CD, Docker, Kuber
9	19	Alex Petrov	alex.petrov@example.com	https://i.pravatar.cc/1507/img=36	Russia	Active	Senior Game Developer	Samsung, Sony	Mobile Developer	Python, TypeScript
10	5	David Kim	david.kim@example.com	https://i.pravatar.cc/1507/img=15	South Korea	Active	Career Coach & Leadership Consultant	Apple, Stripe	Engineering Manager, VP of Engineering	Agile, Communication
11	12	Sophie Dubois	sophie.dubois@example.com	https://i.pravatar.cc/1507/img=24	France	Active	Creative Director & Brand Strategist	Adobe, Apple	Lead Product Designer, VP of Product	Adobe XD, Communi
12	27	Lucas Silva	lucas.silva@example.com	https://i.pravatar.cc/1507/img=48	Portugal	Active	Senior QA Engineer	Atlassian, Spotify	QA Engineer	Git, Project Managerr
13	9	Robert Thompson	robert.thompson@example.com	https://i.pravatar.cc/1507/img=17	Australia	Active	Senior Backend Engineer	Amazon, Netflix	Senior Backend Engineer	MongoDB, Node.js, F
14	7	James Martinez	james.martinez@example.com	https://i.pravatar.cc/1507/img=13	Mexico	Active	Senior Mobile Developer	Apple, Google	Mobile Developer	Flutter, Kotlin, React
15	28	Chloe Martin	chloe.martin@example.com	https://i.pravatar.cc/1507/img=49	Belgium	Active	E-commerce Strategy Lead	Amazon, Shopify	Product Manager	Google Analytics, SEC
16	24	Rachel Cohen	rachel.cohen@example.com	https://i.pravatar.cc/1507/img=44	Israel	Active	Senior Motion Designer	Adobe, Netflix	Lead Product Designer	Adobe XD, UI/UX Des
17	14	Laura Muller	laura.muller@example.com	https://i.pravatar.cc/1507/img=29	Germany	Active	Senior Product Manager	Google, Microsoft	Product Manager, Product Owner	Agile, Project Manag

Hình 37: Kết quả thực thi thủ tục mẫu 1

218

--2. Thong ke Mentor theo doanh thu cao nhat--

219

EXEC dbo.sp_DashboardStatistics

220

@GroupBy = N'mentor',

221

@SortBy = N'total_revenue',

222

@SortOrder = N'DESC';

223

118 %

6

0

↑

↓

Ln: 219

Ch: 1

SPC

Results

Messages

	mentor_id	mentor_name	mentor_email	mentor_avatar_url	mentor_country	mentor_status	mentor_headline	company_names	job_titles	skills
1	2	Sarah Johnson	sarah.johnson@example.com	https://i.pravatar.cc/1507/img=5	United Kingdom	Active	Lead Product Designer	Airbnb, Amazon	Lead Product Designer, Senior UX Designer	Adobe XD, Figma
2	1	John Doe	john.doe@example.com	https://i.pravatar.cc/1507/img=1	United States	Active	Senior Full-Stack Developer & Tech Lead	Google, Microsoft	Senior Software Engineer, Tech Lead	AWS, CI/CD, Git
3	11	Thomas Lee	thomas.lee@example.com	https://i.pravatar.cc/1507/img=18	Japan	Active	Principal Software Architect	Google, Meta	CTO, Senior Software Engineer	Git, Node.js, Python
4	23	Henrik Andersson	henrik.andersson@example.com	https://i.pravatar.cc/1507/img=43	Sweden	Active	Senior Blockchain Developer	PayPal, Spotify	Full Stack Developer	Blockchain, JavaScript
5	15	Daniel Santos	daniel.santos@example.com	https://i.pravatar.cc/1507/img=26	Brazil	Active	AI Research Engineer	Google, Meta	Chief Data Officer, ML Engineer	Machine Learning, TensorFlow
6	6	Lisa Anderson	lisa.anderson@example.com	https://i.pravatar.cc/1507/img=10	Canada	Active	Senior Cloud Architect	Amazon, Microsoft	Cloud Architect, Product Manager	AWS, Azure, Cloud
7	10	Maria Garcia	maria.garcia@example.com	https://i.pravatar.cc/1507/img=23	Argentina	Active	Digital Transformation Consultant	Oracle, Salesforce	Business Analyst	Communication, CRM
8	29	Erik Hansen	erik.hansen@example.com	https://i.pravatar.cc/1507/img=52	Denmark	Active	Senior SRE	Dropbox, Google	Site Reliability Engineer	CI/CD, Docker, Kubernetes
9	19	Alex Petrov	alex.petrov@example.com	https://i.pravatar.cc/1507/img=36	Russia	Active	Senior Game Developer	Samsung, Sony	Mobile Developer	Python, TypeScript, Unity
10	5	David Kim	david.kim@example.com	https://i.pravatar.cc/1507/img=15	South Korea	Active	Career Coach & Leadership Consultant	Apple, Stripe	Engineering Manager, VP of Engineering	Agile, Communication
11	12	Sophie Dubois	sophie.dubois@example.com	https://i.pravatar.cc/1507/img=24	France	Active	Creative Director & Brand Strategist	Adobe, Apple	Lead Product Designer, VP of Product	Adobe XD, Creative Suite
12	27	Lucas Silva	lucas.silva@example.com	https://i.pravatar.cc/1507/img=48	Portugal	Active	Senior QA Engineer	Atlassian, Spotify	QA Engineer	Git, Project Management
13	9	Robert Thompson	robert.thompson@example.com	https://i.pravatar.cc/1507/img=17	Australia	Active	Senior Backend Engineer	Amazon, Netflix	Senior Backend Engineer	MongoDB, NoSQL
14	7	James Martinez	james.martinez@example.com	https://i.pravatar.cc/1507/img=13	Mexico	Active	Senior Mobile Developer	Apple, Google	Mobile Developer	Flutter, Kotlin, Swift

Hình 38: Kết quả thực thi thủ tục mẫu 2

240 --5. Thông kê theo danh mục (Web Development, UI/UX Design,...)--

241 EXEC dbo.sp_DashboardStatistics

242 @GroupBy = N'category',

243 @SortBy = N'total_revenue',

244 @SortOrder = N'DESC';

245

246 --6. Thông kê Web Development category (category_id = 8)--

247 EXEC dbo.sp_DashboardStatistics

118 % 6 0

Results Messages

	category_id	category_name	total_mentors_with_skill	total_bookings	total_revenue
1	10	Cloud & DevOps	8	9	1356.50
2	16	Embedded Systems	5	6	1309.00
3	12	Backend Development	9	10	1162.00
4	8	Web Development	7	8	810.50
5	17	Blockchain Development	5	6	758.00
6	18	UI/UX Design	5	6	703.00
7	46	Team Management	9	8	574.75
8	22	Brand Design	9	8	570.00
9	44	Career Coaching	9	8	570.00
10	27	Product Development	6	6	554.50
11	36	Agile Methodologies	4	4	509.00
12	9	Mobile Development	5	5	503.50
13	24	Business Strategy	10	9	500.75
14	31	Natural Language Processing	7	7	497.50

Hình 39: Kết quả thực thi thủ tục mẫu 3

253 --7. Lọc Mentor có doanh thu >= 50\$ và có ít nhất 1 booking--

254 EXEC dbo.sp_DashboardStatistics

255 @GroupBy = N'mentor',

256 @MinRevenue = 50.00,

257 @MinBookingCount = 1,

258 @SortBy = N'total_revenue',

259 @SortOrder = N'DESC';

18 % 6 0 Ln: 254 Ch: 1 SPC LF

Results Messages

	mentor_id	mentor_name	mentor_email	mentor_avatar_url	mentor_country	mentor_status	mentor_headline	company_names	job_titles	skills
1	2	Sarah Johnson	sarah.johnson@example.com	https://pravatar.cc/150?img=5	United Kingdom	Active	Lead Product Designer	Airbnb, Amazon	Lead Product Designer, Senior UX Designer	Adobe XD, Figma, Protob
2	1	John Doe	john.doe@example.com	https://pravatar.cc/150?img=1	United States	Active	Senior Full-Stack Developer & Tech Lead	Google, Microsoft	Senior Software Engineer, Tech Lead	AWS, CI/CD, Git, JavaScri
3	11	Thomas Lee	thomas.lee@example.com	https://pravatar.cc/150?img=18	Japan	Active	Principal Software Architect	Google, Meta	CTO, Senior Software Engineer	Git, Node.js, Python, Syst
4	23	Henrik Andersson	henrik.andersson@example.com	https://pravatar.cc/150?img=43	Sweden	Active	Senior Blockchain Developer	PayPal, Spotify	Full Stack Developer	Blockchain, Node.js, Pyth
5	15	Daniel Santos	daniel.santos@example.com	https://pravatar.cc/150?img=26	Brazil	Active	AI Research Engineer	Google, Meta	Chief Data Officer, ML Engineer	Machine Learning, Pytho
6	6	Lisa Anderson	lisa.anderson@example.com	https://pravatar.cc/150?img=10	Canada	Active	Senior Cloud Architect	Amazon, Microsoft	Cloud Architect, Product Manager	AWS, Azure, CI/CD, Docke
7	10	Maria Garcia	maria.garcia@example.com	https://pravatar.cc/150?img=23	Argentina	Active	Digital Transformation Consultant	Oracle, Salesforce	Business Analyst	Communication, System
8	29	Erik Hansen	erik.hansen@example.com	https://pravatar.cc/150?img=52	Denmark	Active	Senior SRE	Dropbox, Google	Site Reliability Engineer	CI/CD, Docker, Kubermete
9	19	Alex Petrov	alex.petrov@example.com	https://pravatar.cc/150?img=36	Russia	Active	Senior Game Developer	Samsung, Sony	Mobile Developer	Python, TypeScript
10	5	David Kim	david.kim@example.com	https://pravatar.cc/150?img=15	South Korea	Active	Career Coach & Leadership Consultant	Apple, Stripe	Engineering Manager, VP of Engineering	Agile, Communication, Le
11	12	Sophie Dubois	sophie.dubois@example.com	https://pravatar.cc/150?img=24	France	Active	Creative Director & Brand Strategist	Adobe, Apple	Lead Product Designer, VP of Product	Adobe XD, Communicati
12	27	Lucas Silva	lucas.silva@example.com	https://pravatar.cc/150?img=48	Portugal	Active	Senior QA Engineer	Atlassian, Spotify	QA Engineer	Git, Project Management
13	9	Robert Thompson	robert.thompson@example.com	https://pravatar.cc/150?img=17	Australia	Active	Senior Backend Engineer	Amazon, Netflix	Senior Backend Engineer	MongoDB, Node.js, Post
14	7	James Martinez	james.martinez@example.com	https://pravatar.cc/150?img=13	Mexico	Active	Senior Mobile Developer	Apple, Google	Mobile Developer	Flutter, Kotlin, React Nati
15	28	Chloe Martin	chloe.martin@example.com	https://pravatar.cc/150?img=49	Belgium	Active	E-commerce Strategy Lead	Amazon, Shopify	Product Manager	Google Analytics, SEO, T
16	24	Rachel Cohen	rachel.cohen@example.com	https://pravatar.cc/150?img=44	Israel	Active	Senior Motion Designer	Adobe, Netflix	Lead Product Designer	Adobe XD, UI/UX Design

Hình 40: Kết quả thực thi thủ tục mẫu 4

296

--14. Sắp xếp theo giá trị hoa đơn trung bình--

297

EXEC dbo.sp_DashboardStatistics

298

@GroupBy = N'mentor',

299

@SortBy = N'average_invoice_amount',

300

@SortOrder = N'DESC';

301

118 %

6

0

↑

↓

Ln: 297

Ch: 1

SPC

Results

Messages

	mentor_id	mentor_name	mentor_email	mentor_avatar_url	mentor_country	mentor_status	mentor_headline	company_names	job_titles	skills
1	2	Sarah Johnson	sarah.johnson@example.com	https://i.pravatar.cc/150?img=5	United Kingdom	Active	Lead Product Designer	Airbnb, Amazon	Lead Product Designer, Senior UX Designer	Adobe XD
2	11	Thomas Lee	thomas.lee@example.com	https://i.pravatar.cc/150?img=18	Japan	Active	Principal Software Architect	Google, Meta	CTO, Senior Software Engineer	Git, Node
3	23	Henrik Andersson	henrik.andersson@example.com	https://i.pravatar.cc/150?img=43	Sweden	Active	Senior Blockchain Developer	PayPal, Spotify	Full Stack Developer	Blockchain
4	15	Daniel Santos	daniel.santos@example.com	https://i.pravatar.cc/150?img=26	Brazil	Active	AI Research Engineer	Google, Meta	Chief Data Officer, ML Engineer	Machine Learning
5	6	Lisa Anderson	lisa.anderson@example.com	https://i.pravatar.cc/150?img=10	Canada	Active	Senior Cloud Architect	Amazon, Microsoft	Cloud Architect, Product Manager	AWS, Azure
6	10	Maria Garcia	maria.garcia@example.com	https://i.pravatar.cc/150?img=23	Argentina	Active	Digital Transformation Consultant	Oracle, Salesforce	Business Analyst	Communication
7	29	Erik Hansen	erik.hansen@example.com	https://i.pravatar.cc/150?img=52	Denmark	Active	Senior SRE	Dropbox, Google	Site Reliability Engineer	CI/CD, Docker
8	19	Alex Petrov	alex.petrov@example.com	https://i.pravatar.cc/150?img=36	Russia	Active	Senior Game Developer	Samsung, Sony	Mobile Developer	Python, TypeScript
9	5	David Kim	david.kim@example.com	https://i.pravatar.cc/150?img=15	South Korea	Active	Career Coach & Leadership Consultant	Apple, Stripe	Engineering Manager, VP of Engineering	Agile, Coaching
10	12	Sophie Dubois	sophie.dubois@example.com	https://i.pravatar.cc/150?img=24	France	Active	Creative Director & Brand Strategist	Adobe, Apple	Lead Product Designer, VP of Product	Adobe XD, Branding
11	1	John Doe	john.doe@example.com	https://i.pravatar.cc/150?img=1	United States	Active	Senior Full-Stack Developer & Tech Lead	Google, Microsoft	Senior Software Engineer, Tech Lead	AWS, CI/CD
12	27	Lucas Silva	lucas.silva@example.com	https://i.pravatar.cc/150?img=48	Portugal	Active	Senior QA Engineer	Atlassian, Spotify	QA Engineer	Git, Project Management
13	9	Robert Thompson	robert.thompson@example.com	https://i.pravatar.cc/150?img=17	Australia	Active	Senior Backend Engineer	Amazon, Netflix	Senior Backend Engineer	MongoDB, Python
14	7	James Martinez	james.martinez@example.com	https://i.pravatar.cc/150?img=13	Mexico	Active	Senior Mobile Developer	Apple, Google	Mobile Developer	Flutter, Kotlin

Hình 41: Kết quả thực thi thủ tục mẫu 5

294 --12. Trường hợp lỗi - GroupBy không hợp lệ--
295 EXEC dbo.sp_DashboardStatistics @GroupBy = N'invalid_param';
296

119 % No issues found

Messages

Msg 50000, Level 16, State 1, Procedure dbo.sp_DashboardStatistics, Line 214 [Batch Start Line 294]
Invalid GroupBy param. Supported: mentor, month, category

Completion time: 2025-12-07T14:06:35.3559863+07:00

Hình 42: Kết quả thực thi thủ tục mẫu 6

2.4 Hàm

2.4.1 Hàm tìm mã giảm giá tốt nhất cho Mentee

Mô tả hàm: Hàm FindBestDiscountForMentee được sử dụng để tìm mã giảm giá có lợi nhất cho Mentee khi mua plan. Hàm so sánh tất cả các mã giảm giá đang active và trả về mã có mức tiết kiệm cao nhất dựa trên giá plan.

Usecase:

- Hiển thị gợi ý mã giảm giá tốt nhất khi Mentee checkout.

Các thao tác chính:

1. Validate tham số đầu vào và kiểm tra sự tồn tại của Mentee, Plan.
2. Lấy giá plan từ database.
3. Duyệt qua tất cả discount codes đang active bằng con trỏ.

4. Tính toán mức tiết kiệm cho từng loại discount (Percentage/Fixed).

5. So sánh và lưu discount tốt nhất.

Input:

- @mentee_id (INT): Mã định danh Mentee.
- @plan_id (INT): Mã định danh Plan cần mua.

Output:

- NVARCHAR(100): Tên mã giảm giá tốt nhất.
- Trả về NULL nếu không có mã giảm giá phù hợp hoặc tham số không hợp lệ.

Câu lệnh tạo hàm:

```
1 IF OBJECT_ID('dbo.FindBestDiscountForMentee', 'FN') IS NOT NULL
2     DROP FUNCTION dbo.FindBestDiscountForMentee;
3 GO
4
5 CREATE FUNCTION dbo.FindBestDiscountForMentee
6 (
7     @mentee_id INT,
8     @plan_id INT
9 )
10 RETURNS NVARCHAR(100)
11 AS
12 BEGIN
13     DECLARE @best_discount_name NVARCHAR(100) = NULL;
14     DECLARE @best_savings DECIMAL(10,2) = 0;
15     DECLARE @plan_charge DECIMAL(10,2);
16
17     DECLARE @current_name NVARCHAR(100);
18     DECLARE @current_type NVARCHAR(50);
19     DECLARE @current_value DECIMAL(10,2);
20     DECLARE @current_savings DECIMAL(10,2);
21
22     -- Validate tham so dau vao
23     IF @mentee_id IS NULL OR @mentee_id <= 0
```

```
24      RETURN NULL;
25
26  IF @plan_id IS NULL OR @plan_id <= 0
27      RETURN NULL;
28
29  -- Check mentee co ton tai khong
30  IF NOT EXISTS (SELECT 1 FROM dbo.users
31                  WHERE user_id = @mentee_id AND role = N'Mentee')
32      RETURN NULL;
33
34  -- Lay plan_charge tu plan_id
35  SELECT @plan_charge = plan_charge
36  FROM dbo.plans
37  WHERE plan_id = @plan_id;
38
39  -- Check plan co ton tai khong
40  IF @plan_charge IS NULL
41      RETURN NULL;
42
43  -- Cursor de duyet qua active discount code
44  DECLARE discount_cursor CURSOR FOR
45  SELECT
46      d.discount_name ,
47      d.discount_type ,
48      d.discount_value
49  FROM dbo.discounts d
50  WHERE d.status = N'Active';
51      AND GETDATE() BETWEEN d.start_date AND d.end_date;
52  OPEN discount_cursor;
53  FETCH NEXT FROM discount_cursor INTO
54      @current_name , @current_type , @current_value;
55
56  WHILE @@FETCH_STATUS = 0
57  BEGIN
58      SET @current_savings = 0;
59
```

```
60      -- Tinh toan muc tiet kiem
61      IF @current_type = N'Percentage'
62      BEGIN
63          SET @current_savings = (@plan_charge * @current_value /
100);
64      END
65      ELSE IF @current_type = N'Fixed'
66      BEGIN
67          SET @current_savings = @current_value;
68
69          -- Saving khong the vuot qua plan_charge
70          IF @current_savings > @plan_charge
71              SET @current_savings = @plan_charge;
72      END
73
74      -- So sanh best saving voi saving hien tai
75      IF @current_savings > @best_savings
76      BEGIN
77          SET @best_savings = @current_savings;
78          SET @best_discount_name = @current_name;
79      END
80
81      FETCH NEXT FROM discount_cursor INTO
82          @current_name, @current_type, @current_value;
83      END
84
85      CLOSE discount_cursor;
86      DEALLOCATE discount_cursor;
87
88      RETURN @best_discount_name;
89 END;
90 GO
```

Nhóm câu lệnh thực thi hàm mẫu:

```
1 -- Test 1.1: valid input
2 SELECT
```

```
3      31 AS mentee_id,
4      'Alice Smith' AS mentee_name,
5      p.plan_id,
6      p.plan_charge,
7      dbo.FindBestDiscountForMentee(31, p.plan_id) AS best_discount
8 FROM dbo.plans p
9 WHERE p.plan_id IN (6, 7, 8, 9, 10, 11, 14, 18)
10 ORDER BY p.plan_charge;
11
12 -- Test 1.2 invalid inputs
13 SELECT
14     'NULL mentee_id' AS test_case,
15     dbo.FindBestDiscountForMentee(NULL, 1) AS result;
16
17 SELECT
18     'Negative mentee_id' AS test_case,
19     dbo.FindBestDiscountForMentee(-1, 1);
20
21 SELECT
22     'Zero mentee_id' AS test_case,
23     dbo.FindBestDiscountForMentee(0, 1);
24
25 SELECT
26     'NULL plan_id' AS test_case,
27     dbo.FindBestDiscountForMentee(31, NULL);
28
29 SELECT
30     'Negative plan_id' AS test_case,
31     dbo.FindBestDiscountForMentee(31, -1);
32
33 SELECT
34     'Non-existent mentee' AS test_case,
35     dbo.FindBestDiscountForMentee(9999, 1);
36
37 SELECT
38     'Non-existent plan' AS test_case,
```

```

39     dbo.FindBestDiscountForMentee(31, 9999);
40
41 SELECT
42     'Mentor user_id (not mentee)' AS test_case,
43     dbo.FindBestDiscountForMentee(1, 1);

```

Kết quả màn hình minh họa:

	mentee_id	mentee_name	plan_id	plan_charge	best_discount
1	31	Alice Smith	11	45.00	FIRST50
2	31	Alice Smith	14	70.00	FIRST50
3	31	Alice Smith	7	75.00	FIRST50
4	31	Alice Smith	18	85.00	FIRST50
5	31	Alice Smith	8	130.00	FIRST50
6	31	Alice Smith	6	349.00	MENTOR25
7	31	Alice Smith	9	399.00	MENTOR25
8	31	Alice Smith	10	699.00	MENTOR25

Hình 43: Kết quả tìm mã giảm giá khi input hợp lệ

	test_case	result
1	NULL mentee_id	NULL
	test_case	(No column name)
1	Negative mentee_id	NULL
	test_case	(No column name)
1	Zero mentee_id	NULL
	test_case	(No column name)
1	NULL plan_id	NULL
	test_case	(No column name)
1	Negative plan_id	NULL
	test_case	(No column name)
1	Non-existent mentee	NULL
	test_case	(No column name)
1	Non-existent plan	NULL
	test_case	(No column name)
1	Mentor user_id (not mentee)	NULL

Hình 44: Kết quả tìm mã giảm giá khi input không hợp lệ

2.4.2 Hàm tính thời gian phản hồi trung bình của Mentor

Mô tả hàm: Hàm CalculateMentorResponseTime được sử dụng để tính thời gian phản hồi trung bình của Mentor đối với tin nhắn từ Mentee. Hàm phân tích lịch sử tin nhắn và trả về mô tả thời gian phản hồi theo các mức độ khác nhau.

Usecase:

- Hiển thị badge thời gian phản hồi trên profile Mentor.

- Giúp Mentee lựa chọn Mentor dựa trên tốc độ phản hồi.

Các thao tác chính:

1. Validate tham số đầu vào và kiểm tra sự tồn tại của Mentor.
2. Duyệt qua các tin nhắn từ Mentee gửi cho Mentor bằng CURSOR.
3. Tìm tin nhắn phản hồi đầu tiên của Mentor cho mỗi tin nhắn Mentee.
4. Tính khoảng cách thời gian (gap).
5. Tính thời gian phản hồi trung bình và phân loại theo 7 mức độ.

Input:

- @mentor_id (INT): Mã định danh Mentor cần tính response time.

Output:

- NVARCHAR(50): Mô tả thời gian phản hồi.
- Trả về 'Invalid ID' nếu mentor_id không hợp lệ.
- Trả về 'Mentor not found' nếu Mentor không tồn tại.
- Trả về 'No responses yet' nếu Mentor chưa phản hồi tin nhắn nào.

Câu lệnh tạo hàm:

```
1 IF OBJECT_ID('dbo.CalculateMentorResponseTime', 'FN') IS NOT NULL
2     DROP FUNCTION dbo.CalculateMentorResponseTime;
3 GO
4
5 CREATE FUNCTION dbo.CalculateMentorResponseTime
6 (
7     @mentor_id INT
8 )
9 RETURNS NVARCHAR(50)
10 AS
11 BEGIN
12     DECLARE @avg_hours DECIMAL(10,2) = 0;
13     DECLARE @total_gap_hours DECIMAL(10,2) = 0;
```

```
14 DECLARE @response_count INT = 0;
15 DECLARE @result NVARCHAR(50);
16
17 DECLARE @mentee_message_id INT;
18 DECLARE @mentee_sent_time DATETIME;
19 DECLARE @mentor_reply_time DATETIME;
20 DECLARE @gap_hours DECIMAL(10,2);
21
22 -- Validate tham so dau vao
23 IF @mentor_id IS NULL OR @mentor_id <= 0
24     RETURN N'Invalid ID';
25
26 -- Kiem tra mentor ton tai
27 IF NOT EXISTS (SELECT 1 FROM dbo.mentors WHERE user_id =
28 @mentor_id)
29     RETURN N'Mentor not found';
30
31 -- Cursor duyet tin nhan tu mentee
32 DECLARE message_cursor CURSOR FOR
33 SELECT
34     m1.message_id,
35     m1.sent_time
36 FROM dbo.messages m1
37 WHERE m1.receiver_id = @mentor_id
38     AND m1.sender_id IN (SELECT user_id FROM dbo.users
39     WHERE role = N'Mentee')
40 ORDER BY m1.sent_time;
41
42 OPEN message_cursor;
43 FETCH NEXT FROM message_cursor INTO
44     @mentee_message_id, @mentee_sent_time;
45
46 WHILE @@FETCH_STATUS = 0
47 BEGIN
48     -- Tim tin nhan phan hoi dau tien
49     SELECT TOP 1 @mentor_reply_time = m2.sent_time
```

```
49      FROM dbo.messages m2
50      WHERE m2.sender_id = @mentor_id
51             AND m2.receiver_id = (SELECT sender_id FROM dbo.messages
52                                     WHERE message_id =
@mentee_message_id)
53             AND m2.sent_time > @mentee_sent_time
54      ORDER BY m2.sent_time ASC;
55
56      -- Tinh gap neu co reply
57      IF @mentor_reply_time IS NOT NULL
58      BEGIN
59          SET @gap_hours = DATEDIFF(MINUTE, @mentee_sent_time,
60                                     @mentor_reply_time) / 60.0;
61
62          -- Chi tinh phan hoi trong 7 ngay
63          IF @gap_hours <= 168
64          BEGIN
65              SET @total_gap_hours = @total_gap_hours + @gap_hours;
66              SET @response_count = @response_count + 1;
67          END
68
69          SET @mentor_reply_time = NULL;
70      END
71
72      FETCH NEXT FROM message_cursor INTO
73          @mentee_message_id, @mentee_sent_time;
74  END
75
76  CLOSE message_cursor;
77  DEALLOCATE message_cursor;
78
79  -- Tinh trung binh
80  IF @response_count = 0
81      RETURN N'No responses yet';
82
83  SET @avg_hours = @total_gap_hours / @response_count;
```



```
84
85  -- Phan loai response time
86  IF @avg_hours < 1
87      SET @result = N'Within 1 hour';
88  ELSE IF @avg_hours < 3
89      SET @result = N'Within 3 hours';
90  ELSE IF @avg_hours < 6
91      SET @result = N'Within 6 hours';
92  ELSE IF @avg_hours < 12
93      SET @result = N'Within 12 hours';
94  ELSE IF @avg_hours < 24
95      SET @result = N'Within 24 hours';
96  ELSE IF @avg_hours < 36
97      SET @result = N'Within 36 hours';
98  ELSE IF @avg_hours < 48
99      SET @result = N'Within 48 hours';
100 ELSE
101     SET @result = N'More than 48 hours';
102 RETURN @result;
103 END;
104 GO
```

Nhóm câu lệnh thực thi hàm mẫu:

```
1  -- Scenario 1: Mentor ID 1 (John Doe) - Response nhanh (45 phút)
2  INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
3  VALUES
4  (N'John, I need help with async/await in JavaScript', '2024-12-08
   14:00:00', 31, 1),
5  (N'Let me explain async/await with a practical example', '2024-12-08
   14:45:00', 1, 31);
6  PRINT 'Scenario 1: Added - John Doe (45 min response)';
7
8  -- Scenario 2: Mentor ID 2 (Sarah Johnson) - Response trung bình (2 gi
   ờ)
9  INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
10 VALUES
```

```
11 (N'Sarah, can you review my new design mockups?', '2024-12-08 09:00:00', 32, 2),
12 (N'Absolutely! Your mockups look great, here's my feedback', '2024-12-08 11:00:00', 2, 32);
13 PRINT 'Scenario 2: Added - Sarah Johnson (2 hours response)';
14
15 -- Scenario 3: Mentor ID 3 (Michael Chen) - Response chậm (8 giờ)
16 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
17 VALUES
18 (N'Michael, question about neural network architecture', '2024-12-07 09:00:00', 33, 3),
19 (N'Great question! Let's discuss the architecture patterns', '2024-12-07 17:00:00', 3, 33);
20 PRINT 'Scenario 3: Added - Michael Chen (8 hours response)';
21
22 -- Scenario 4: Mentor ID 4 (Emily Rodriguez) - Response rất chậm (24 giờ)
23 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
24 VALUES
25 (N'Emily, need advice on email marketing campaign', '2024-12-06 10:00:00', 34, 4),
26 (N'I've reviewed your campaign, here are my suggestions', '2024-12-07 10:00:00', 4, 34);
27 PRINT 'Scenario 4: Added - Emily Rodriguez (24 hours response)';
28
29 -- Scenario 5: Mentor ID 5 (David Kim) - Response 30 giờ
30 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
31 VALUES
32 (N'David, can we discuss my career transition plan?', '2024-12-05 09:00:00', 35, 5),
33 (N'Let's set up a session to discuss your career path', '2024-12-06 15:00:00', 5, 35);
34 PRINT 'Scenario 5: Added - David Kim (30 hours response)';
35
36 -- Scenario 6: Mentor ID 6 (Lisa Anderson) - Không có response
37 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
```

```
38 VALUES
39 (N'Lisa, I'm interested in learning AWS architecture', '2024-12-08
    16:00:00', 31, 6);
40 PRINT 'Scenario 6: Added - Lisa Anderson (no response)';
41
42 -- Scenario 7: Mentor ID 7 (James Martinez) - Response quá chậm (>7 ng
    ày - không tính)
43 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
44 VALUES
45 (N'James, need help with mobile app development', '2024-11-25 10:00:00
    ', 32, 7),
46 (N'Sorry for the delay, I was on vacation', '2024-12-03 10:00:00', 7,
    32);
47 PRINT 'Scenario 7: Added - James Martinez (>7 days response)';
48
49 -- Scenario 8: Mentor ID 8 (Anna Kowalski) - Multiple responses (2h,
    30min, 5h)
50 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
51 VALUES
52 -- Response 1: 2 hours
53 (N'Anna, feedback on my UI design please?', '2024-12-07 10:00:00', 33,
    8),
54 (N'Your UI design is clean, here's what I think', '2024-12-07
    12:00:00', 8, 33),
55 -- Response 2: 30 min
56 (N'Thanks! One more question about color scheme', '2024-12-07 15:00:00
    ', 33, 8),
57 (N'For color scheme, try using this palette', '2024-12-07 15:30:00',
    8, 33),
58 -- Response 3: 5 hours
59 (N'Perfect! Can you review the final version?', '2024-12-08 08:00:00',
    33, 8),
60 (N'Final version looks excellent!', '2024-12-08 13:00:00', 8, 33);
61 PRINT 'Scenario 8: Added - Anna Kowalski (multiple responses)';
62
63 -- Scenario 9: Mentor ID 10 (Maria Garcia) - Response 4 giờ
```

```
64 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
65 VALUES
66 (N'Maria, advice on digital transformation strategy?', '2024-12-08
    10:00:00', 36, 10),
67 (N'Let's discuss your digital transformation roadmap', '2024-12-08
    14:00:00', 10, 36);
68 PRINT 'Scenario 9: Added - Maria Garcia (4 hours response)';
69
70 -- Scenario 10: Mentor ID 1 (John Doe) - Mentee gửi nhiều tin, Mentor
    reply nhiều tin
71 INSERT INTO dbo.messages (content, sent_time, sender_id, receiver_id)
72 VALUES
73 -- Mentee gửi 3 tin nhắn
74 (N'Question 1: How to use React hooks?', '2024-12-08 09:00:00', 31, 1)
    ,
75 (N'Question 2: What about custom hooks?', '2024-12-08 10:00:00', 31,
    1),
76 (N'Question 3: Performance optimization?', '2024-12-08 12:00:00', 31,
    1),
77 -- Mentor reply nhiều lần cho Q1 (2 tin)
78 (N'A1 part 1: useState explanation', '2024-12-08 09:30:00', 1, 31),
79 (N'A1 part 2: useEffect example', '2024-12-08 09:45:00', 1, 31),
80 -- Mentor reply nhiều lần cho Q2 (2 tin)
81 (N'A2 part 1: Custom hooks intro', '2024-12-08 11:00:00', 1, 31),
82 (N'A2 part 2: Complete example', '2024-12-08 11:15:00', 1, 31),
83 -- Mentor reply nhiều lần cho Q3 (3 tin)
84 (N'A3 part 1: Use React.memo', '2024-12-08 13:00:00', 1, 31),
85 (N'A3 part 2: useMemo/useCallback', '2024-12-08 13:10:00', 1, 31),
86 (N'A3 part 3: Avoid re-renders', '2024-12-08 13:20:00', 1, 31);
87 PRINT 'Scenario 10: Added - John Doe (multiple Q&A)';
88 GO
89 -- Test 2.1: valid mentor và có response
90 SELECT
91     m.user_id AS mentor_id,
92     u.first_name + ' ' + u.last_name AS mentor_name,
93     dbo.CalculateMentorResponseTime(m.user_id) AS response_time
```

```

94 FROM dbo.mentors m
95 JOIN dbo.users u ON m.user_id = u.user_id
96 WHERE m.user_id IN (1, 2, 3, 4, 5, 8, 10)
97 ORDER BY m.user_id;
98
99 -- Test 2.2: Mentor không có response nào
100 SELECT
101     6 AS mentor_id,
102     'Lisa Anderson' AS mentor_name,
103     dbo.CalculateMentorResponseTime(6) AS response_time
104
105 -- Test 2.3: Invalid input
106 SELECT
107     'NULL' AS test_case,
108     dbo.CalculateMentorResponseTime(NULL) AS result;
109
110 SELECT
111     'Negative ID' AS test_case,
112     dbo.CalculateMentorResponseTime(-1) AS result;
113
114 SELECT
115     'id = 0' AS test_case,
116     dbo.CalculateMentorResponseTime(0) AS result;
117
118 SELECT
119     'mentor không tồn tại' AS test_case,
120     dbo.CalculateMentorResponseTime(9999) AS result;

```

Kết quả màn hình minh họa:

	mentor_id	mentor_name	response_time
1	1	John Doe	Within 1 hour
2	2	Sarah Johnson	Within 3 hours
3	3	Michael Chen	Within 6 hours
4	4	Emily Rodriguez	Within 24 hours
5	5	David Kim	Within 24 hours
6	8	Anna Kowalski	Within 3 hours
7	10	Maria Garcia	Within 6 hours

Hình 45: Kết quả tính thời gian response khi mentor tồn tại và có response

	mentor_id	mentor_name	response_time
1	6	Lisa Anderson	No responses yet

Hình 46: Kết quả tính thời gian response khi mentor tồn tại không có response nào

	test_case	result
1	NULL	Invalid ID

	test_case	result
1	Negative ID	Invalid ID

	test_case	result
1	id = 0	Invalid ID

	test_case	result
1	mentor không tồn tại	Mentor not found

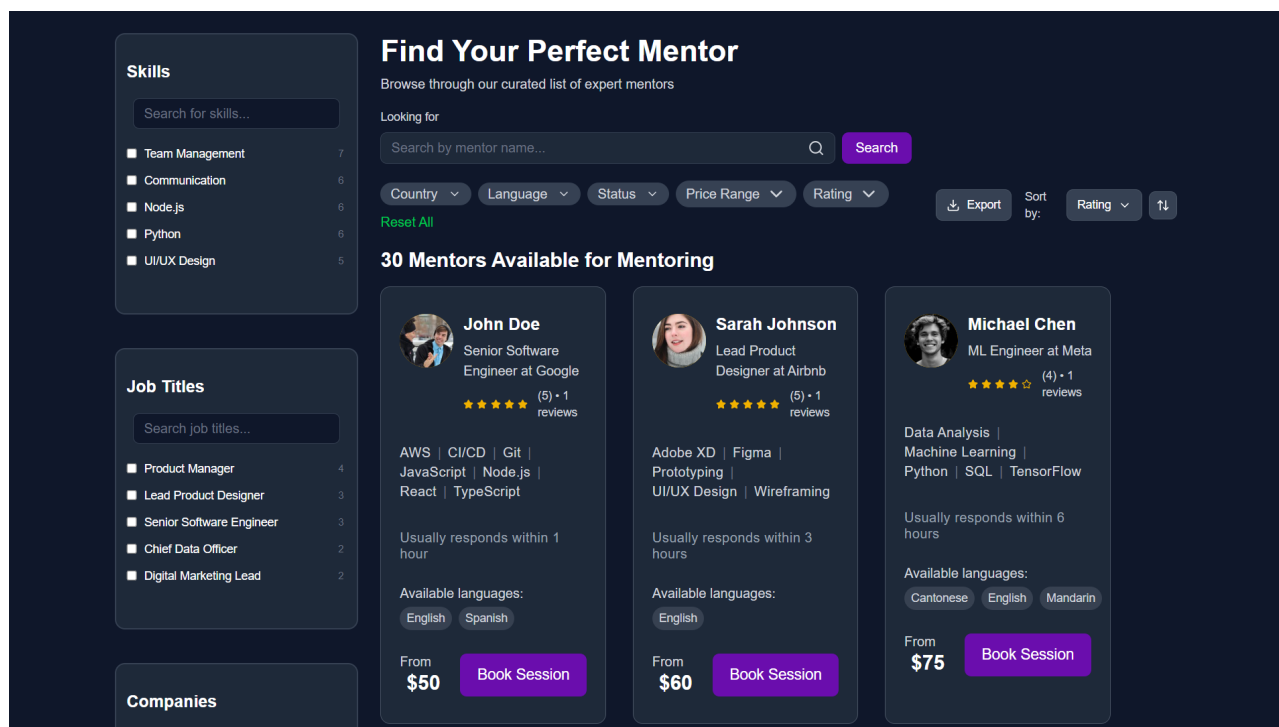
Hình 47: Kết quả khi input đầu và không hợp lệ

3 Hiện thực ứng dụng

Hệ thống được xây dựng theo kiến trúc Client–Server gồm ba lớp:

- Database Layer (SQL Server): Chịu trách nhiệm lưu trữ dữ liệu và đảm bảo tính toàn vẹn thông qua các Constraints, Triggers, đồng thời cung cấp một phần logic nghiệp vụ thông qua các Stored Procedures.
- Backend Layer (Node.js + Express): Đảm nhiệm việc xây dựng RESTful API, sử dụng thư viện mssql để kết nối và tương tác với cơ sở dữ liệu. Lớp này xử lý các nghiệp vụ trung gian như xác thực (Authentication) và kiểm tra dữ liệu đầu vào (Validation).
- Frontend Layer (Next.js): Giao diện người dùng được phát triển bằng React, giao tiếp với Backend thông qua các HTTP Requests (Axios hoặc Fetch) nhằm hiển thị và xử lý dữ liệu một cách trực quan.

3.1 Giao diện tìm kiếm Mentor



Hình 48: Giao diện tìm kiếm Mentor

Màn hình này cho phép người dùng tìm kiếm, lọc và xem danh sách mentor được lưu trữ trong cơ sở dữ liệu. Các chức năng chính bao gồm:

- **Thanh tìm kiếm Mentor:** Cho phép người dùng nhập tên mentor để tìm nhanh.
- **Bộ lọc nâng cao:**
 - Lọc theo quốc gia (*Country*).
 - Lọc theo ngôn ngữ (*Language*).
 - Lọc theo ngôn ngữ (*Status*).
 - Lọc theo mức giá (*Price Range*).
 - Lọc theo đánh giá (*Rating*).

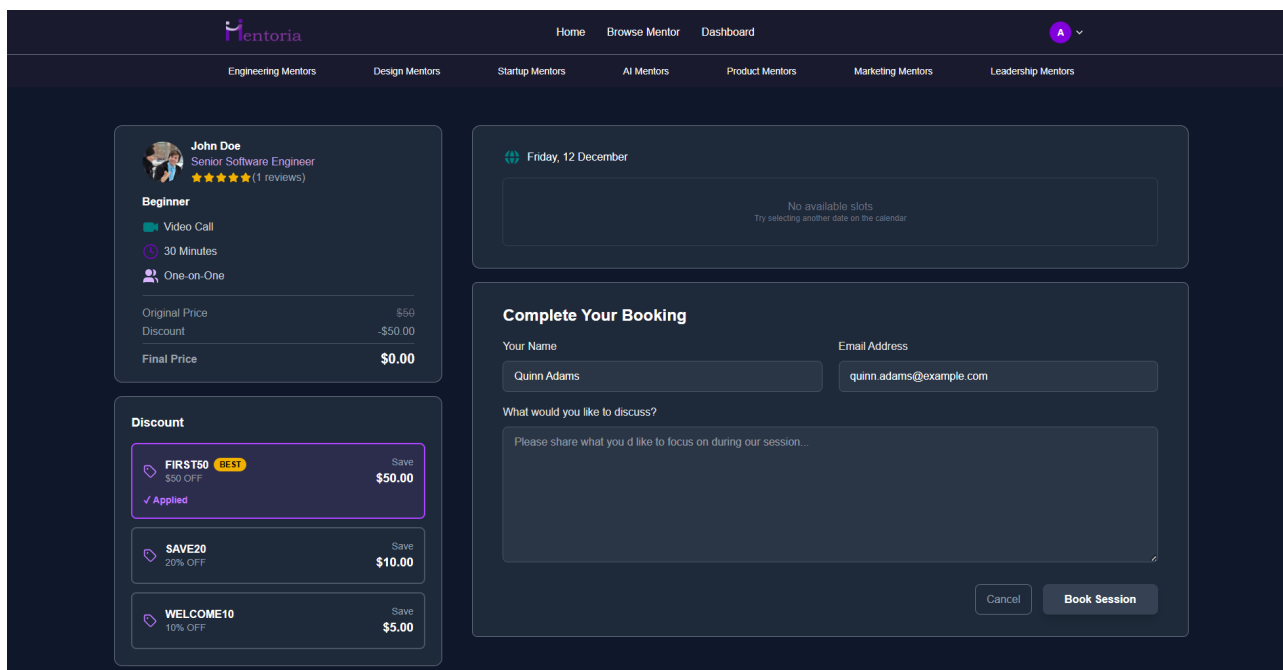
Kèm theo nút **Reset All** để xóa toàn bộ bộ lọc.

- **Bảng lọc bên trái màn hình:**
 - **Skills:** Lọc mentor theo kỹ năng như Python, Node.js, Team Management,...
 - **Job Titles:** Lọc theo chức danh như Product Manager, Lead Designer,...
 - **Companies:** Lọc theo công ty như Google, Amazon,...

Mỗi mục hiển thị số lượng mentor tương ứng (dựa trên truy vấn COUNT từ CSDL).

- **Chức năng sắp xếp (Sort By):** Cho phép sắp xếp danh sách mentor theo: Rating, Reviews, Price và Name.
- **Chức năng xuất dữ liệu (Export):** Hỗ trợ xuất danh sách thông tin mentor ra tệp CSV, giúp thuận tiện cho việc lưu trữ, chia sẻ hoặc xử lý dữ liệu.

3.2 Giao diện áp dụng mã giảm giá

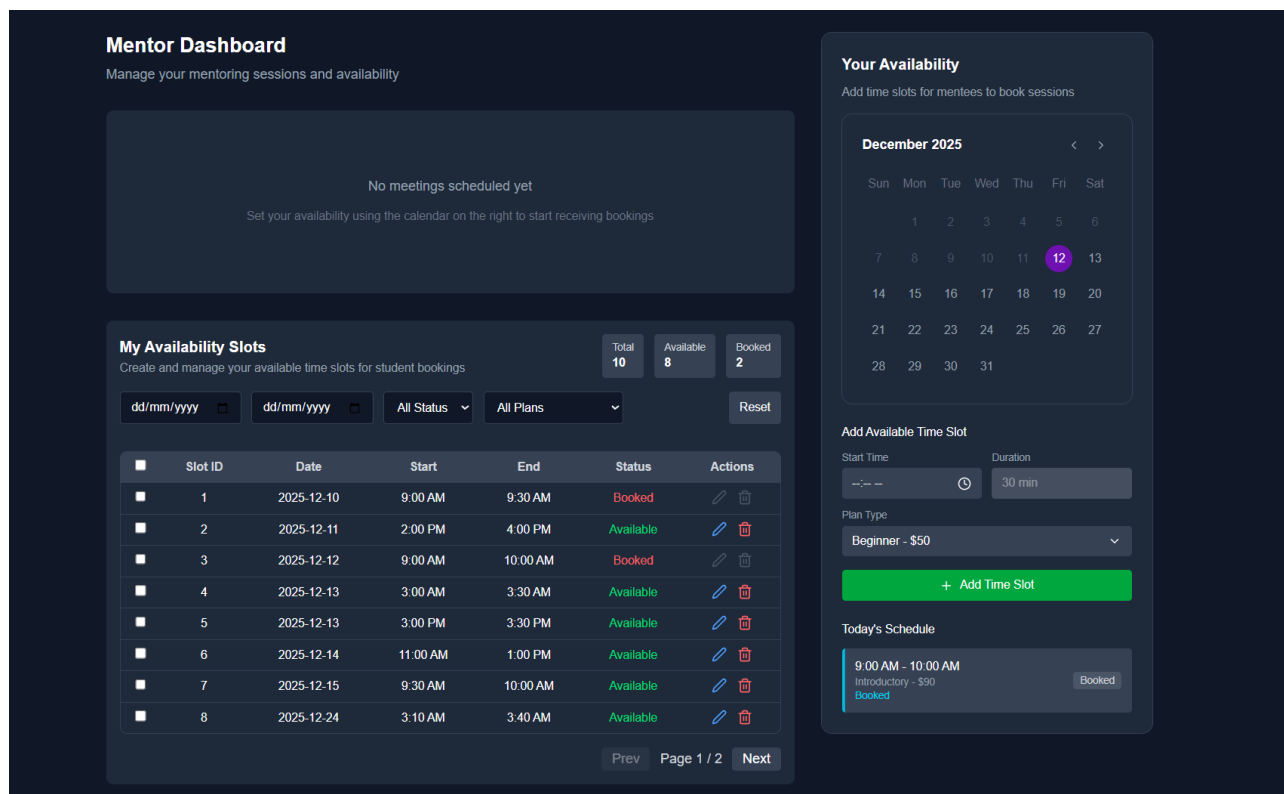


Hình 49: Giao diện áp dụng mã giảm giá

Màn hình này cho phép người dùng hoàn tất việc đặt lịch hẹn với mentor và áp dụng mã giảm giá cho buổi mentoring. Dữ liệu về mentor, phiên làm việc và mã giảm giá được lấy từ cơ sở dữ liệu.

- Danh sách các mã giảm giá (ví dụ: FIRST50, SAVE20, WELCOME10) được tải từ bảng Discounts trong CSDL.
- Mỗi mã hiển thị:
 - Tên mã giảm giá.
 - Loại giảm giá: theo số tiền cố định (ví dụ -\$50.00) hoặc theo phần trăm (10%, 20%...).
 - Số tiền tiết kiệm tương ứng (*Save \$50.00, Save \$10.00...*).
 - Trạng thái: mã đang được áp dụng (*Applied*), gợi ý mã tốt nhất (*BEST*)...

3.3 Giao diện quản lý slot cho Mentor



Hình 50: Giao diện quản lý slot cho Mentor

Màn hình này hỗ trợ mentor quản lý toàn bộ thời gian rảnh để nhận booking từ học viên. Toàn bộ dữ liệu về slot, trạng thái, kế hoạch (plan type) và lịch đã được đặt đều được truy xuất và cập nhật trực tiếp trong CSDL.

- **Khu vực thông báo lịch hẹn:**

- Hiện thị danh sách các buổi meeting được đặt trong ngày.
- Nếu chưa có buổi nào, màn hình hiện thị thông báo “No meetings scheduled yet”.

- **Bảng quản lý các khung giờ khả dụng (My Availability Slots):**

- Danh sách các slot mentor đã tạo, bao gồm:
 - * **Slot ID:** mã slot do hệ thống tạo.
 - * **Date:** ngày mentor có thể nhận booking.

* **Start** và **End**: giờ bắt đầu và kết thúc.

* **Status**:

- Booked - slot đã được học viên đặt.
- Available - slot còn trống.

* **Actions**: Mentor có thể chỉnh sửa hoặc xóa slot:

- **Edit icon**: cập nhật giờ hoặc loại plan.
- **Delete icon**: xóa slot khỏi CSDL (nếu chưa được đặt).

– Bộ lọc danh sách:

- * Lọc theo ngày (**from – to**).
- * Lọc theo trạng thái (*All Status, Available, Booked*).
- * Lọc theo loại buổi mentoring (*Beginner, Standard, Advanced...*).
- * Nút **Reset** để xóa toàn bộ bộ lọc.

– Bộ đếm tổng quan slot:

- * **Total**: tổng số slot mentor đã tạo.
- * **Available**: số slot còn trống.
- * **Booked**: số slot đã được học viên đặt.

● **Lịch tháng (Calendar)**:

- Mentor chọn ngày cần tạo slot mới hoặc xem các slot đã có trong ngày đó.
- Dữ liệu hiển thị dựa trên truy vấn slot theo tháng.

● **Tạo mới khung giờ khả dụng (Add Available Time Slot)**:

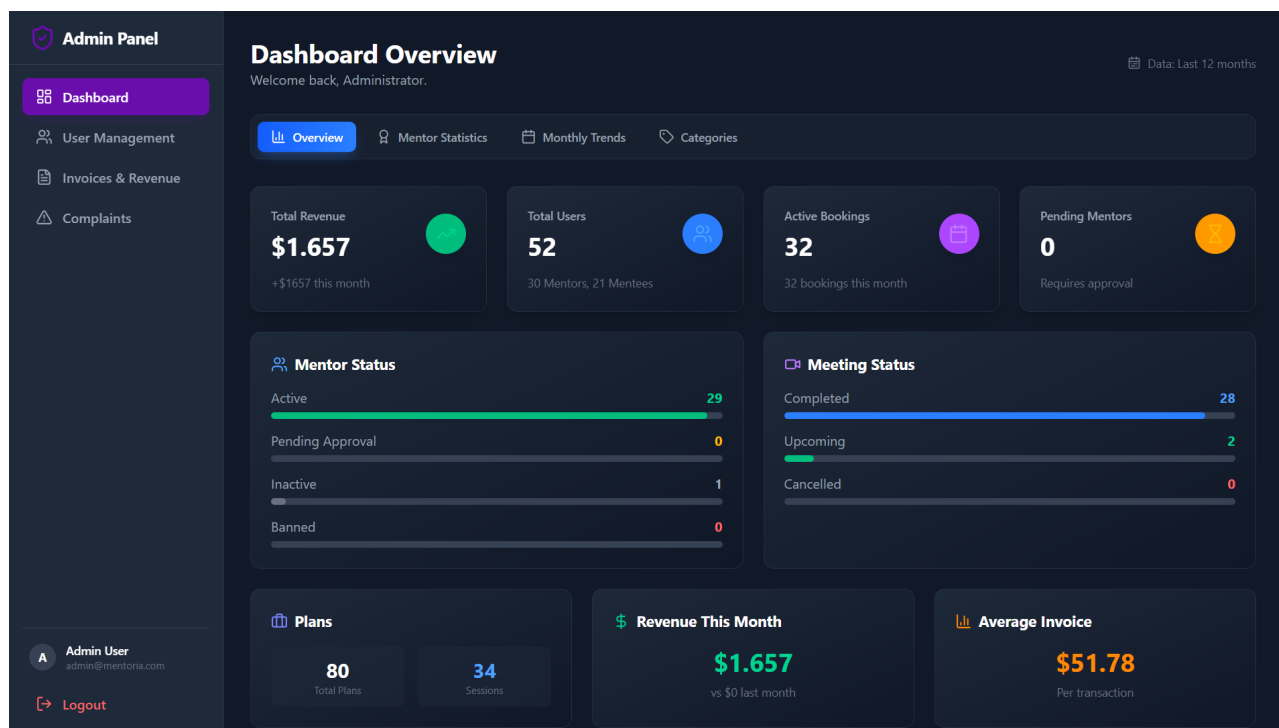
– Mentor chọn:

- * **Start Time**: thời gian bắt đầu.
- * **Plan Type**: loại buổi mentoring kèm giá (*Beginner – \$50, v.v.*).

– Khi bấm **Add Time Slot**, hệ thống:

- * Tạo slot mới.
- * Gắn slot với mentor đang đăng nhập.
- * Kiểm tra xung đột thời gian (không cho tạo nếu trùng slot).

3.4 Giao diện quản lý cho Admin

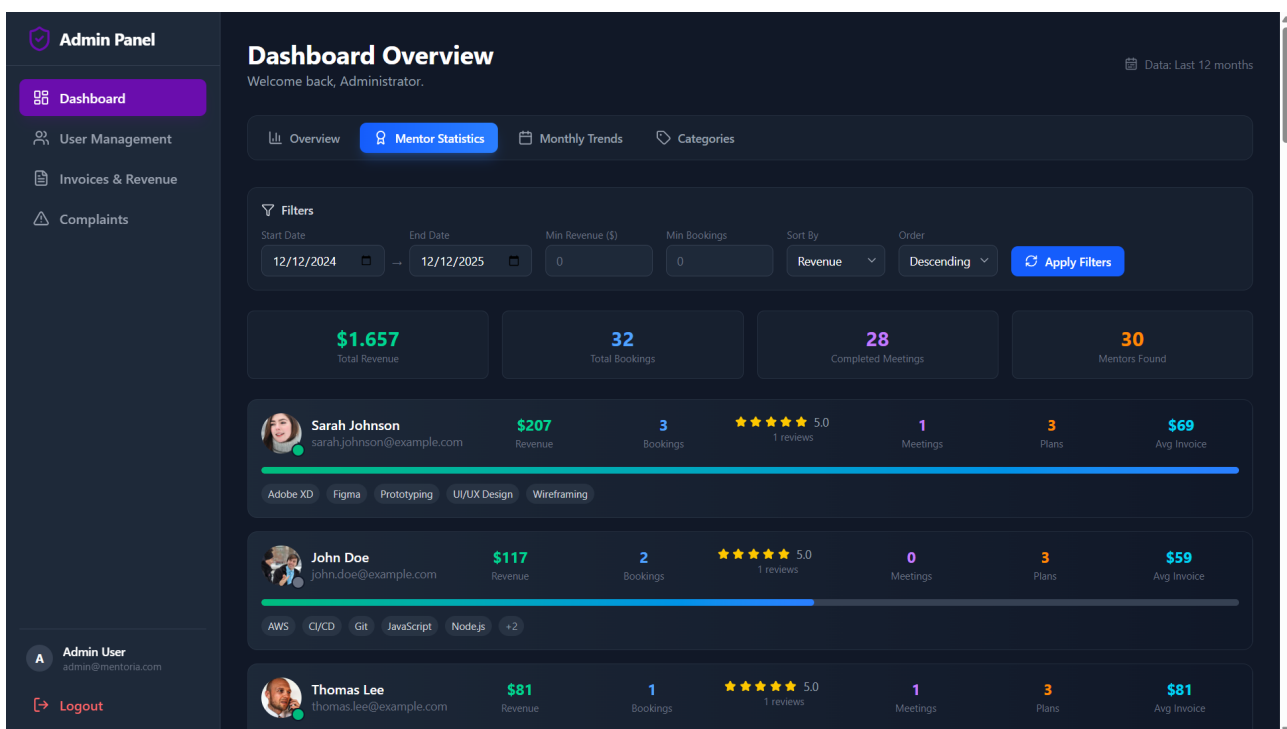


Hình 51: Giao diện Dashboard Overview

Màn hình này cung cấp cái nhìn tổng quan về toàn bộ hoạt động của hệ thống. Dữ liệu hiển thị được truy xuất từ thủ tục thống kê Dashboard.

- **Thống kê tổng quan:**
 - **Total Revenue:** tổng doanh thu trong 12 tháng gần nhất.
 - **Total Users:** tổng số mentor và mentee.
 - **Active Bookings:** số lượng booking trong tháng.
 - **Pending Mentors:** số mentor đang chờ duyệt.
- **Mentor Status:** Hiển thị số lượng mentor đang ở từng trạng thái: *Active*, *Pending Approval*, *Inactive*, *Banned*.

- **Meeting Status:** Thống kê số lượng meeting: *Completed, Upcoming, Cancelled*.
- **Plans Overview:**
 - Số lượng gói mentoring đang được mentor cung cấp.
 - Số lượng gói mentoring theo dạng sessions.
- **Revenue This Month:** So sánh doanh thu tháng hiện tại với tháng trước.
- **Average Invoice:** Giá trị trung bình mỗi hoá đơn.



Hình 52: Giao diện Mentor Statistics

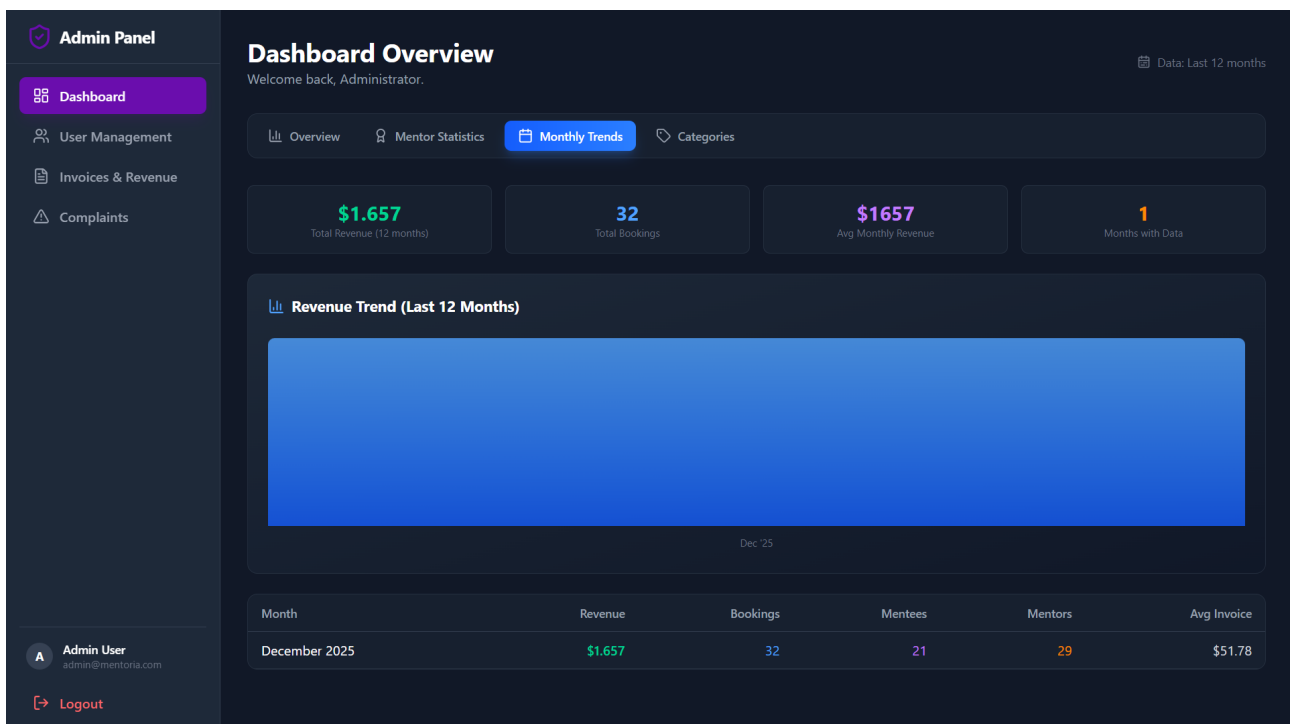
Màn hình cho phép admin xem thống kê chi tiết theo từng mentor.

- **Bộ lọc nâng cao:**
 - **Start Date / End Date:** giới hạn thời gian thống kê.
 - **Minimum Revenue, Minimum Bookings.**
 - **Sort By:** Revenue, Bookings, Rating, Meeting, Avg Invoice và Name.
 - **Order:** tăng dần hoặc giảm dần.
- **Thống kê tổng quan:**

- Tổng doanh thu.
- Tổng số booking.
- Số meeting đã hoàn thành.
- Số mentor tìm được.

• **Danh sách mentor chi tiết:** Mỗi mentor hiển thị:

- Tên, email, kỹ năng.
- Total Revenue.
- Số bookings.
- Rating và reviews.
- Số meeting.
- Số plan đang cung cấp.
- Average Invoice.



Hình 53: Giao diện Monthly Trends

Màn hình thể hiện xu hướng hoạt động của hệ thống theo từng tháng.

• **Chỉ số tổng quan:**

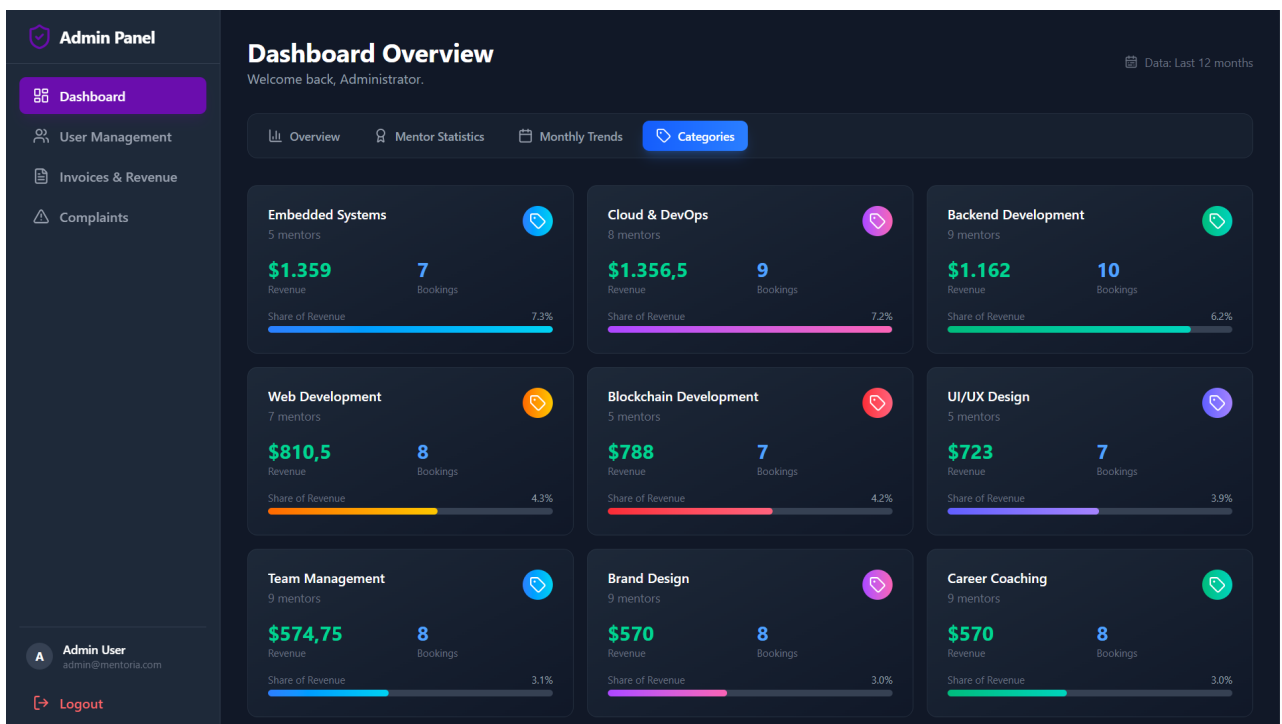
- Total Revenue (12 months).
- Total Bookings.
- Avg Monthly Revenue.
- Months with Data.

• **Biểu đồ Revenue Trend:**

- Trục X: tháng.
- Trục Y: doanh thu.

• **Bảng thống kê chi tiết:** Hiển thị theo từng tháng:

- Revenue.
- Số Booking.
- Số Mentee.
- Số Mentor.
- Average Invoice.



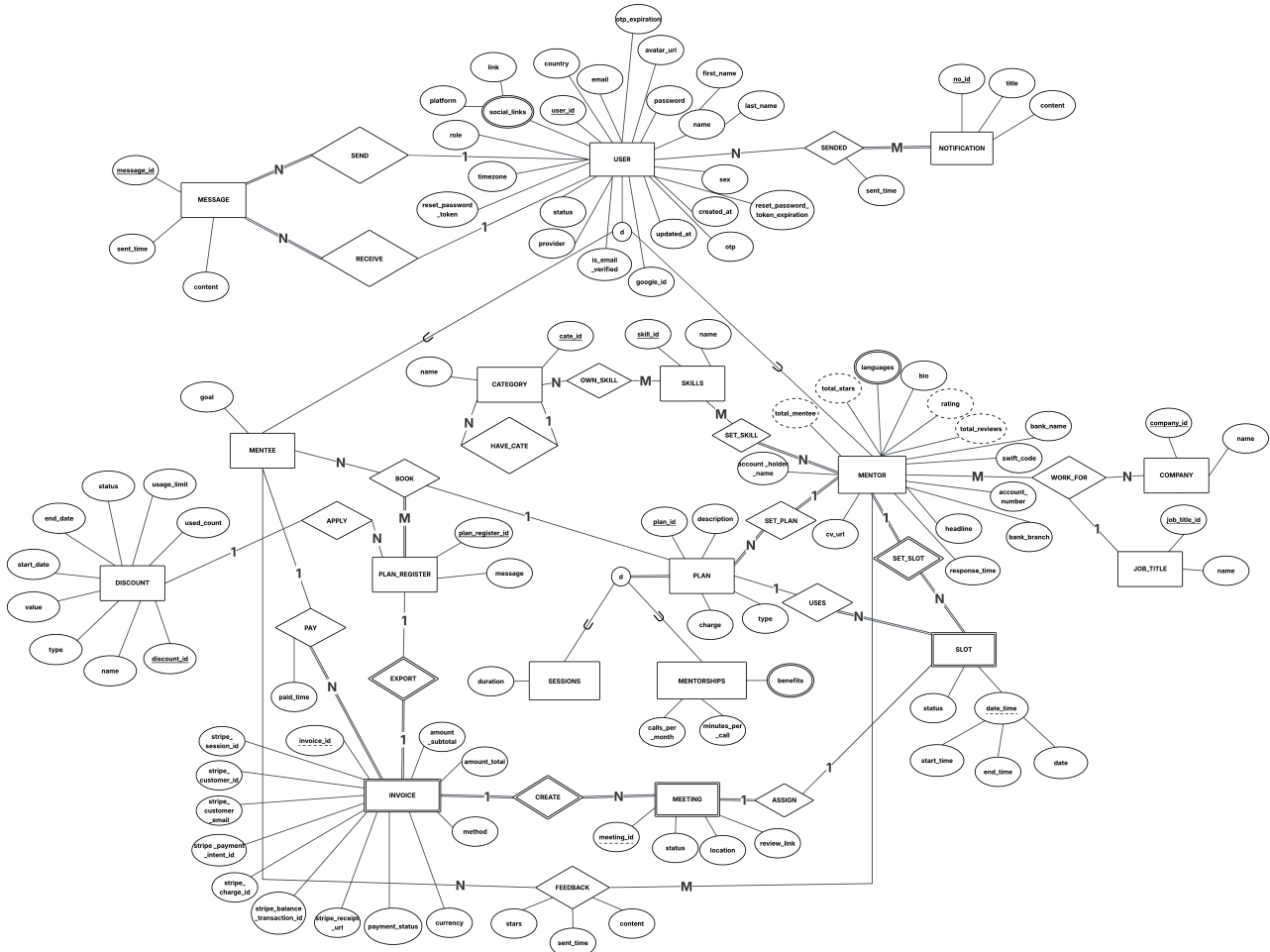
Hình 54: Giao diện Categories



Màn hình này phân tích doanh thu theo từng nhóm kỹ năng. Mỗi danh mục hiển thị:

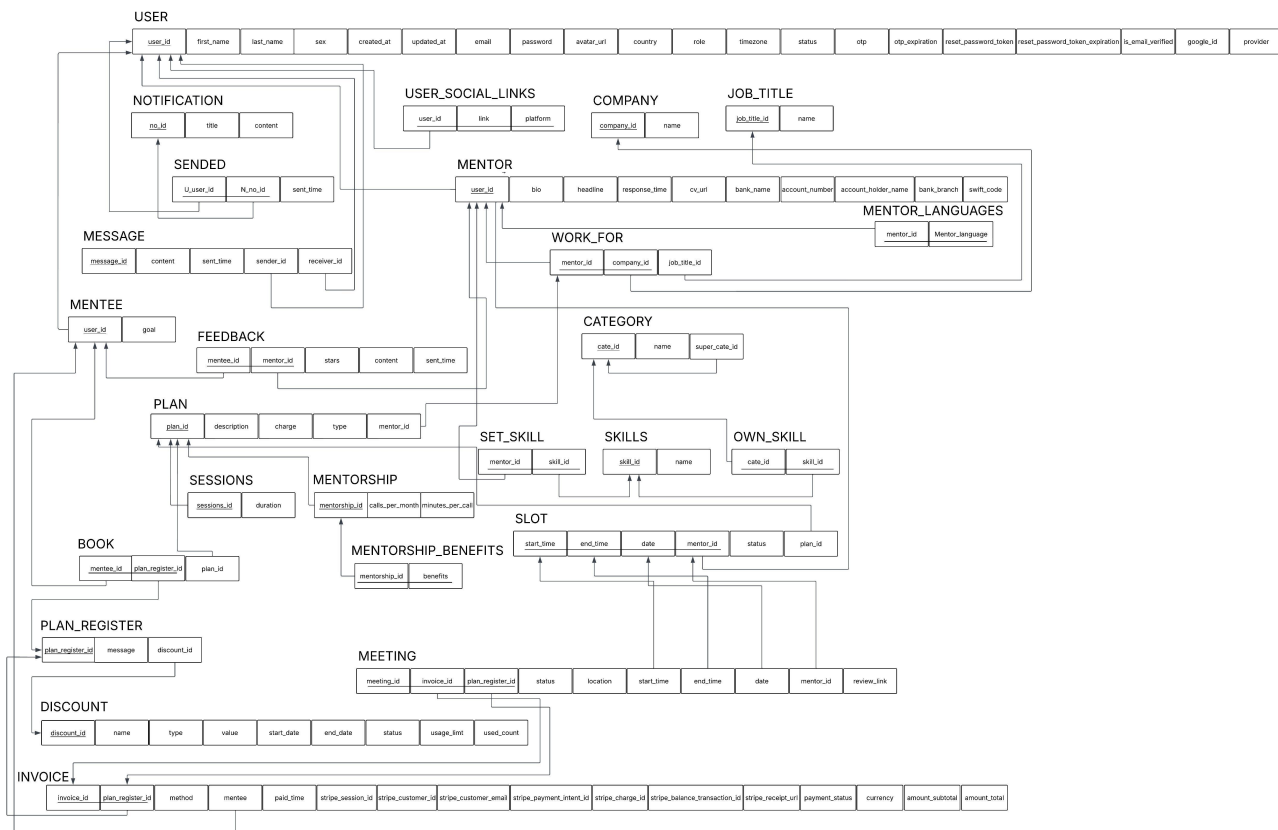
- Tên category.
- Số mentor thuộc category.
- Tổng revenue của category.
- Booking count của category.
- Share of revenue (tính phần trăm).

4 Thiết kế EERD



Hình 55: Mô hình EERD của hệ thống

5 Ảnh xạ lược đồ



Hình 56: Lược đồ CSDL quan hệ ánh xạ từ EERD đã thiết kế

6 Tổng kết

Trong Bài tập lớn 2, nhóm đã hoàn thiện quá trình hiện thực hoá cơ sở dữ liệu cho hệ thống *Hỗ trợ tìm kiếm mentor cho sinh viên* dựa trên EERD và lược đồ quan hệ đã xây dựng ở Bài tập lớn 1. Toàn bộ cấu trúc bảng, ràng buộc, thủ tục, trigger và hàm đều được triển khai trên SQL Server theo đúng yêu cầu kỹ thuật của đề bài.

Các nội dung kỹ thuật chính đã thực hiện bao gồm:

- **Tạo bảng và ràng buộc toàn vẹn:** Hoàn chỉnh các bảng dữ liệu, thiết lập CHECK, UNIQUE, FOREIGN KEY và các ràng buộc ngữ nghĩa nhằm bảo đảm tính nhất quán dữ liệu.
- **Chèn dữ liệu mẫu:** Xây dựng bộ dữ liệu thử nghiệm giúp kiểm chứng logic vận hành của hệ thống.
- **Stored Procedures:** Hiện thực bộ thủ tục INSERT/UPDATE/DELETE cho bảng `slots`, kèm các bước kiểm tra điều kiện nghiệp vụ như kiểm tra thời gian hợp lệ, xác nhận plan thuộc mentor, và tránh trùng slot.
- **Trigger:**
 - Trigger kiểm tra ràng buộc xác thực email trước khi kích hoạt tài khoản.
 - Trigger tính toán và cập nhật thuộc tính dẫn xuất: `total_reviews`, `total_stars`, `rating`.
 - Trigger cập nhật `total_mentee` dựa trên số buổi gặp ở trạng thái `Completed`.
- **Hàm:** Xây dựng hàm tính điểm đánh giá trung bình của mentor để tái sử dụng trong hệ thống.
- **Thủ tục truy vấn nâng cao:** Bao gồm tìm kiếm mentor theo nhiều tiêu chí và thống kê hiển thị dạng dashboard.

Source code đầy đủ của nhóm có tại đường dẫn:

<https://github.com/BaoDuong254/mentoria-database>. Toàn bộ các file SQL phục vụ xây dựng database (DDL, DML, stored procedures, triggers, functions, seed data) được lưu trong thư mục: `server/src/database/`.